



Crackz Documentation

AI Image Analysis

Version: 0.69.0

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Home

Crackz



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crackz

crackz-gui



```
my-project/
├─ images/      # Source images by split (raw/train/val/test)
├─ masks/      # Ground truth for training
├─ artifacts/   # Detection results and training metrics
├─ logs/        # Complete audit trail
└─ project.yaml # Configuration snapshot
```

```
# Create project
crackz init --name demo

# Import images
crackz import-images --project demo --src ./photos --type raw --size 512 512

# Run detection
crackz detect run --project demo

# View results in demo/artifacts/detections/
```

```
crackz-gui
# File → New Project → Import Images → Run Detection
```

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Getting Started

Getting Started with Crackz

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```
# Install UV package manager
curl -Lsf https://astral.sh/uv/install.sh | sh # macOS/Linux
# OR
irm https://astral.sh/uv/install.ps1 | iex # Windows PowerShell

# Authenticate with GitHub
gh auth login

# Download and install Crackz
gh release download --repo anaxiatech/crackz --pattern '*.whl'
uv pip install crackz-*.whl

# Verify installation
crackz --version
```

```
# Pull the Docker image
docker pull ghcr.io/anaxiatech-ab/crackz:latest

# Verify it works
docker run --rm ghcr.io/anaxiatech-ab/crackz:latest --version
```

```
# Create a new project
crackz init --name my-first-project

# This creates a structured directory:
# my-first-project/
```

```
# |— images/      # Your photos organized by type
# |— masks/      # Ground truth (for training)
# |— artifacts/   # Results and metrics
# |— logs/        # Audit trail
```

```
crackz-gui
# Then: File → New Project (Ctrl/Cmd+N)
```

```
# Import images for crack detection
crackz import-images \
  --project my-first-project \
  --src ./your-photos \
  --type raw \
  --size 512 512
```

```
# Run detection using the pre-trained model
crackz detect run --project my-first-project
```

```
my-first-project/artifacts/detections/
project/artifacts/detections/metrics.json my-first-
```

```
# 1. Create project
crackz init --name harbor-inspection

# 2. Get pre-trained model checkpoint
# (Contact your Crackz provider or use a partner model)
# Place it in: harbor-inspection/artifacts/training/checkpoints/best.ckpt

# 3. Import images
crackz import-images --project harbor-inspection --src ./photos --type raw --size 512 512

# 4. Run detection (uses the pre-trained model automatically)
crackz detect run --project harbor-inspection
```

```
# 1. Create project
crackz init --name custom-harbor

# 2. Import labeled training data
crackz import-images \
  --project custom-harbor \
  --src ./labeled/images \
  --src-masks ./labeled/masks \
  --type train \
  --size 512 512

# 3. Import validation data
crackz import-images \
  --project custom-harbor \
  --src ./labeled/val_imgs \
  --src-masks ./labeled/val_masks \
  --type val \
  --size 512 512

# 4. Train your custom model
crackz detect train --project custom-harbor --epochs 20

# 5. Run detection with your trained model
crackz detect run --project custom-harbor
```

```
my-first-project/
├── images/
│   ├── raw/      # Original images for detection
│   ├── train/    # Training images (if training a model)
│   ├── val/      # Validation images
│   └── test/     # Test images
├── masks/
│   ├── train/    # Ground truth masks for training
│   ├── val/      # Validation masks
│   └── test/     # Test masks
├── artifacts/
│   ├── training/ # Model checkpoints and training metrics
│   └── detections/ # Detection results (images + JSON)
├── logs/         # Project-specific logs
└── project.yaml  # Configuration file
```

```
my-first-project/my-first-project_project.yaml
```

```
ai:
  model:
    threshold: 0.5 # Increase to 0.7 for fewer false positives
                  # Decrease to 0.3 to catch more cracks
```

```
# Process several inspection campaigns
for project in site-A site-B site-C; do
  crackz detect run --project $project
done
```

```
# Run detection in container
docker run --rm \
  -v $(pwd)/my-first-project:/home/app \
  ghcr.io/anaxiatech-ab/crackz:latest \
  detect run --project /home/app
```

--	--	--	--

```
# Verify CUDA installation
nvidia-smi

# Check PyTorch CUDA support
python -c "import torch; print(torch.cuda.is_available())"

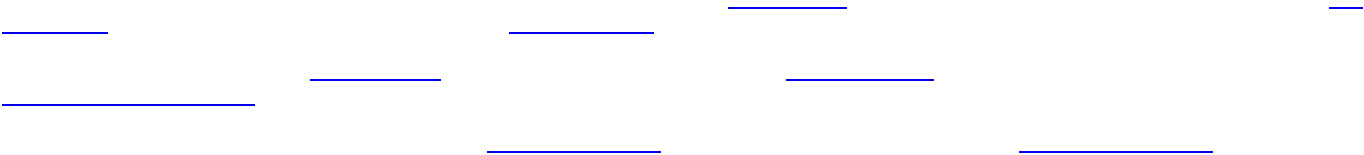
# Reinstall with CUDA support (see Installation Guide)
```

images/raw/ .jpg .jpeg .png .bmp .tif .tiff
project.yaml

```
# Reduce batch size
crackz detect train --project my-project --batch-size 1

# Or reduce image size during import
crackz import-images --project my-project --src ./data --type train --size 256 256
```

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```
crackz init --name my-first-project
```

Installation

Installation

uv sync

LICENSE

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gh auth login

curl -LsSf https://astral.sh/uv/install.sh | sh

irm https://astral.sh/uv/install.ps1 | iex

Login to GitHub
gh auth login

Download the latest wheel
gh release download --repo anaxiatech/crackz --pattern '*.whl'

Install with uv
uv pip install crackz-*.whl

Verify installation
crackz --version

crackz-VERSION-py3-none-any.whl

uv pip install crackz-VERSION-py3-none-any.whl

crackz

crackz-gui


```
# Using pip with GitHub authentication
pip install git+https://github.com/OWNER/crackz.git@vVERSION

# Or download the wheel from the private repository's Releases page
# (Requires GitHub authentication - use personal access token)
pip install https://USERNAME:TOKEN@github.com/OWNER/crackz/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl
```

```
pip install git+https://github.com/OWNER/crackz.git
```

```
# Login with GitHub CLI
gh auth login

# Download release asset
gh release download vVERSION --repo OWNER/crackz --pattern '*.whl'

# Install the downloaded wheel
pip install crackz-VERSION-py3-none-any.whl
```

```
# Create a personal access token with 'repo' scope at:
# https://github.com/settings/tokens

# Set your token (keep it secure!)
export GITHUB_TOKEN="ghp_XXXXXXXXXX" # gitleaks:allow

# Install using token authentication
pip install git+https://${GITHUB_TOKEN}@github.com/OWNER/crackz.git@vVERSION
```

```
https://github.com/OWNER/crackz
crackz-VERSION-py3-none-any.whl
```

```
pip install crackz-VERSION-py3-none-any.whl
```

```
# Download wheel using GitHub token
curl -L \
  -H "Authorization: token ${GITHUB_TOKEN}" \
  -o crackz.whl \
  https://github.com/OWNER/crackz/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl

# Install the downloaded wheel
pip install crackz.whl
```

```
# If your organization has configured a private index
pip install --index-url https://pypi.company.com/simple/ crackz
```

```
# Using the provided index URL and credentials
pip install crackz --index-url https://INDEX_URL/simple/

# Example with authentication (if required):
pip install crackz --index-url https://USERNAME:TOKEN@pypi.company.com/simple/
```

```
~/.config/pip/pip.conf
```

```
%APPDATA%\pip\pip.ini
```

```
[global]
index-url = https://USERNAME:TOKEN@pypi.company.com/simple/
```

```
pip install crackz
pip install --upgrade crackz # For updates
```

```
# Set credentials via environment variables
export PIP_INDEX_URL="https://pypi.company.com/simple/"
export PIP_EXTRA_INDEX_URL="https://pypi.org/simple" # Fallback to PyPI for dependencies

pip install crackz
```

```
https://dl.cloudsmith.io/TOKEN/org/repo/python/simple/
https://artifactory.company.com/artifactory/api/pypi/pypi-local/simple/
https://pypi.fury.io/TOKEN/USERNAME/
https://aws:TOKEN@domain.d.codeartifact.region.amazonaws.com/pypi/repo/simple/
https://pkgs.dev.azure.com/org/_packaging/feed/pypi/simple/ https://devpi.company.com/username/index/+simple/
```

```
chmod 600 ~/.config/pip/pip.conf # Linux/macOS
```

```
pip install --trusted-host pypi.company.com crackz
# Or add CA cert:
pip install --cert /path/to/ca-bundle.crt crackz
```

```
pip install crackz --extra-index-url https://pypi.org/simple/
```

crackz-gui.exe

crackz

crackz-gui

crackz

crackz-gui

crackz.exe

```
REM Download the executables from GitHub Releases
REM Move them to a permanent location
mkdir C:\Program Files\Crackz
move crackz.exe "C:\Program Files\Crackz\"
move crackz-gui.exe "C:\Program Files\Crackz\"

REM Add to PATH (optional, for CLI access from anywhere)
setx PATH "%PATH%;C:\Program Files\Crackz"
```

```
# Download the executables from GitHub Releases
chmod +x crackz crackz-gui

# Move to a system directory (optional)
sudo mv crackz /usr/local/bin/
sudo mv crackz-gui /usr/local/bin/

# Or keep in a local directory and add to PATH
mkdir -p ~/Applications/crackz
mv crackz crackz-gui ~/Applications/crackz/
echo 'export PATH="$HOME/Applications/crackz:$PATH"' >> ~/.bashrc
source ~/.bashrc
```

```
# Run CLI directly (no Python needed)
crackz --help
crackz init --name myproject

# Run GUI by double-clicking crackz-gui.exe (Windows)
# Or run from command line:
crackz-gui
```

cuda.exe

crackz-cuda.exe

crackz-gui-

```
crackz --help
python -c "import crackz; import importlib.metadata as m; print(m.version('crackz'))"
```

- crackz-VERSION-py3-none-any.whl
- crackz_cu118-VERSION-py3-none-any.whl
- crackz_cu128-VERSION-py3-none-any.whl

```
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl
```

```
# Step 1: Install PyTorch with CUDA 11.8
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu118

# Step 2: Install crackz CUDA variant
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz_cu118-VERSION-py3-none-any.whl
```

```
# Step 1: Install PyTorch with CUDA 12.8
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu128

# Step 2: Install crackz CUDA variant
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz_cu128-VERSION-py3-none-any.whl
```

```
nvidia-smi
```

```
nvcc --version
# Or check driver
nvidia-smi | grep "CUDA Version"
```

crackz_cu118

crackz_cu128

crackz

```
# Login first
gh auth login

# Download and install CPU version (one step)
gh release download vVERSION --repo OWNER/crackz --pattern 'crackz-*.whl'
pip install crackz-VERSION-py3-none-any.whl
```

```
# Login first
gh auth login
```

```
# Step 1: Install PyTorch with CUDA 12.8
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu128

# Step 2: Download and install crackz CUDA variant
gh release download vVERSION --repo OWNER/crackz --pattern 'crackz_cu128-*.whl'
pip install crackz_cu128-VERSION-py3-none-any.whl
```

```
# Check CUDA availability
python -c "import torch; print(f'CUDA available: {torch.cuda.is_available()}')"

# Check PyTorch version (should include +cu118 or +cu128 suffix)
python -c "import torch; print(f'PyTorch version: {torch.__version__}')"

# Check GPU device name
python -c "import torch; print(f'Device: {torch.cuda.get_device_name(0)} if torch.cuda.is_available() else \"CPU\"')"
```

```
CUDA available: True
PyTorch version: 2.5.0+cu128
Device: NVIDIA GeForce RTX 4090
```

```
# Step 1: Install specific PyTorch version
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu121

# Step 2: Install crackz CUDA variant (cu118 or cu128, either works)
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz_cu118-VERSION-py3-none-any.whl
```

cpu cu128

```
# This installs CPU PyTorch even with the cu128 extra - BROKEN!
pip install "crackz-VERSION-py3-none-any.whl[cu128]"
```

```
# Step 1: Install PyTorch with CUDA
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu128

# Step 2: Install crackz
pip install crackz_cu128-VERSION-py3-none-any.whl
```

keyring

```
# With uv
uv add keyring

# With pip
pip install keyring
```

```

from crackz_gui.state.secure_storage import (
    store_api_key,
    retrieve_api_key,
    is_secure_storage_available,
)
# Check availability
if is_secure_storage_available():
    store_api_key("your-api-key")
    api_key = retrieve_api_key()

```

1.

```

mkdir crackz_wheels
cd crackz_wheels

# Download Crackz wheel
wget https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl

# Download all dependencies
pip download -r <(pip show -f crackz | grep Requires) -d .
# Or download from the wheel itself:
pip download crackz-VERSION-py3-none-any.whl -d .

```

2.

3.

```
pip install --no-index --find-links crackz_wheels crackz-VERSION-py3-none-any.whl
```

```
pip install .
```

```
pyproject.toml
```

```

# Download the new version from GitHub Releases
pip install --upgrade https://github.com/Anaxiatech-AB/crackz/releases/download/vNEW_VERSION/crackz-NEW_VERSION-py3-none-any.whl

# Or if using a private index:
pip install -U --index-url https://pypi.company.com/simple/ crackz

```

```
pip uninstall crackz
```

```

crackz --version
crackz init --name sanity-test
crackz detect run --project sanity-test # (Will use default config; may download model weights)

```

```
torch.__version__
```

```
No module named crackz
```

```
CUDA not available
```

```
torch.__version__
```

```
ModuleNotFoundError: No module named
'torch'
```

```
CRACKZ_LOG_DIR
```

```
pip install --force-reinstall crackz-VERSION.whl
```

```
pyproject.toml
```

```
# Check if CUDA is available
python -c "import torch; print(f'CUDA available: {torch.cuda.is_available()}')"
python -c "import torch; print(f'CUDA device: {torch.cuda.get_device_name(0)}')"
```

```
CUDA available: True
CUDA device: NVIDIA GeForce RTX 3080
```

```
# These commands automatically use GPU if available:
crackz detect train --project my_project
crackz detect run --project my_project
crackz-gui # GUI will use GPU automatically
```

```
cpu
```

```
cuda
```

```
mps
```

```
# Force GPU usage (will error if CUDA not available)
crackz detect train --project my_project --device cuda

# Force CPU usage (useful for debugging)
crackz detect train --project my_project --device cpu

# Let Crackz auto-detect (default)
crackz detect train --project my_project
```

```
<project>/<project>_project.yaml
```

```
ai:
  model:
    device: cuda # or cpu, or mps
```

```
crackz detect train --project my_project
```

```
GPU available: True, used: True
TPU available: False, using: 0 TPU cores
IPU available: False, using: 0 IPUs
HPU available: False, using: 0 HPUs
```

```
# NVIDIA GPUs
watch -n 1 nvidia-smi
```

```
# Should show Python process using GPU memory and compute
```

```
crackz_cu118-VERSION-py3-none-any.whl
```

```
crackz_cu128-VERSION-py3-none-any.whl
```

```
crackz_cu118-VERSION-py3-none-any.whl
```

```
nvcc --version # If CUDA toolkit installed
nvidia-smi # Shows driver version (different from CUDA toolkit)
```

1.

```
python -c "import torch; print(torch.__version__)"
# Should show: 2.5.0+cu118 or 2.5.0+cu128
# If it shows just 2.5.0, you have CPU-only version!
```

2.

```
pip uninstall torch torchvision
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu118
```

3.

```
python -c "import torch; print(f'CUDA available: {torch.cuda.is_available()}')"
```

```
# GUI: Project → Settings → Training → Batch Size
# CLI:
crackz detect train --project my_project --batch-size 2
```

- `nvidia-smi`
-
-



```
CRACKZ_LOG_DIR=my_logs crackz detect run --project myproj
```

```
$Env:CRACKZ_LOG_DIR='my_logs'
```

- `pip install --require-hashes -r requirements.txt`
-

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```
crackz migrate --project ./my_project # applies changes + backup
crackz migrate --project ./my_project --dry-run # preview only
```

```
CRACKZ_AUTO_MIGRATE=1
```

User Guide

User Guide

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crackz detect train

crackz detect run

```
graph LR
  A[Create Project] --> B[Import Images]
  B --> C{Need Custom Model?}
  C -->|Yes| D[Train Model]
  C -->|No| E[Run Detection]
  D --> E
  E --> F[View Results]
  F --> G{Satisfied?}
  G -->|No| H[Adjust Config]
  H --> D
  G -->|Yes| I[Export/Package]
```

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```
nvidia-smi
```

```
# Install from GitHub Release (replace VERSION with actual version)
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl

# Or download first, then install
wget https://github.com/Anaxiatech-AB/crackz/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl
pip install crackz-0.30.2-py3-none-any.whl

# Verify installation
crackz --help
```

```
# Install PyTorch with CUDA support first
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu121

# Then install Crackz from GitHub Release
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl
```

```
# CPU version
docker pull ghcr.io/anaxiatech-ab/crackz:latest

# GPU version (requires nvidia-docker)
docker pull ghcr.io/anaxiatech-ab/crackz:gpu

# Slim version (minimal runtime)
docker pull ghcr.io/anaxiatech-ab/crackz:slim
```

```
crackz --version
python -c "import crackz; print('Crackz installed successfully')"
```

```
crackz init --name my-first-project
cd my-first-project
```

```
my-first-project/
├── images/           # Image storage by split
│   ├── raw/         # Unprocessed images for inference
│   ├── train/       # Training images
│   ├── val/         # Validation images
│   └── test/        # Test images
├── masks/           # Ground truth masks (optional for raw)
│   ├── train/
│   ├── val/
│   └── test/
├── artifacts/       # Model outputs (metrics, plots)
├── detections/      # Detection results (annotated images, JSON)
├── models/          # Saved model weights (or checkpoints/ in older projects)
├── logs/            # Project-specific logs
└── project.yaml     # Project configuration
```

checkpoints/

models/

crackz migrate --project <project> --migrate-files

```
# Import sample images for detection (no training)
crackz import-images \
  --project my-first-project \
  --src ./samples \
  --type raw \
  --size 512 512
```

images/raw/

```
# Use pre-trained model
crackz detect run --project my-first-project
```

detections/

artifacts/metrics.json

```
# Import training images with masks
crackz import-images \
  --project my-first-project \
  --src ./training-data/images \
  --src-masks ./training-data/masks \
  --type train \
  --size 512 512
```

```
# Quick training (1 epoch for testing)
crackz detect train --project my-first-project --epochs 1 --batch-size 2
```

```
# Check detection outputs
ls my-first-project/detections/
```

```
# View metrics
cat my-first-project/artifacts/metrics.json
```

```
# Open annotated images
open my-first-project/detections/*.png # macOS
xdg-open my-first-project/detections/*.png # Linux
```

```
# Launch Streamlit dashboard for advanced analysis
crackz dashboard --project my-first-project
```

```
# Open project in GUI and launch dashboard
crackz-gui
# Then: Tools → Launch Dashboard
```

train/val/test/

images/

masks/

raw/

detections/

logs/

project.yaml

models/

artifacts/

checkpoints/

confusion_matrix.png

metrics.json

loss_curve.png

detections/

metrics.json

CRACKZ_LOG_DIR

<project>/logs/

```
# Set central log directory
export CRACKZ_LOG_DIR=~/.crackz-logs

# Both central and project logs will be written
crackz detect run --project my-project

--debug --quiet
```

```
schema_version: "1.0"
project:
  name: my-project
  # ... rest of config
```

```
crackz migrate --project old-project
```

[_____](#)

num_classes: 1

num_classes: 3

efficientnet-b3

```
efficientnet-b0
resnet34
efficientnet-b7
resnet50
resnet101
mobilenet_v2
resnet18
densenet121
```

<project>/<project>_project.yaml

```
ai:
  model:
    encoder: resnet34          # Choose encoder backbone
    encoder_weights: imagenet  # Use transfer learning (recommended)
    in_channels: 3            # RGB images (use 1 for grayscale)
    num_classes: 1            # Binary segmentation
    threshold: 0.5            # Detection sensitivity (0.3=more sensitive, 0.7=less)

  training:
    epochs: 20                # Number of training iterations
    batch_size: 8              # Images per batch (reduce if out of memory)
    learning_rate: 0.001       # Optimizer learning rate
```

```
crackz model import --project ./my_project --model pretrained_model.zip
```

```
crackz model export --project ./my_project --out my_model.zip
```

```
project <name>                                project.yaml  ai.model.encoder  crackz detect train --
```

```
crackz init --name harbor-inspection-2025 --description "Q1 2025 harbor crack survey"
```

```
crackz-gui
```

```
# Create with custom settings
crackz init --name custom-project --force # Overwrite if exists

# Initialize from template (future)
crackz init --name new-project --template harbor-config.yaml
```

```
crackz import-images \
  --project harbor-inspection-2025 \
  --src ~/Pictures/harbor-photos \
  --type raw \
  --size 512 512
```

```
crackz import-images \
  --project harbor-inspection-2025 \
  --src ~/labeled-data/images \
  --src-masks ~/labeled-data/masks \
  --type train \
  --size 512 512
```

```
# Training data
crackz import-images --project my-project --src train_imgs --src-masks train_masks --type train --size 512 512

# Validation data
crackz import-images --project my-project --src val_imgs --src-masks val_masks --type val --size 512 512

# Test data
crackz import-images --project my-project --src test_imgs --src-masks test_masks --type test --size 512 512
```

```
.jpg .jpeg .png .bmp .tif .tiff .png
```

```
crackz-gui
```

```
images/train/
```

```
B
```

```
E
```

```
masks/train/
```

-
-
-
-
-

B

E

Ctrl+Z

B

E

Ctrl+Z

```
crackz detect train --project my-project
```

```
crackz detect train \  
--project my-project \  
--epochs 20 \  
--batch-size 8 \  
--learning-rate 0.001
```

project.yaml

```
# Automatically uses GPU if available  
crackz detect train --project my-project --epochs 50  
  
# Verify GPU is being used (check logs)  
tail -f my-project/logs/crackz.log | grep -i cuda
```

```
# Watch loss in real-time  
tail -f my-project/artifacts/metrics.json  
  
# Or use TensorBoard (if enabled)  
tensorboard --logdir my-project/artifacts/tensorboard
```

project.yaml

```
crackz detect run --project my-project
```

```
# Uses GPU automatically if available
crackz detect run --project my-project

# Force CPU (even with GPU available)
CUDA_VISIBLE_DEVICES="" crackz detect run --project my-project
```

```
--model
```

```
# Auto-select (default): best.ckpt > last.ckpt > final.ckpt
crackz detect run --project my-project

# Explicitly use best checkpoint
crackz detect run --project my-project --model best

# Use most recent checkpoint (useful during iterative development)
crackz detect run --project my-project --model last

# Use final checkpoint
crackz detect run --project my-project --model final

# Use specific epoch checkpoint (e.g., epoch-0.9234.ckpt)
crackz detect run --project my-project --model epoch-0.9234
```

```
--model
```

```
--model last
```

```
--model best
```

```
--model final
```

```
--model epoch-X
```

```
# Run detection with different checkpoints to compare results
crackz detect run --project my-project --model best
mv my-project/detections my-project/detections-best

crackz detect run --project my-project --model last
mv my-project/detections my-project/detections-last

crackz detect run --project my-project --model epoch-0.8532
mv my-project/detections my-project/detections-epoch-0.8532

# Compare detection outputs side by side
```

```
# Detect on raw images (default)
crackz detect run --project my-project

# Detect on test set (with metrics if masks available)
crackz detect run --project my-project --split test
```

```
# Process multiple projects
for project in project1 project2 project3; do
  crackz detect run --project $project
done
```

```
# CPU
docker run --rm \
  -v $(pwd)/my-project:/home/app/my-project \
  ghcr.io/anaxiatech-ab/crackz-cli:latest \
  detect run --project /home/app/my-project

# GPU
docker run --gpus all --rm \
  -v $(pwd)/my-project:/home/app/my-project \
  ghcr.io/anaxiatech-ab/crackz-cli:gpu \
  detect run --project /home/app/my-project
```

```
project.yaml
```

```
training:
  tensorboard: true
```

```
# tensorboard-config.yaml
training:
  tensorboard: true
```



```
# After training with tensorboard enabled
tensorboard --logdir my-project/artifacts/tensorboard

# Open browser to http://localhost:6006
```

```
# Quick summary
cat my-project/artifacts/metrics.json | python -m json.tool

# Extract specific metrics
cat my-project/artifacts/metrics.json | jq '.final_val_iou'
```

```
crackz clean --project my-project --dry-run
```

```
crackz clean --project my-project
```

```
crackz clean --project my-project --logs-only
```

```
*.log
```

```
project.yaml
```

```
training:
  checkpoint:
    save_top_k: 3 # Keep top 3 checkpoints by validation loss
```

```
# Remove all but the last checkpoint
rm my-project/checkpoints/*.ckpt
# (except last-*.ckpt)

# Or keep only best checkpoint
rm my-project/checkpoints/epoch*.ckpt
# (keep best-*.ckpt)
```

```
# Train on 50-100 manually labeled images
crackz detect train --project harbor-project --epochs 10
```

```
# Prioritize top 50 images from raw/unlabeled pool
uv run crackz active-learning run \
  --project harbor-project \
  --split raw \
  --top-n 50 \
  --method entropy \
  --out priorities.json
```

variance

entropy

mean_confidence

max_probability

```
# Use GUI mask editor or external tools
# Focus on top-ranked images from priorities.json
```

```
# Move annotated images to train split
crackz import-images --project harbor-project \
  --src annotated_batch \
  --src-masks annotated_masks \
  --type train

# Retrain with expanded dataset
crackz detect train --project harbor-project --epochs 15
```

wall_section_01.jpg

```
# Preview similarity scores
uv run crackz propagate-masks find-similar \
  --project harbor-project \
  --reference wall_section_01.jpg \
  --split raw \
  --out similarities.json
```

```
# Copy mask to similar images (>80% similarity)
uv run crackz propagate-masks run \
  --project harbor-project \
  --reference wall_section_01.jpg \
  --mask wall_section_01_mask.png \
  --split raw \
  --threshold 0.8 \
  --out propagated_masks/
```

```
# Import propagated masks
crackz import-images --project harbor-project \
  --src images_with_propagated \
  --src-masks propagated_masks \
  --type train

# Review in GUI mask editor - correct errors before training
```

```
# More permissive (more propagations, less reliable)
--threshold 0.6 --min-matches 5

# Stricter (fewer propagations, higher quality)
--threshold 0.9 --min-matches 20 --sift
```

```
crackz detect train --project harbor-project
```

```
uv run crackz active-learning run \
  --project harbor-project \
  --top-n 20 \
  --method entropy \
  --out uncertain.json
```

```
# For each reference image
uv run crackz propagate-masks run \
  --reference refl.jpg --mask refl_mask.png \
  --project harbor-project --threshold 0.8

# Propagate remaining references...
```

```
# Import all propagated masks
crackz import-images --project harbor-project \
  --src propagated_images \
  --src-masks propagated_masks \
  --type train

# Retrain with 50 original + ~40 propagated = 90 annotations
crackz detect train --project harbor-project --epochs 15
```

```
crackz export-config \
  --project my-project \
  --out my-project-snapshot.yaml
```

```
# Create archive (excluding large files)
tar -czf my-project.tar.gz \
  --exclude='checkpoints' \
  --exclude='detections' \
  my-project/

# Send to collaborator
# They extract and run:
tar -xzf my-project.tar.gz
crackz detect run --project my-project # Downloads default checkpoint if missing
```

```
latest
slim
gpu
```

```
# Full image
docker build -t crackz-cli .

# Slim image
docker build -f Dockerfile.slim -t crackz-cli:slim .

# GPU image
docker build -f Dockerfile.gpu -t crackz-cli:gpu .
```

```
# CPU
docker run --rm \
-v $(pwd)/my-project:/home/app/my-project \
crackz-cli detect run --project /home/app

# GPU
docker run --gpus all --rm \
-v $(pwd)/my-project:/home/app/my-project \
crackz-cli:gpu detect run --project /home/app
```

```
docker run -it --rm \
-v $(pwd)/my-project:/home/app/my-project \
crackz-cli bash
```

Makefile

```
# Build documentation
make docs

# Run tests
make test

# Build Docker image
make docker-build

# Clean artifacts
make clean
```

```
# Add to $PROFILE
function Invoke-Crackz {
    docker run --rm -v $(pwd):/home/app ghcr.io/anaxiatech-ab/crackz-cli:latest @args
}
Set-Alias crackz Invoke-Crackz

# Usage
crackz detect run --project /home/app/my-project
```

```
# process-all.ps1
$projects = Get-ChildItem -Directory -Filter "*-project"
foreach ($project in $projects) {
    Write-Host "Processing $($project.Name)..."
    crackz detect run --project $project.FullName
}
```

```
crackz-gui
```

```
python -m crackz_gui.qt.app
```

`project.yaml``images/raw``detections/`

-
-
-

```
# Install from GitHub Release
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl
crackz init --name local-project
```

```
docker pull ghcr.io/anaxiatech-ab/crackz-cli:latest
docker run --rm -v $(pwd)/project:/home/app crackz-cli detect run --project /home/app
```

```
# Create Batch pool
./scripts/azure/azure_batch.sh \
  --batch-account myaccount \
  --storage-account mystorage
```

```
# Submit job
./scripts/azure/submit_batch_job.sh \
--project my-project \
--pool-id crackz-pool
```

Solution:

```
# Update pip and setuptools
pip install --upgrade pip setuptools

# Download and install from GitHub Release
wget https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl
pip install crackz-VERSION-py3-none-any.whl
```

Solution:

```
# Verify CUDA installation
nvidia-smi

# Check PyTorch CUDA support
python -c "import torch; print(torch.cuda.is_available())"

# Reinstall PyTorch with CUDA
pip uninstall torch torchvision
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu121
```

No images found in training split

Solution:

```
images/train/      images/raw/      .jpg .jpeg .png .bmp .tif .tiff
--type train
```

Solution:

```
masks/train/
```

Solution:

```
# Reduce batch size
crackz detect train --project my-project --batch-size 1

# Or reduce image size
crackz import-images --project my-project --src ./data --type train --size 256 256

# Monitor GPU memory
nvidia-smi -l 1 # Update every second
```

Solution:

```
tail my-project/logs/crackz.log      raw/
ls my-project/models/      my-project/checkpoints/      crackz detect run --project my-project --debug
```

Solution: project.yaml

```
detection:
  threshold: 0.5 # Pixel-level binarization threshold
                # Increase to reduce false positives (0.7)
                # Decrease to catch more cracks (0.3)

# Severity categorization thresholds (crack area %)
min_confidence: 0.005 # Min to save (0.5% crack area)
medium_confidence: 0.02 # "Probable Crack" threshold (2%)
high_confidence: 0.05 # "Crack Detected" threshold (5%)
min_crack_area_px: 100 # Filter out detections < 100 pixels
```

Solution:

```
# Check permissions
ls -la $(python -c "import tempfile; print(tempfile.gettempdir())")/crackz_logs

# Set explicit log directory
export CRACKZ_LOG_DIR=~/.crackz-logs
mkdir -p ~/.crackz-logs
crackz detect run --project my-project
```

Solution:

```
# Clean old logs
rm -rf $CRACKZ_LOG_DIR/*.log

# Or set retention policy (future feature)
# See configuration.md for log rotation settings
```

Missing --project or --cfg	--project <name>	--cfg <file>
CUDA out of memory	--batch-size	
Schema version mismatch	crackz migrate --project <name>	
Permission denied writing logs		CRACKZ_LOG_DIR
No checkpoint found	crackz detect train --project <name>	



<project>/logs/ \$CRACKZ_LOG_DIR /tmp/crackz_logs

```
crackz detect train --project my-project --epochs 50 --batch-size 8
```

project.yaml

```
crackz detect run --cfg my-custom-config.yaml
```

```
crackz export-config --project my-project --out snapshot.yaml
crackz detect run --cfg snapshot.yaml
```

.jpg .jpeg .png .bmp .tif .tiff

segmentation-models-pytorch
--checkpoint <path>

project.yaml model.encoder

512 512 --batch-size mobilenet_v2 resnet50 --size 256 256

&

```
for p in project1 project2 project3; do
  crackz detect run --project $p &
done
wait
```

CRACKZ_LOG_DIR

.ckpt

project.yaml

raw train val test

1. _____
2. _____
3. _____

1. _____
2. _____
3. _____

1. _____
2. _____
3. _____

1. _____
2. _____
3. _____

• _____
• _____
• _____

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CLI Cookbook

CLI Cookbook

```
crackz init --name demo-project
```

```
demo-project/
images/
  raw/ train/ val/ test/
masks/
artifacts/
detections/
checkpoints/
project.yaml
```

- `crackz --help` `--help`
- `--project` `--cfg`
- `--debug` `--quiet`
-

```
crackz init --name demo-project
```

```
crackz init --name demo-project --force
```

```
# Preview what will be cleaned (safe)
crackz clean --project demo-project --dry-run

# Clean logs and artifacts
crackz clean --project demo-project

# Clean only logs, keep checkpoints and detections
crackz clean --project demo-project --logs-only
```

```
--logs-only *.log --logs-only
```

```
--type
```

```
crackz import-images \
--project demo-project \
--src ./samples \
--type raw \
--size 512 512 \
--quality 90
```

```
crackz import-images --project demo-project --src ./samples --type train --size 512 512
crackz import-images --project demo-project --src ./samples --type val --size 512 512
crackz import-images --project demo-project --src ./samples --type test --size 512 512
```

```
crackz import-images \
--project demo-project \
--src ./samples \
--split \
--train-ratio 0.7 \
--val-ratio 0.15 \
--test-ratio 0.15 \
--size 512 512
```

```
--split
```

```
--split
```

```
--type
```

```
--random-seed
```

```
crackz import-images --project demo-project --src ./samples --split --random-seed 42
```

```
--src/masks
```

```
crackz import-images --project demo-project --src ./samples --masks_src ./samples/masks --type train
```

```
crackz import-images --project demo-project --src ./samples --type train --overwrite
```

```
train
```

```
images/train
train/
```

```
val test
crackz detect train
```

```
raw/
--split
```

```
--type
```

```
scripts/import_masks.py
```

```
# Basic usage - import from organized mask directories
```

```
python scripts/import_masks.py <project_dir> <masks_source_dir>
```

```
# Example: Import masks for skytteren_projekt from smasks directory
```

```
python scripts/import_masks.py ./skytteren_projekt ./smasks
```

```
# Custom size and quality
```

```
python scripts/import_masks.py ./my_project ./mask_sources --size 1024 1024 --quality 95
```

```
masks_source_dir
```

```
masks_source_dir/
train/
  mask_001.jpg
  mask_002.png
  ...
val/
  mask_101.jpg
  ...
test/
  mask_201.png
  ...
```

```
data.mask_threshold
_mask
```

```
--size WIDTH HEIGHT
```

```
--quality QUALITY
```

```
data.mask_threshold
```

```
val/
```

```
# Basic usage - edit an image (mask path auto-derived)
```

```
crackz masks edit images/train/crack_001.jpg
```

```
# With custom mask path
```

```
crackz masks edit photo.jpg --mask output/my_mask.png

# With project context (smart path derivation)
crackz masks edit images/raw/inspection.jpg --project my_harbor_project
```

--mask

1.

--project

```
Image: project/images/train/crack_001.jpg
Mask:  project/masks/train/crack_001.png ← Auto-derived
```

2.

```
Image: data/images/train/photo.jpg
Mask:  data/masks/train/photo.png ← Auto-derived
```

3.

```
Image: /tmp/photo.jpg
Mask:  /tmp/masks/photo.png ← Auto-derived
```

pyside6

```
for img in images/train/*.jpg; do
    crackz masks edit "$img" --project my_project
done
```

```
ssh -X user@server
crackz masks edit /data/harbor/crack.jpg
```

```
# Convert all masks in a project (uses project threshold)
crackz masks convert --project my_harbor_project

# Convert with custom threshold
crackz masks convert --project my_harbor_project --threshold 128
```

-
- train val test raw
- raw/

1. train val test image_01.jpg
image_01.png
2. raw/
3. crackz detect train
4. crackz detect run --project <p> raw/
5. test

```
--type train
```

1. labelme labelme
2. pip install labelme
3. image_001.png images/train/image_001.jpg
4. masks/train/image_001.png masks/<split>/ masks/train/ masks/val/ masks/test/
- 5.
- 6.
7. crackz import-images --masks_src masks/

```
cv2.threshold
```

```
crackz detect run --project demo-project
```

```
--project
```

```
crackz detect run --cfg config/detect.yaml
```

```
--model
```

```
best.ckpt last.ckpt final.ckpt
```

```
# Use specific checkpoint by name
crackz detect run --project demo-project --model best
crackz detect run --project demo-project --model last
crackz detect run --project demo-project --model final

# Use specific epoch checkpoint (e.g., epoch-0.9234.ckpt)
crackz detect run --project demo-project --model epoch-0.9234
```

```
--model
final.ckpt
```

```
best.ckpt
```

```
last.ckpt
```

```
--model last
```

```
--model best
```

```
--model epoch-X
```

```
crackz detect train --project demo-project
train/val/test
```

```
crackz detect train --project demo-project --epochs 1 --batch-size 1
```

```
# Auto-selects best checkpoint, outputs to <project-name>_model.zip
crackz model export --project demo-project

# Specify output path
crackz model export --project demo-project --out models/harbor_v1.zip

# Export specific checkpoint (best, last, or final)
crackz model export --project demo-project --checkpoint best --out best_model.zip
```

```
crackz model export -p demo-project -o my_model.zip
crackz model export -p demo-project -c final -o final_model.zip
```

```
.ckpt
```

```
best.ckpt last.ckpt final.ckpt
```

```
<project_name>_model.zip --out
```

```
# Import model package
crackz model import --project demo-project --model harbor_v1.zip

# Overwrite existing checkpoint with same name
crackz model import --project demo-project --model harbor_v1.zip --force
```

```
crackz model import -p demo-project -m my_model.zip
crackz model import -p demo-project -m my_model.zip --force
```

```
checkpoints/
```

```
--force
```

```
# Team member A: Export trained model
crackz model export -p training-project -o shared/harbor_crack_v1.zip

# Team member B: Import into their project
crackz model import -p deployment-project -m shared/harbor_crack_v1.zip
```

```
# Export base model trained on general cracks
crackz model export -p general-cracks -o pretrained/base_model.zip

# Import into specialized project for fine-tuning
crackz model import -p harbor-specific -m pretrained/base_model.zip

# Fine-tune on harbor-specific data
crackz detect train -p harbor-specific --epochs 10
```

```
# Backup current best model before trying new hyperparameters
crackz model export -p demo-project -c best -o backups/before_experiment.zip

# Run experiment
crackz detect train -p demo-project --epochs 50

# If results are worse, can import the backup
crackz model import -p demo-project -m backups/before_experiment.zip --force
```

```
# Export with timestamp
crackz model export -p demo-project -o models/model_2025-10-11.zip

# Or use git-style versioning
crackz model export -p demo-project -o models/v1.0.0.zip
```

```
# Export from training environment
crackz model export -p training-project -o production/model.zip

# Transfer to production server and import
crackz model import -p production-project -m production/model.zip

# Run detection in production
crackz detect run -p production-project
```

<checkpoint_name>.ckpt

metadata.json

```
# Linux/Mac
unzip -l my_model.zip

# Windows PowerShell
Expand-Archive -Path my_model.zip -DestinationPath temp -Force
cat temp\metadata.json
```

metadata.json

```
{
  "project_name": "harbor_detection",
  "checkpoint_name": "best.ckpt",
  "model_config": {
    "encoder": "resnet34",
    "encoder_weights": "imagenet",
    "in_channels": 3,
    "num_classes": 1,
    "threshold": 0.5,
    "loss_name": "combo",
    "dice_weight": 0.7,
    "focal_weight": 0.3
  }
}
```

```
# Generate map for raw split (default)
crackz geo create-map --project demo-project --output harbor_map.html

# Map for specific split (train/val/test)
crackz geo create-map --project demo-project --output train_map.html --split train
```

```
crackz geo create-map -p demo-project -o map.html -s raw
```

```
# Export all images with GPS data
crackz geo export-geojson --project demo-project --output detections.geojson

# Export specific split
crackz geo export-geojson --project demo-project --output train.geojson --split train
```

```
crackz geo export-geojson -p demo-project -o data.geojson -s raw
```

```
{
  "type": "FeatureCollection",
  "features": [
    {
      "type": "Feature",
      "geometry": {
        "type": "Point",
        "coordinates": [longitude, latitude]
      },
      "properties": {
        "image_name": "IMG_0001.jpg",
        "image_path": "/path/to/image.jpg",
        "has_detection": true,
        "latitude": 59.123456,
        "longitude": 10.123456
      }
    }
  ],
  "metadata": {
    "project": "demo-project",
    "split": "raw",
    "total_images": 42,
    "images_with_detections": 15
  }
}
```

```
# Linux/Mac
exiftool image.jpg | grep GPS

# Python (Pillow)
from PIL import Image
img = Image.open("image.jpg")
exif = img.getexif()
gps_ifd = exif.get_ifd(0x8825) # 0x8825 is GPS IFD tag
print(gps_ifd)
```

```
# 1. Import drone photos (GPS data preserved in EXIF)
crackz import-images --project harbor --src ./drone_photos --type raw --size 512 512

# 2. Run detection
crackz detect run --project harbor

# 3. Generate map to visualize results
```



```
crackz geo create-map --project harbor --output harbor_results.html

# 4. Export to GeoJSON for GIS analysis
crackz geo export-geojson --project harbor --output harbor.geojson

# 5. Open map in browser to review
# (or import GeoJSON into QGIS/ArcGIS)
```

```
crackz --debug detect run --project demo-project
crackz --quiet detect run --project demo-project
crackz --log-dir run_logs detect run --project demo-project
```

```
CRACKZ_LOG_DIR=logs_debug crackz detect run --project demo-project
```

```
--debug
```

```
# Start training
crackz detect train --project demo-project

# Press Ctrl+C to cancel at any time
# The operation will be cancelled gracefully
# Partial results (e.g., checkpoints) are preserved
```

```
crackz detect run --cfg configs/detect.yaml
crackz detect train --cfg configs/train.yaml
```

```
for P in demo-project other-project; do
  crackz detect run --project "$P" || echo "Failed: $P" >&2
done
```

```
/home/app
```

```
# Init project (host ./project mapped to /home/app)
mkdir -p project
docker run --rm -v %cd%\project:/home/app crackz-cli init --name demo-project

# Import images (mount samples to /data)
docker run --rm -v %cd%\project:/home/app -v %cd%\samples:/data \
  crackz-cli import-images --project /home/app/demo-project --src /data --type raw --size 512 512

# Detection
docker run --rm -v %cd%\project:/home/app crackz-cli detect run --project /home/app/demo-project
```

```
docker run --gpus all --rm -v %cd%\project:/home/app crackz-cli:gpu detect run --project /home/app/demo-project
```

```
--project --cfg
```

```
--src
```

```
images/train
```

```
raw
```

```
torch.cuda.is_available()
```

```
# Generate report with default name (project_name_report.docx)
crackz report generate --project demo-project

# Specify custom output path
crackz report generate --project demo-project --output inspection_report.docx

# Use config file instead of project path
crackz report generate --cfg project.yaml --output report.docx
```

```
crackz detect run
```

```
# Find top 50 most uncertain images
uv run crackz active-learning run \
  --project demo-project \
  --top-n 50 \
  --method entropy \
  --out priorities.json
```

```
entropy
variance
```

```
mean_confidence
```

```
max_probability
```

```
uv run crackz active-learning info
```

```
# Prioritize images in 'raw' split (default)
uv run crackz active-learning run --project demo-project --split raw

# Prioritize images in 'train' split
uv run crackz active-learning run --project demo-project --split train
```

```
{
  "method": "entropy",
  "total_images": 500,
  "prioritized_count": 50,
  "images": [
    {
      "path": "image_042.jpg",
      "uncertainty_score": 0.8234,
      "rank": 1
    }
  ]
}
```

```

    ]
  }

```

```

# Find similar images and propagate mask
uv run crackz propagate-masks run \
  --project demo-project \
  --reference wall_01.jpg \
  --mask wall_01_mask.png \
  --split raw \
  --threshold 0.8 \
  --out ./propagated_masks

```

```

--threshold
sift --orb --min-matches --

```

```

uv run crackz propagate-masks find-similar \
  --project demo-project \
  --reference wall_01.jpg \
  --split raw \
  --out similarities.json

```

```

# More permissive matching (more propagations, less accurate)
uv run crackz propagate-masks run \
  --reference ref.jpg \
  --mask ref.png \
  --project demo-project \
  --threshold 0.6 \
  --min-matches 5

# Stricter matching (fewer propagations, more reliable)
uv run crackz propagate-masks run \
  --reference ref.jpg \
  --mask ref.png \
  --project demo-project \
  --threshold 0.9 \
  --min-matches 20 \
  --sift

```

```

uv run crackz propagate-masks run \
  --reference ref.jpg \
  --mask ref.png \
  --project demo-project \
  --orb

```

- [api.md](#) [detect\(\)](#) [train\(\)](#)

[crackz export-config](#)

GUI

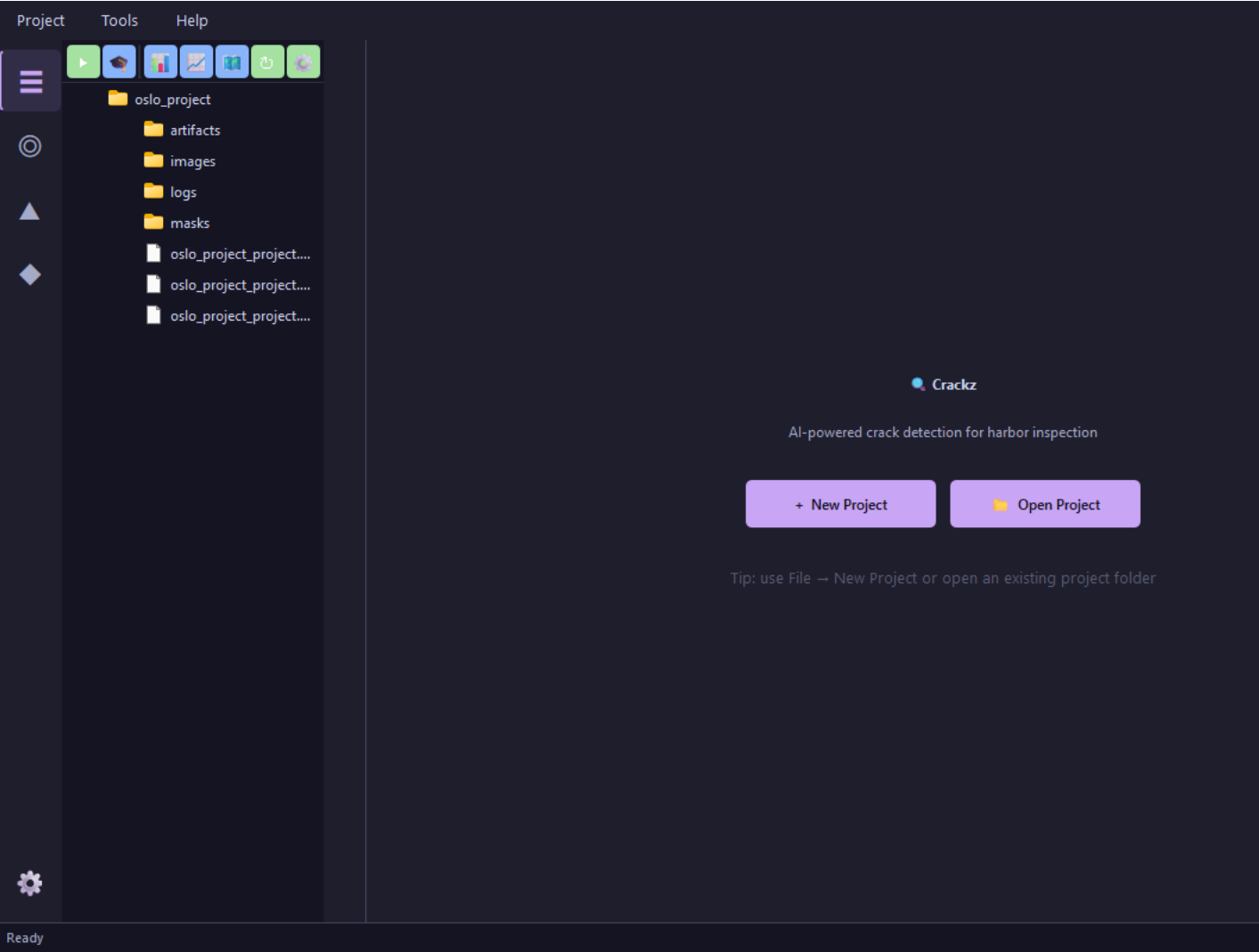
GUI Guide

```
crackz-gui
```

```
# After installing Crackz
crackz-gui
```

```
gui crackz-gui.exe ./crackz-
```

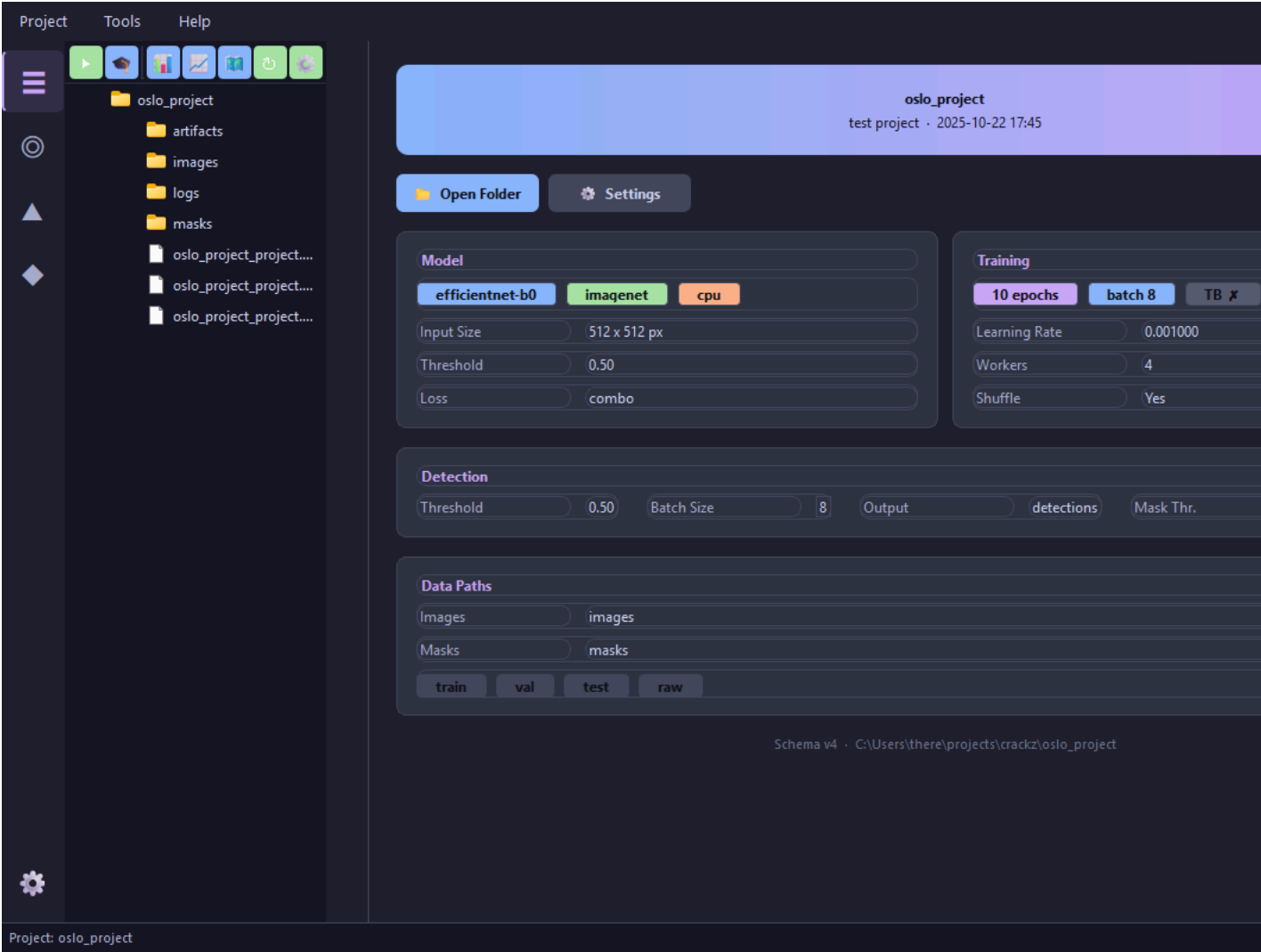
```
uv run python scripts/capture_qt_screenshots.py --project oslo_project
```



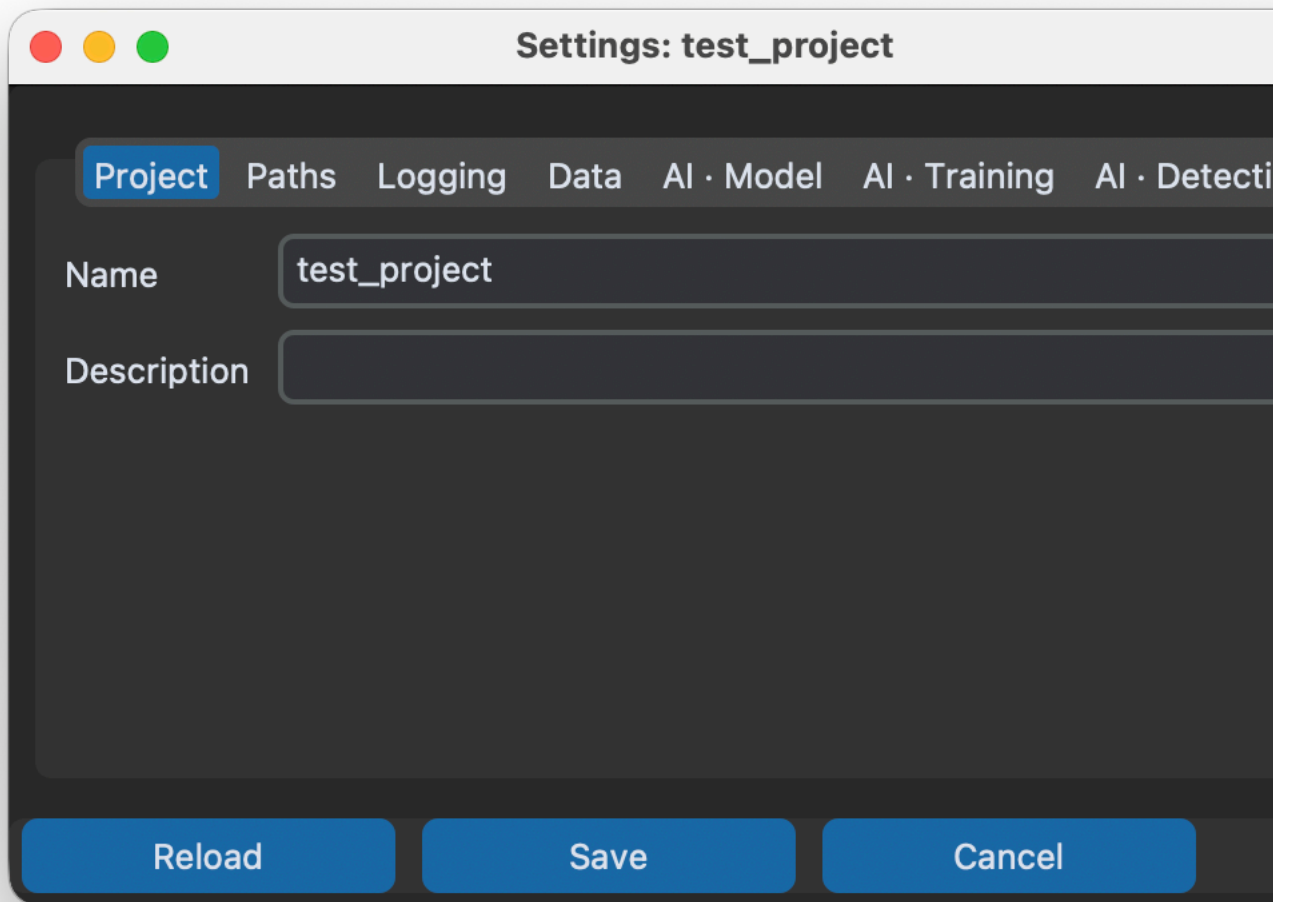
- 1.
- 2.
- 3.
- 4.

- 1.
- 2.
- 3.

project.yaml



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A macOS-style settings dialog box titled "Settings: test_project". It features a tabbed interface with tabs for "Project", "Paths", "Logging", "Data", "AI · Model", "AI · Training", and "AI · Detecti". The "Project" tab is selected. Below the tabs, there are two text input fields: "Name" (containing "test_project") and "Description" (empty). At the bottom, there are three blue buttons: "Reload", "Save", and "Cancel".

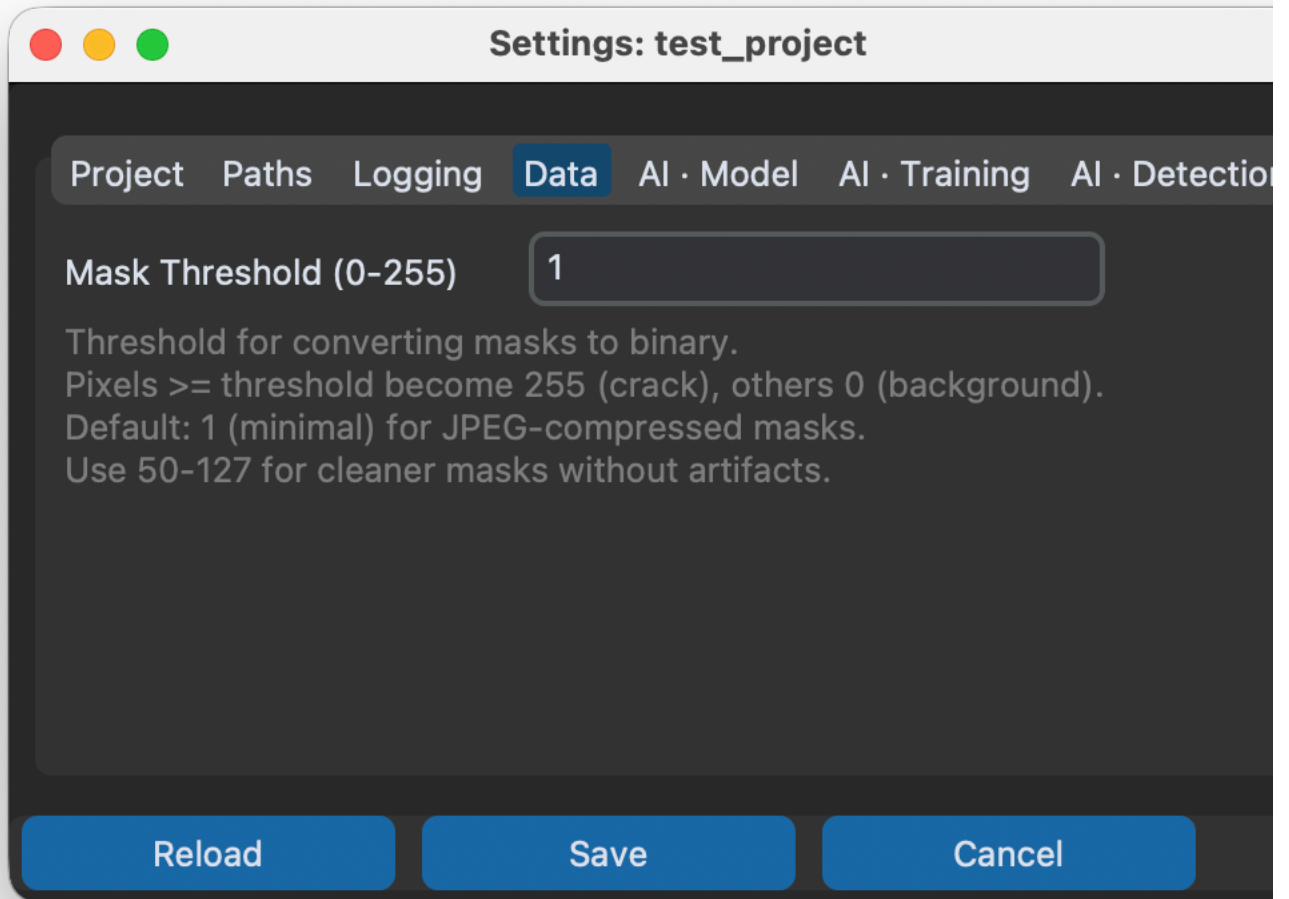
Settings: test_project

Project Paths Logging Data AI · Model AI · Training AI · Detecti

Name test_project

Description

Reload Save Cancel



Settings: test_project

Project Paths Logging Data AI · Model AI · Training

Encoder

resnet34

Encoder Weights

imagenet

Device

mps

 Recommended: 'mps' - Apple Silicon GPU detected! Up to 5-20x faster than CPU.

Available devices: cpu, mps

Input Channels

3

Num Classes

1

Checkpoint Path

Threshold

0.5

Norm Mean (comma)

0.485,0.456,0.406

Norm Std (comma)

0.229,0.224,0.225

Loss Function

combo

Dice Weight

0.7

Focal Weight

0.3

Input Size (width,height)

512,512

Reload

Save

Cancel

Settings: test_project

Project

Paths

Logging

Data

AI · Model

AI · Training

AI · Detection

Batch Size

2

Num Workers

4

Epochs

2

Learning Rate

0.001

Shuffle (True/False)

True

Loss

Metrics

Enable TensorBoard

☐

Random Seed (optional)

Precision (optional)

Devices (optional)

Strategy (optional)

Save Top K Checkpoints

1

Save Last Checkpoint

☒

Auto-Prune After Training

☐

Checkpoints to Keep (when pruning)

3

ReloadSaveCancel

⋮

Settings: test_project

ProjectPathsLoggingDataAI · ModelAI · TrainingAI · Detection

Threshold

0.5

Learning Rate

0.001

Batch Size

8

Shuffle (True/False)

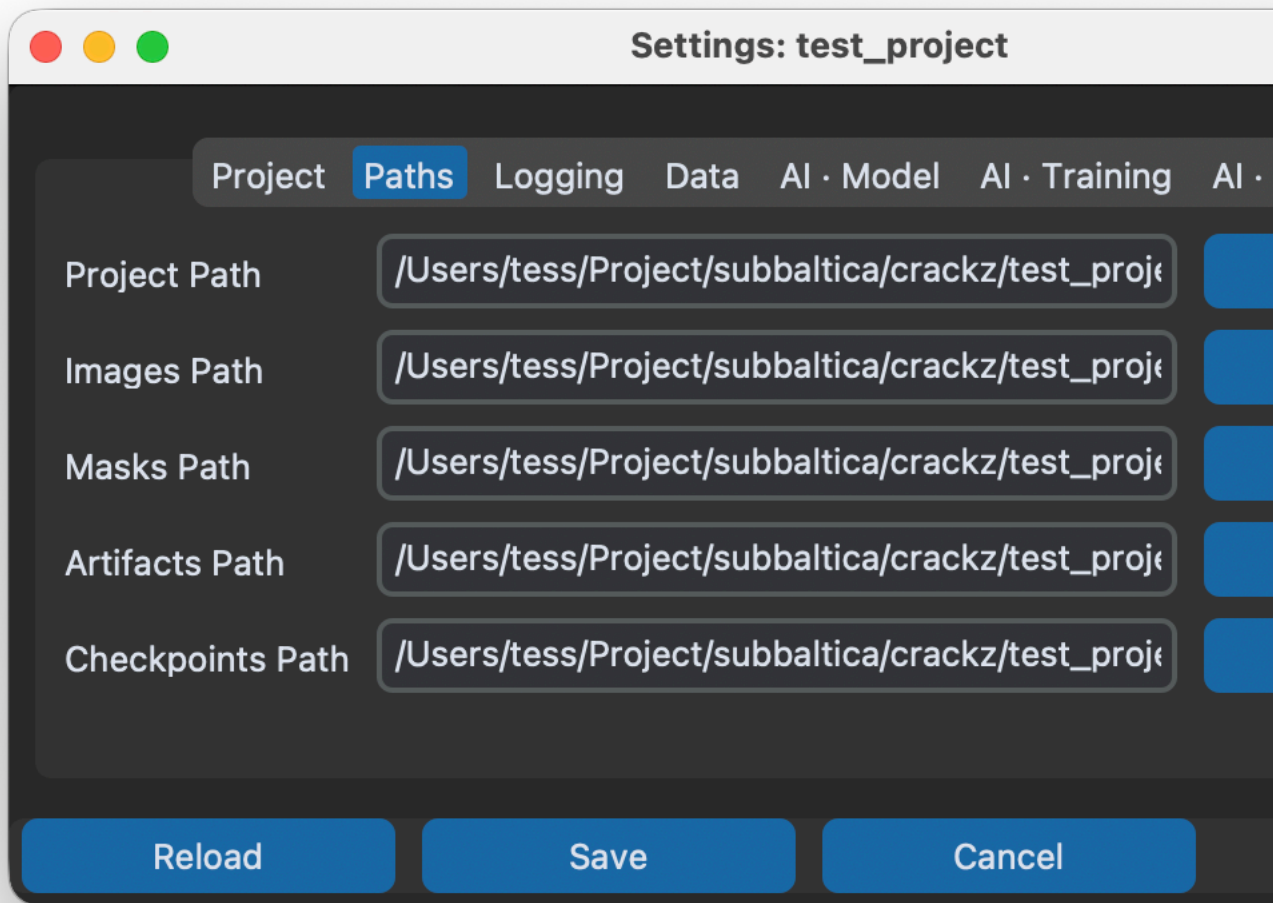
True

Output Dir Name

detections

Random Seed (optional)

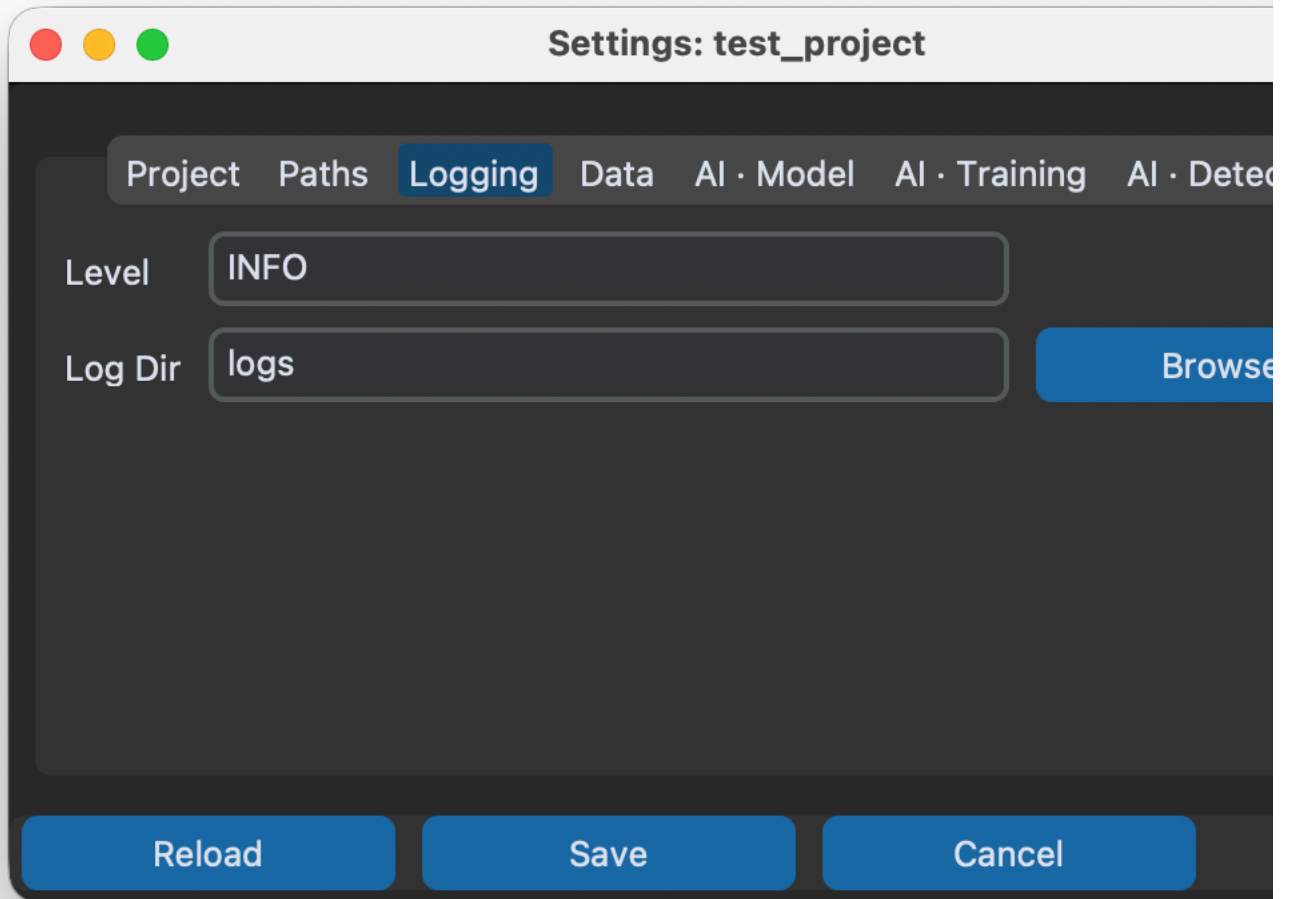
ReloadSaveCancel

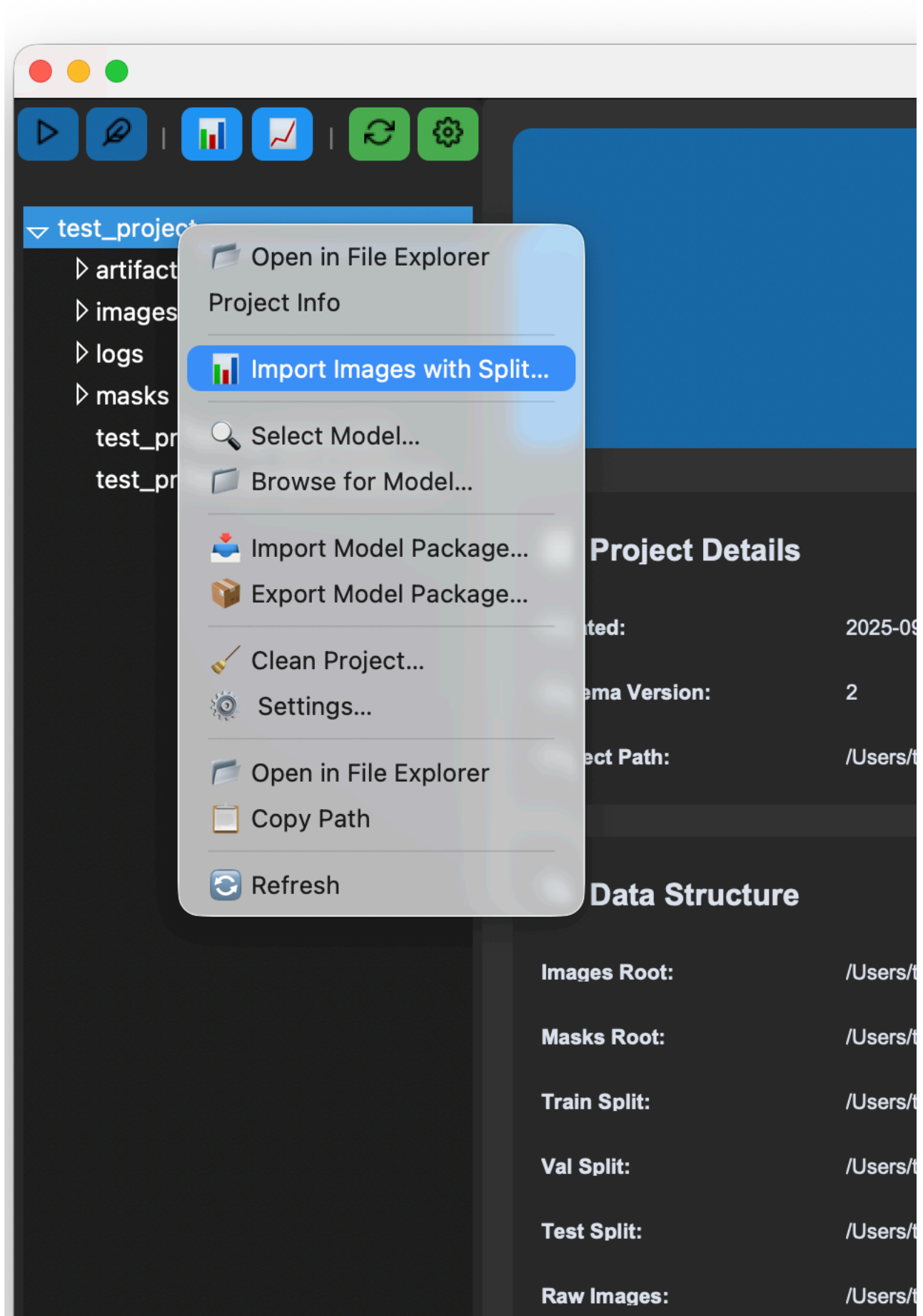


Reload

Save

Cancel







AI Model

Encoder:	resnet3
Encoder Weights:	imagen
Device:	mps
Input Size:	512 x 5
Threshold:	0.50
Loss Function:	combo



Training Configuration

Batch Size:	2
Epochs:	2
Learning Rate:	0.00100
Workers:	4
TensorBoard:	☒ Disabled
Shuffle:	Yes

Project: /Users/tess/Project/subbaltica/crackz/test_project

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.

Import Images with Split



Import Images with Automatic Split

Select a source directory with images and masks.
They will be automatically split into train/val/test sets based on the ratios below.

Source Directory:

Select a directory containing images/ and optionally masks/ subdirectories.
Masks will be automatically matched to images by filename.

No directory selected

 Select Source Directory...

Split Ratios (auto-adjusting)

Train:

Validation:

Test:

Total: 1.00 ✓

Random Seed (Optional)

Set a seed for reproducible splits

Import

Cancel

Import Images with Split

Split Ratios (auto-adjusting)

Train: 0.7

Validation: 0.2

Test: 0.1

Total: 1.00 ✓

Random Seed (Optional)

Set a seed for reproducible splits

☐ Use random seed:

Image Processing Settings

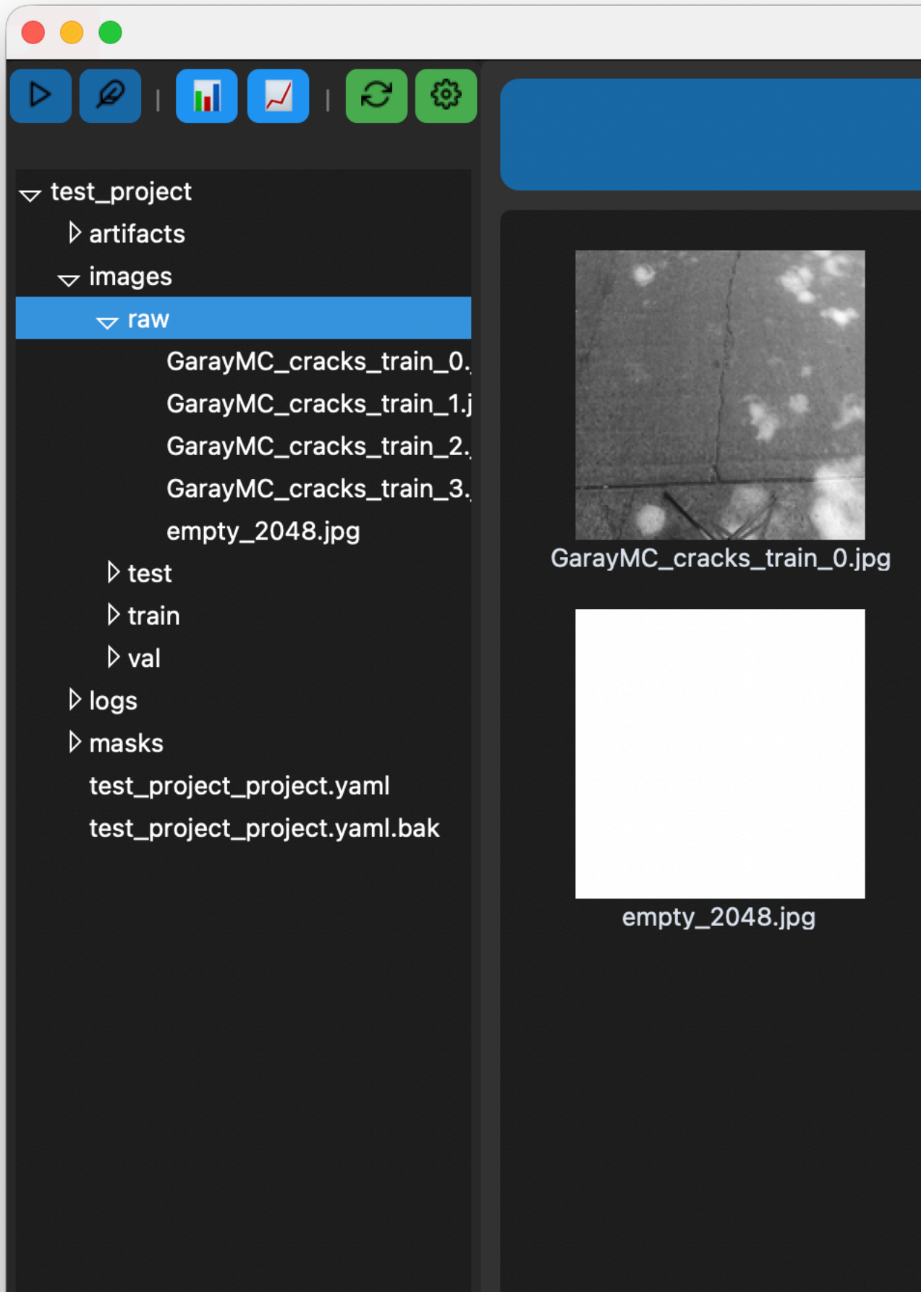
Image Size (width x height):

x pixels

Image Quality: 90

Import

Cancel



Page 1 of 1

 Previous

Project: /Users/tess/Project/subbaltica/crackz/test_project

images/

images/

images/

masks/

-
-
-
-
-



B
E
Ctrl+Z Cmd+Z
Ctrl+Y Cmd+Shift+Z

Ctrl+S Cmd+S
Esc

```
project/
├── images/
│   └── train/
│       └── crack_001.jpg      # Original image
├── masks/
│   └── train/
│       └── crack_001.png     # Binary mask (auto-created)
```

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	Left Arrow	

masks/

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E

A

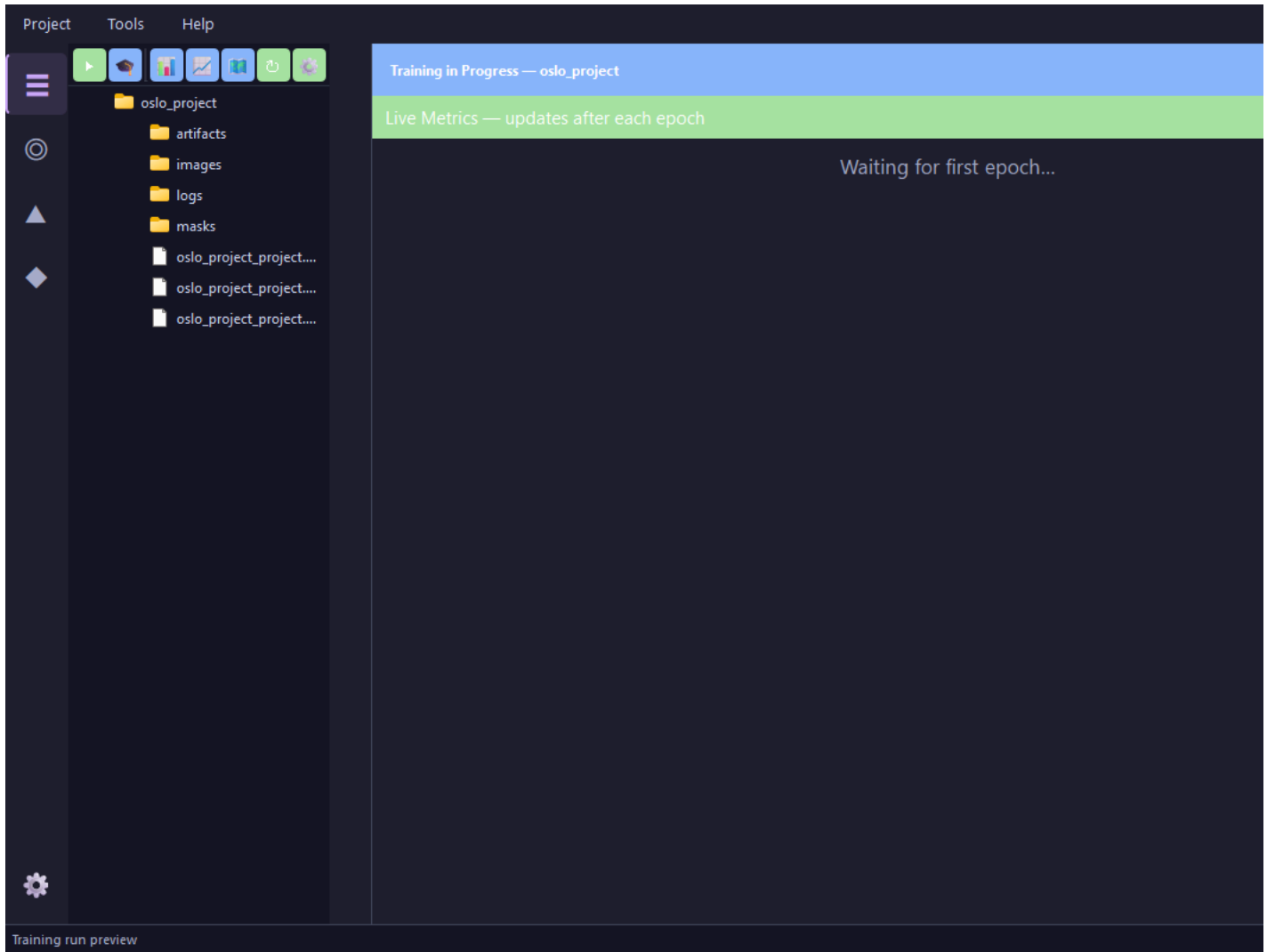
raw/

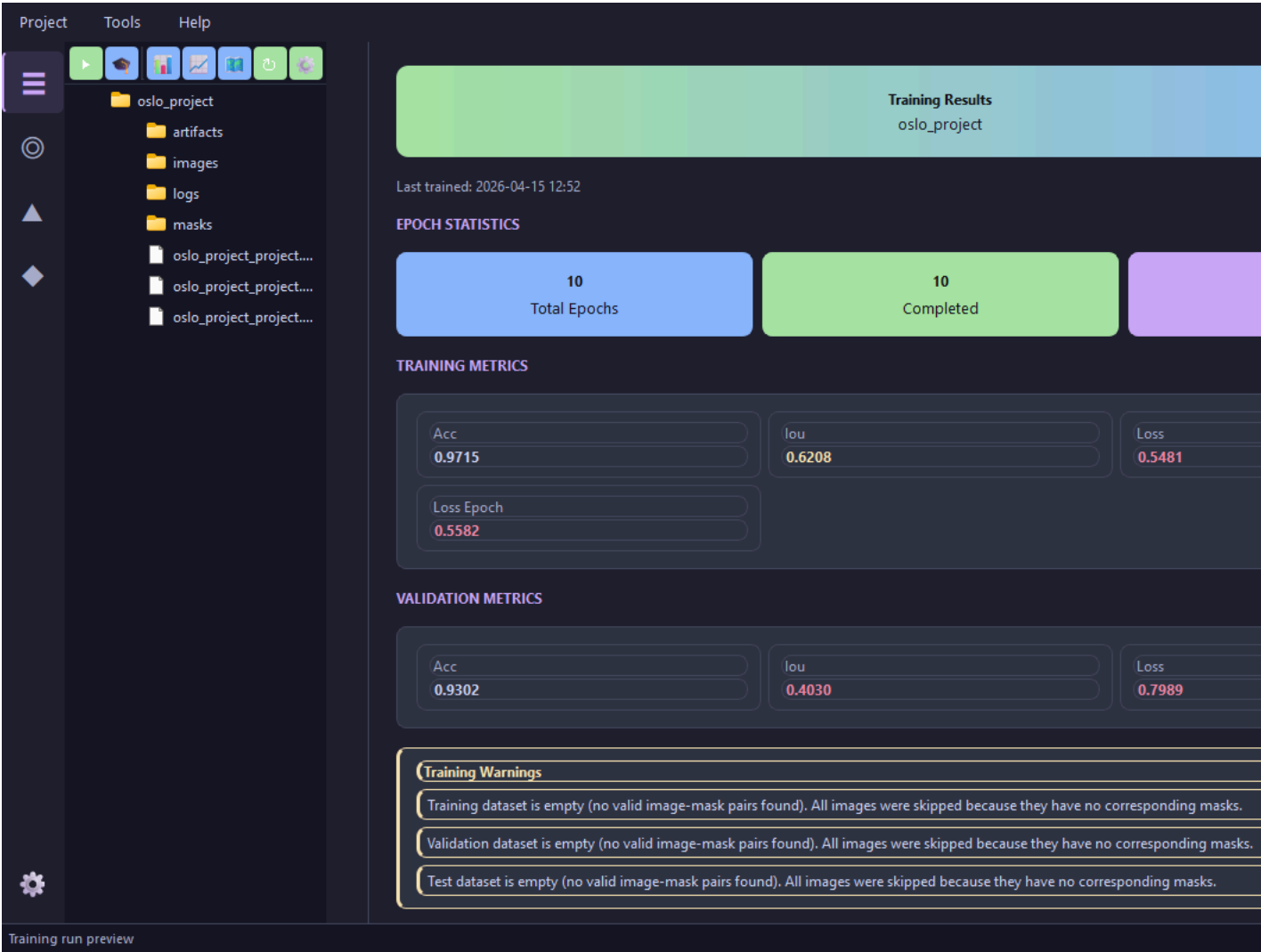
raw/

```
# Move annotated images from raw to train split
crackz move-images --project ./my-project --source raw --target train --pattern "crack_*.jpg"
```

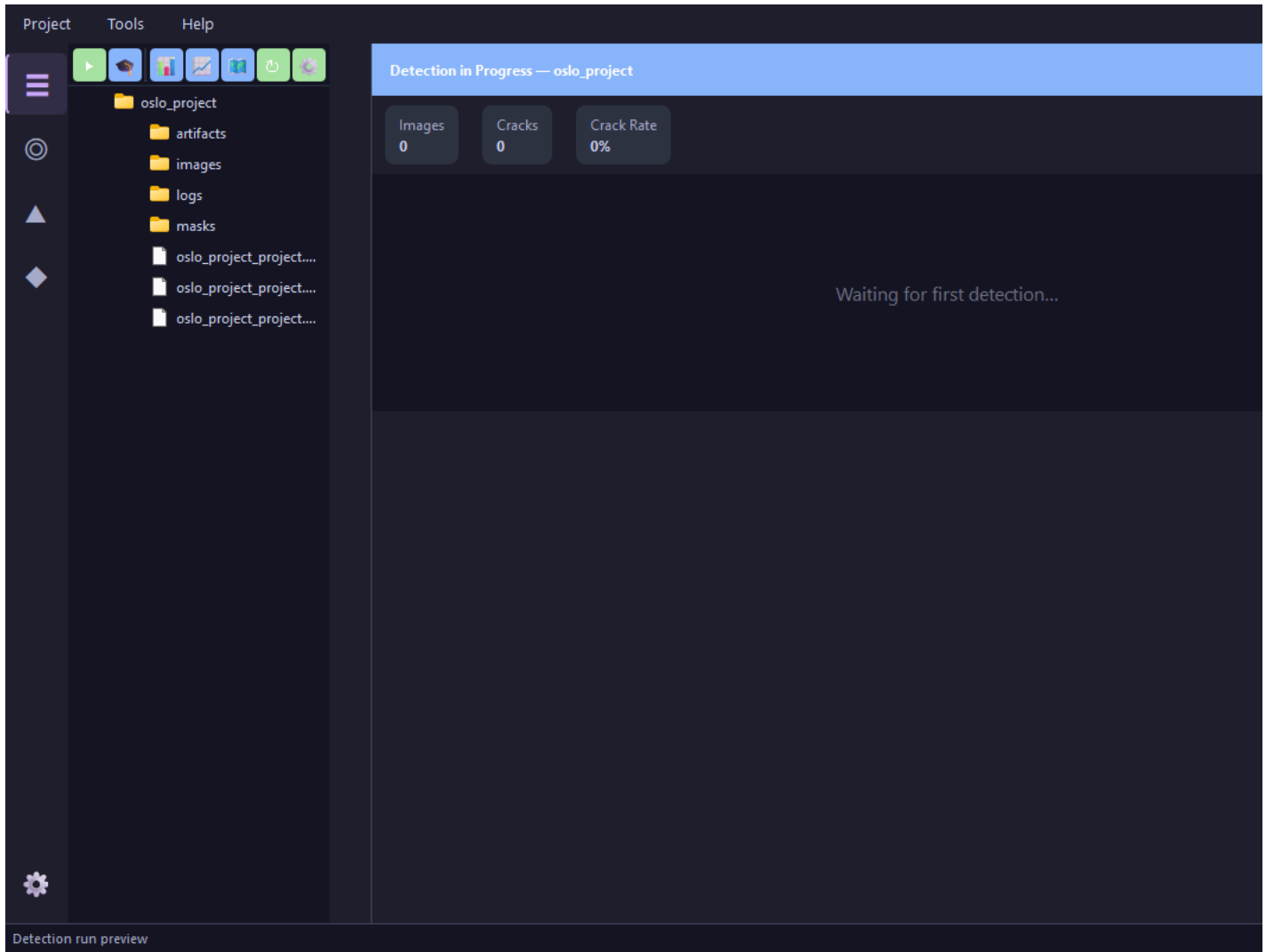
images/raw/
train val test

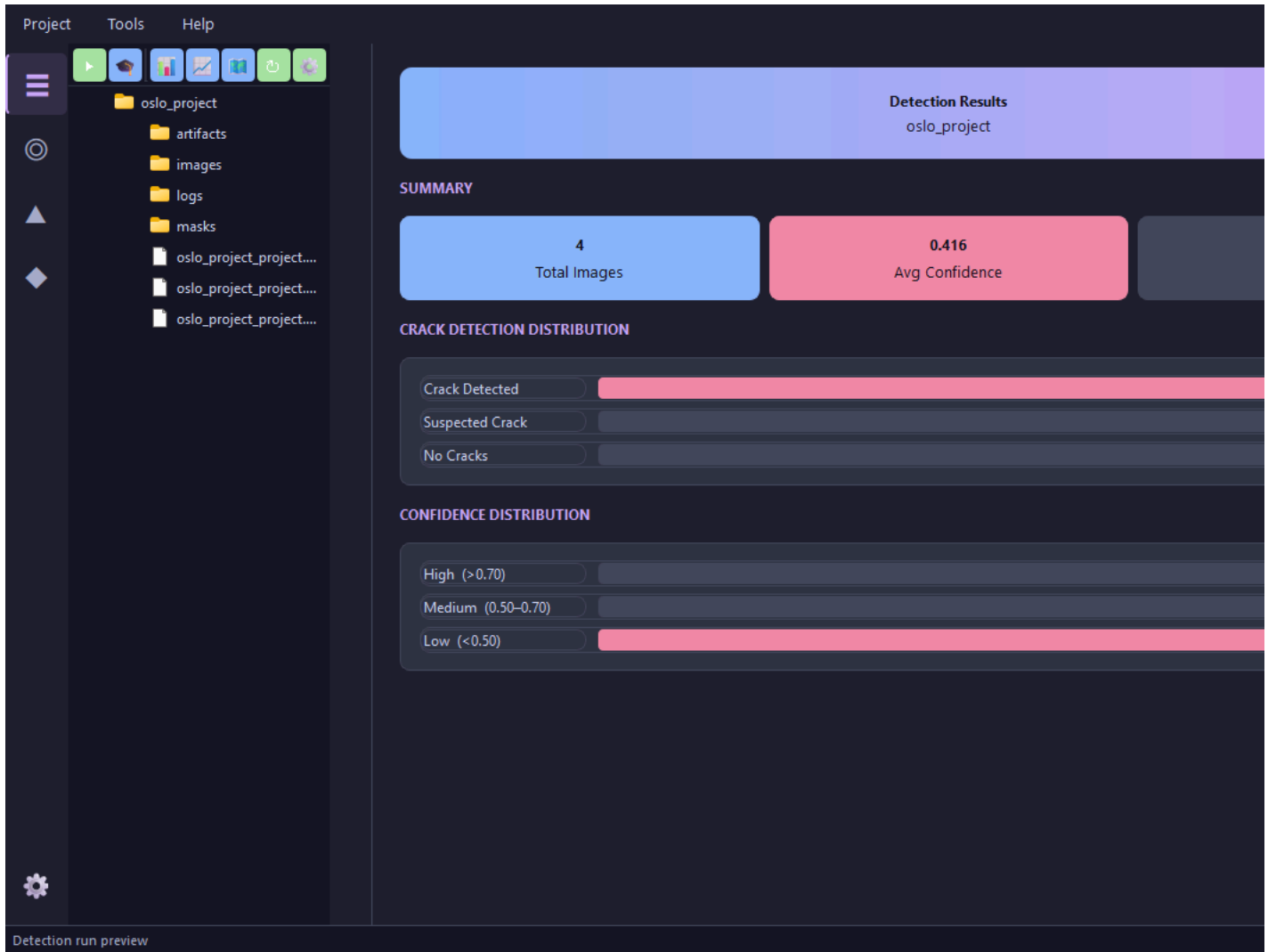
1.

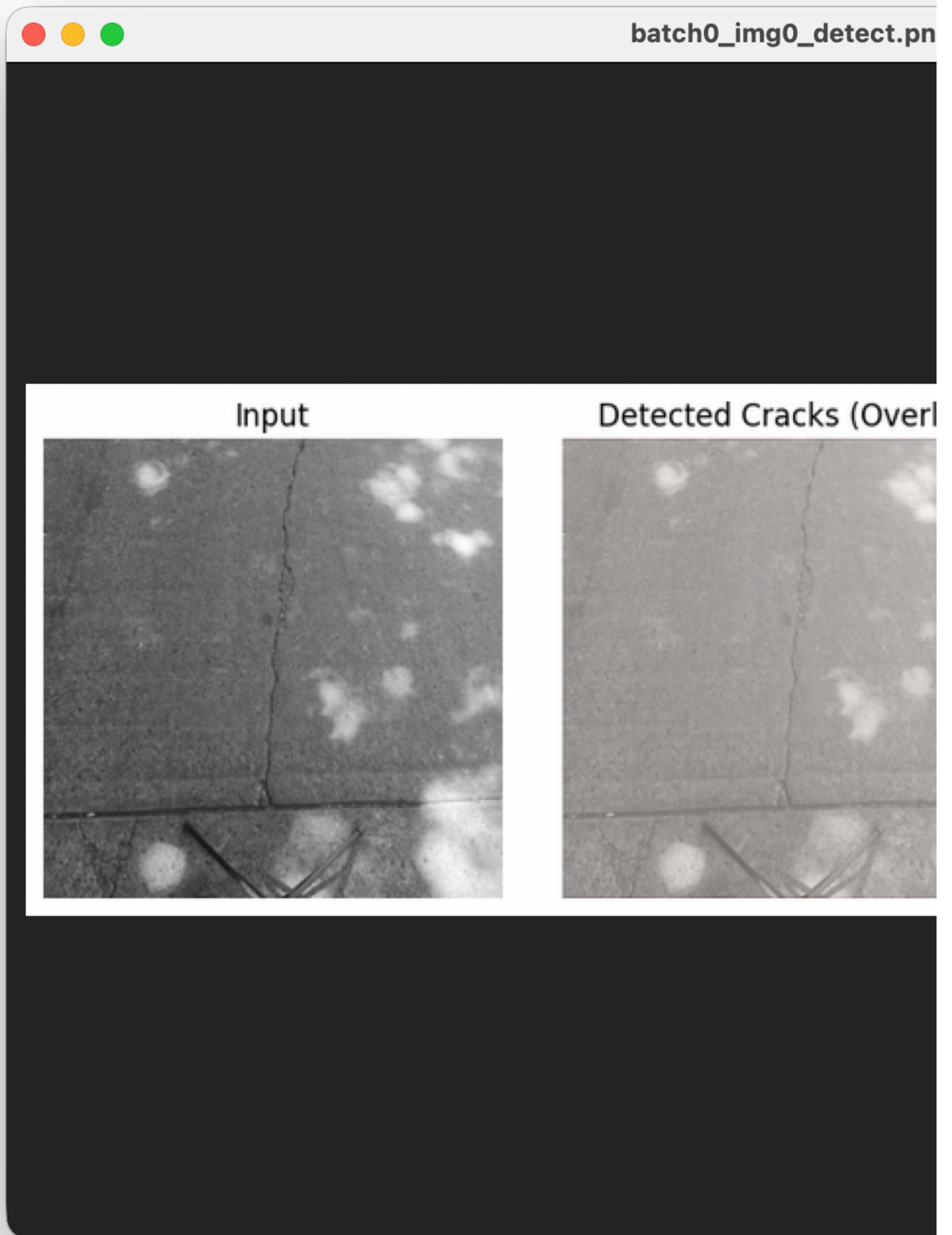




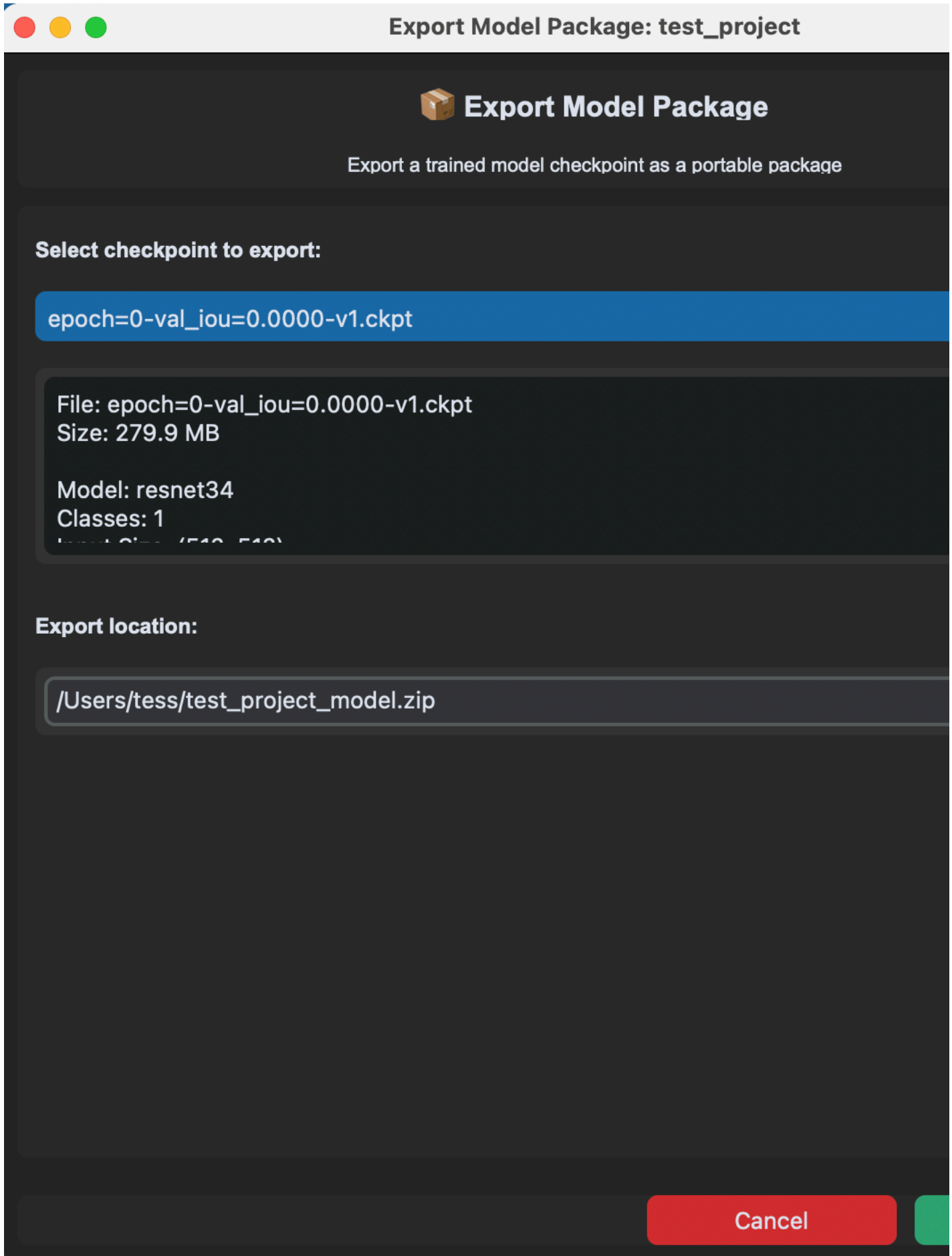
- 1.
 - 2.
 - 3.
 - 4.
- raw







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 - 4.



- 1.
- 2.
- 3.
- 4.

.ckpt

Import Model Package: test_project

 **Import Model Package**

Import a trained model checkpoint from a package file

Select model package file:

Browse...

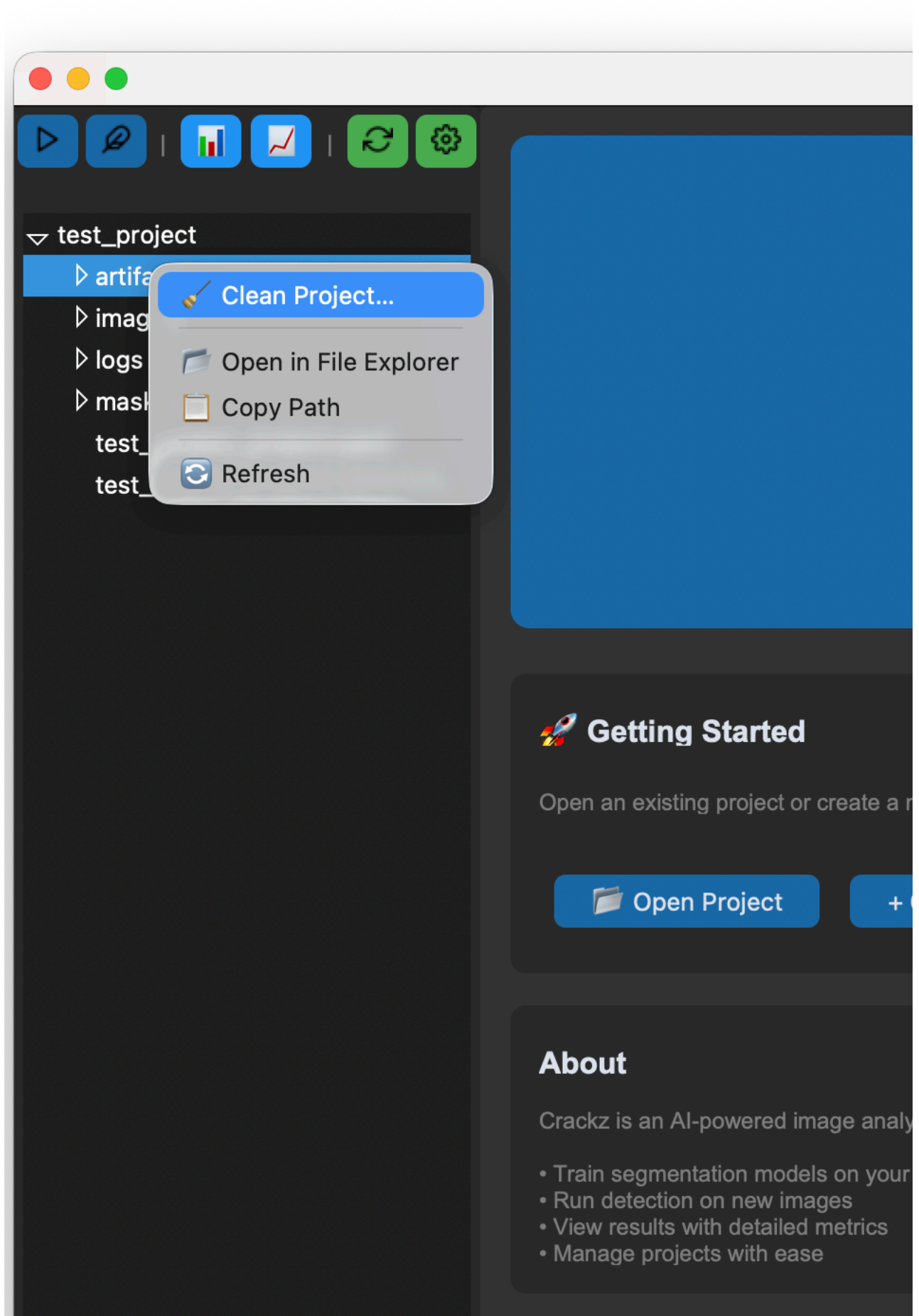
Package Information:

Select a model package file to view details...

☐ Overwrite existing checkpoint if it exists

- 1.
- 2.
- 3.
- 4.

- 5.
- 6.





System Information

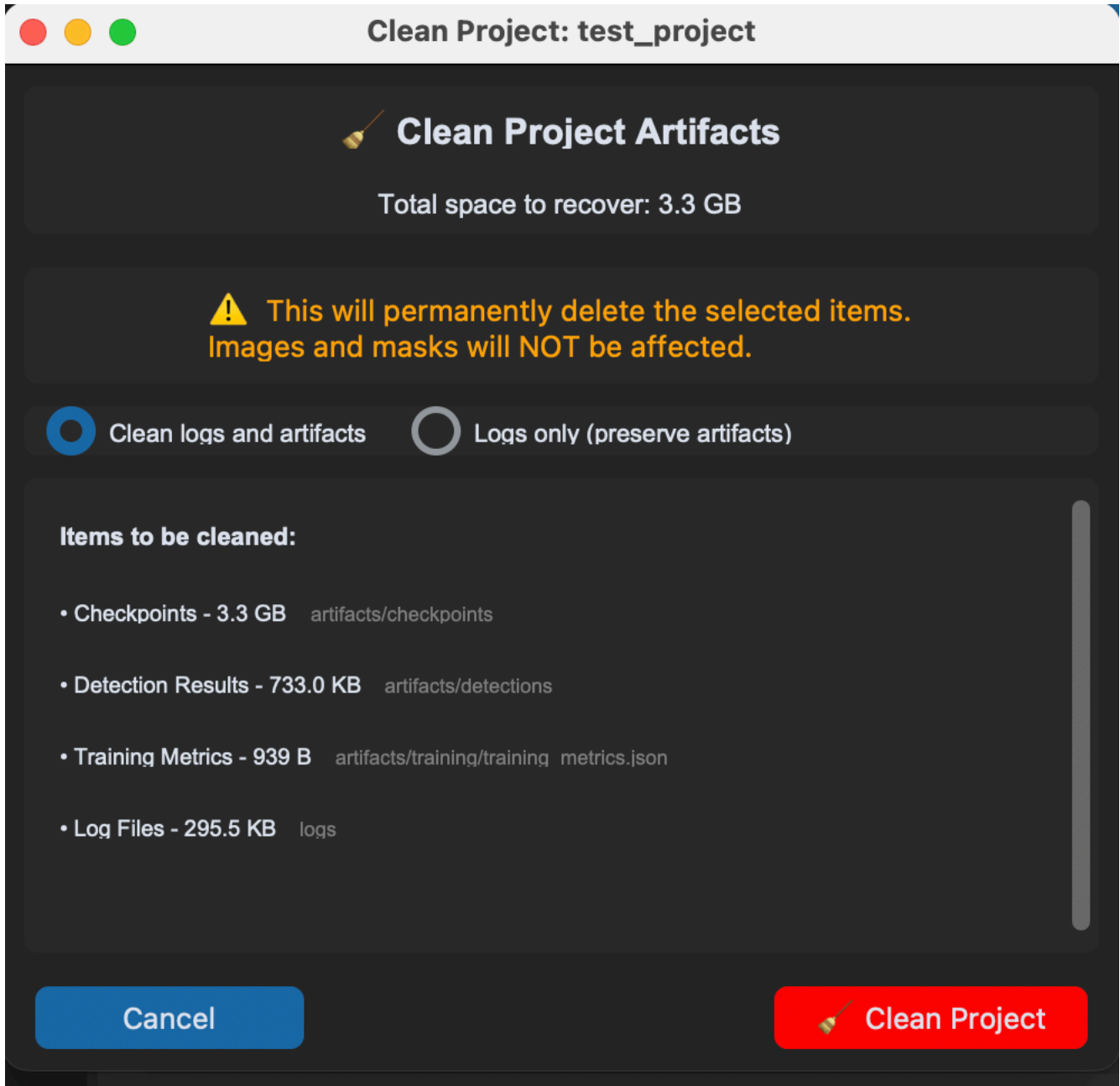
Operating System: Darwin 25.0.0
Python: 3.13.7



Recommended: 'mps' - Apple Silicon

Note: Device can be selected in project settings

Project: /Users/tess/Project/subbaltica/crackz/test_project



- 1.
- 2.

Project Info - test_project

 **test_project**

 **Project Details**

Name:

test project

Description:

N/A

Created:

2025-09-30 08:38

Schema Version:

2

 **Paths**

Project Path:

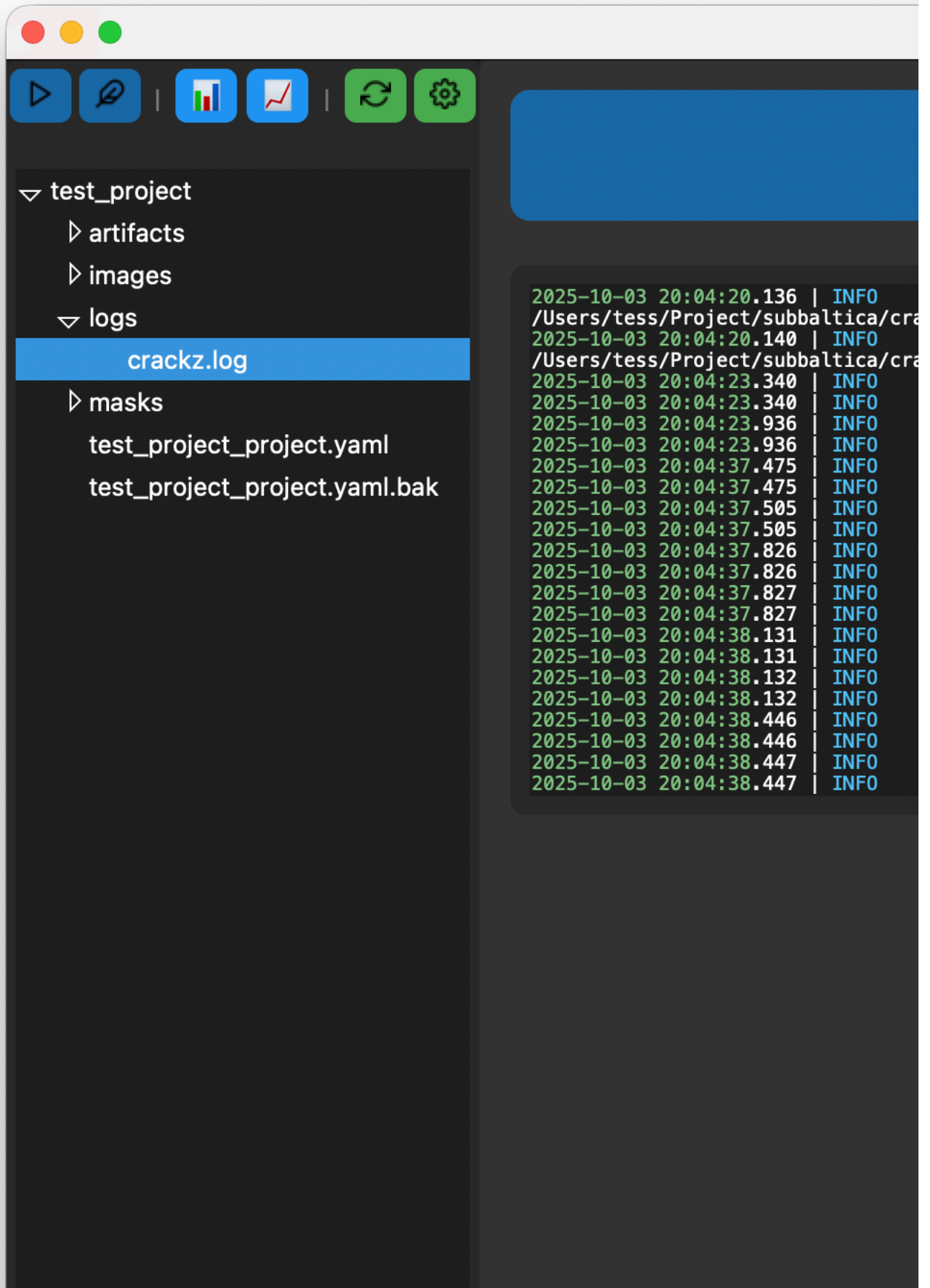
/Users/tess/Project/subbaltica/crackz/test project

Images:

/Users/tess/Project/subbaltica/crackz/test project/images

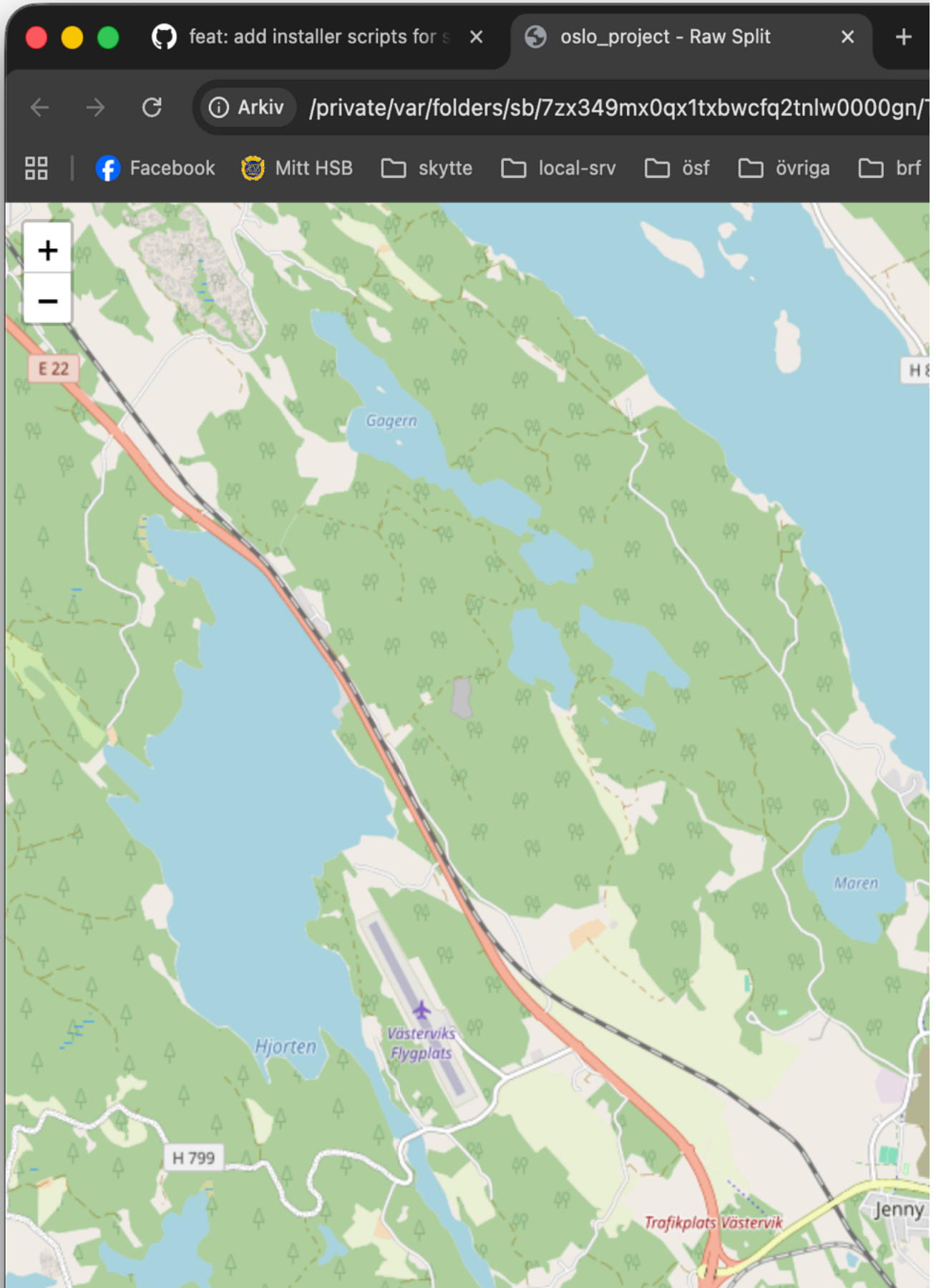
 Settings

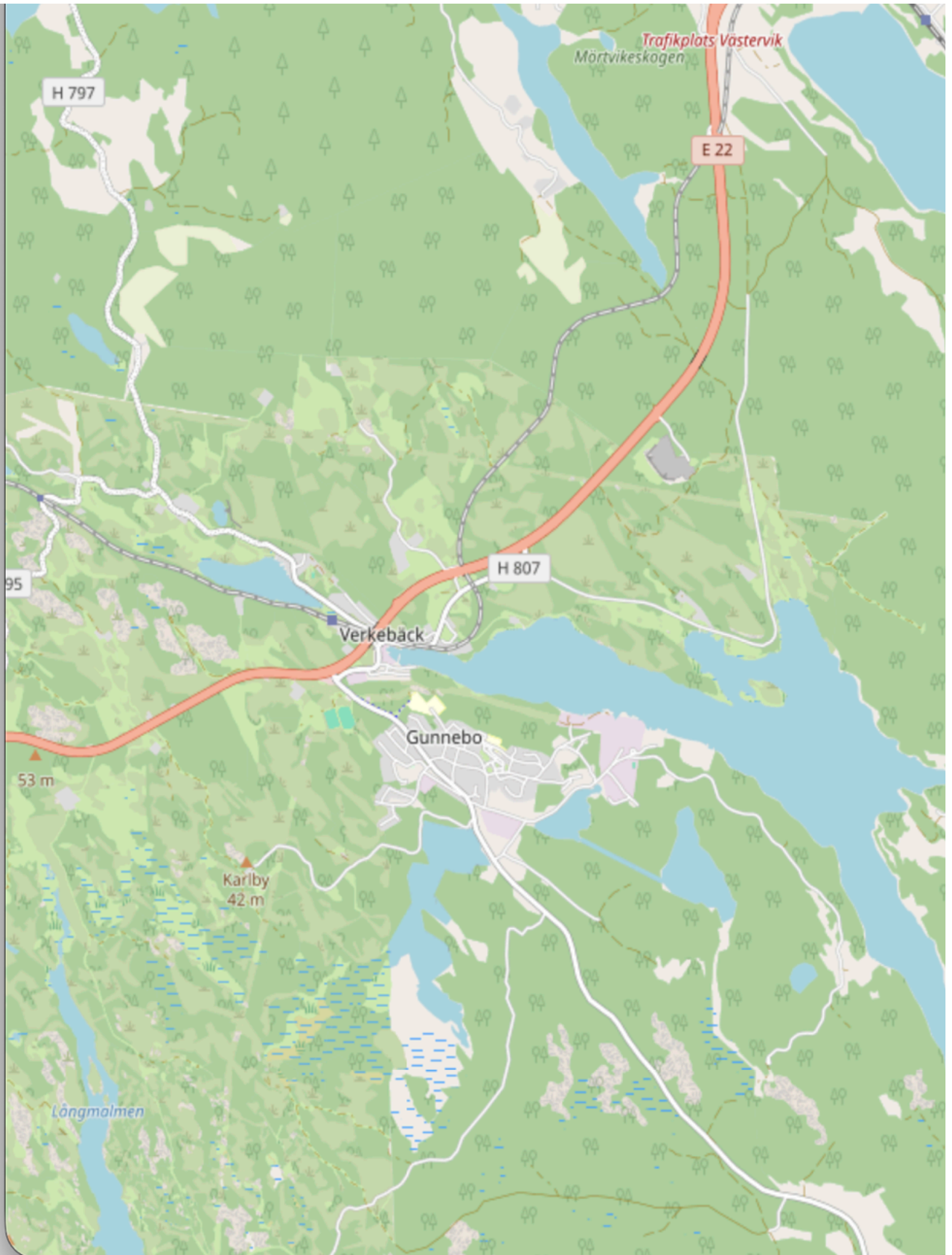
Clo



Project: /Users/tess/Project/subbaltica/crackz/test_project

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- 1.
 - 2.
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- 1.
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\$ ~ Cmd/Ctrl+/

assets/gui/chat_history_panel.png

assets/gui/chat_prompt_templates_panel.png

~/.crackz/chat_history.db

```
# Check installation
python -c "from PySide6.QtWidgets import QApplication; print('PySide6 OK')"
```

```
# Launch with debug output
crackz-gui --debug
```

-
-
-

-
-

CRACKZ_LOG_DIR/gui_app.log

<project>/logs/gui_operations.log

```
# Create and configure in GUI
crackz-gui
```

```
# Batch process via CLI
crackz detect run --project my-project --batch-size 16
```

```
# Review results in GUI
crackz-gui
```

-
-
-
-

Dashboard

Streamlit Dashboard

app

 Detection Results

 GPS Map

 Training Metrics

 Parameter Tuning

Project Selection

Project Path

oslo_project

✓ Loaded: oslo_project

Navigation

Use the sidebar to navigate between:

 **Detection Results** -

View crack detections

 **GPS Map** -

Geographic visualization

 **Training Metrics** -

Compare training runs

 **Parameter Tuning** -

Adjust detection settings



Crackz - Crack Detection Dashboard

AI-powered analysis for harbor and bridge inspection

Project Overview

Project Name

oslo_project

Crackz Version

0.58.0

 Use the **Pages** menu in the sidebar to navigate to different views

app

 **Detection Results**

 GPS Map

 Training Metrics

 Parameter Tuning



Detection Results

View crack detection results with visualizations and statistics

 No project loaded. Please select a project from the home page.

app

 Detection Results

 **GPS Map**

 Training Metrics

 Parameter Tuning



GPS Map Visualization

View crack detections on an interactive map

 No project loaded. Please select a project from the home page.

app

 Detection Results

 GPS Map

 **Training Metrics**

 Parameter Tuning



Training Metrics

Monitor training progress and model performance

 No project loaded. Please select a project from the home page.

app

 Detection Results

 GPS Map

 Training Metrics

 **Parameter Tuning**



Parameter Tuning

Interactively adjust detection parameters and preview results

 No project loaded. Please select a project from the home page.

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```
# Launch with a specific project
crackz dashboard --project my-project

# Launch with custom port
crackz dashboard --project my-project --port 8502
```

1. `crackz-gui`
- 2.
- 3.
- 4.
- 5.
- 6.

```
streamlit run core/src/crackz/dashboard/app.py -- --project my-project
```

```
from crackz.dashboard import launch_dashboard
launch_dashboard(project_name="my-project", port=8501)
```

```
crackz geo extract-gps --project <name>
```

```
artifacts/training/training_metrics.json
```

```
crackz train --project <name>
```

```
core/src/crackz/dashboard/
├── init .py          # Launch function
├── app.py            # Main dashboard app
├── pages/           # Dashboard pages
│   ├── 1_0_Detection_Results.py
│   ├── 2_0_GPS_Map.py
│   ├── 3_0_Training_Metrics.py
│   └── 4_0_Parameter_Tuning.py
├── utils/           # Utility modules
│   ├── data_loader.py # Data loading functions
│   ├── image_utils.py # Image processing utilities
│   ├── map_utils.py   # GPS mapping utilities
│   └── viz_utils.py   # Visualization functions
```

```
artifacts/detections/visualizations/
artifacts/detections/detection_results.json
artifacts/training/training_metrics.json
data/{test,val,train,raw}/
```

```
crackz dashboard --project my-project --port 8080
```

```
.streamlit/config.toml
```

```
[server]
port = 8501
headless = true

[theme]
primaryColor = "#FF4B4B"
backgroundColor = "#FFFFFF"
secondaryBackgroundColor = "#F0F2F6"
```

```
textColor = "#262730"  
font = "sans serif"
```

- streamlit>=1.30.0
- streamlit-folium>=0.15.0
- folium>=0.15.0
- plotly>=5.18.0
- pandas>=2.0.0

1.

```
crackz detect --project my-project
```

2.

3.

1.

```
crackz geo extract-gps --project my-project
```

2.

3.

1.

2.

3.

1.

2.

3.

8502

```
uv sync
```

```
pip install streamlit
```

```
--port
```



```
crackz detect --project <name>
```

```
artifacts/detections/visualizations/
```

```
crackz geo extract-gps
```

```
crackz train --project <name>
```

```
artifacts/training/training_metrics.json
```

- 1.
- 2.
- 3.
- 4.

```
# 1. Initialize project
crackz init my-project

# 2. Import images
crackz import-images --source ./photos --project my-project

# 3. Train model
crackz train --project my-project

# 4. Run detection
crackz detect --project my-project

# 5. Launch dashboard to view results
crackz dashboard --project my-project
```

```
from crackz.dashboard import launch_dashboard

# Launch dashboard programmatically
launch_dashboard(
    project_name="my-project", # Optional: project to load
    port=8501                 # Optional: server port
)
```

```
# Show help
crackz dashboard --help

# Launch with project
crackz dashboard --project my-project

# Launch with custom port
crackz dashboard --project my-project --port 8080

# Launch without initial project
crackz dashboard
```

```
# 1. Create and setup project
crackz init harbor-inspection
crackz import-images --source ./drone-photos --project harbor-inspection

# 2. Train model
crackz train --project harbor-inspection --epochs 50

# 3. View training progress
crackz dashboard --project harbor-inspection
# Navigate to "Training Metrics" page to review

# 4. Run detection on test set
crackz detect --project harbor-inspection --split test

# 5. Analyze results
crackz dashboard --project harbor-inspection
# Navigate to "Detection Results" page
# Navigate to "GPS Map" page to see locations

# 6. Tune parameters if needed
# Use "Parameter Tuning" page to optimize thresholds

# 7. Re-run with optimized parameters
crackz detect --project harbor-inspection --threshold 0.6

# 8. Generate final report
crackz report generate --project harbor-inspection --format pdf
```

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1. _____
2. _____
3. _____ dashboard

AI Chat Assistant

AI Chat Assistant

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```
# Install keyring support
uv add keyring

# Or with pip
pip install keyring
```

```
from crackz_gui.state.secure_storage import (
    store_api_key,
    retrieve_api_key,
    is_secure_storage_available,
)
```

```
# Check if keyring is available
if is_secure_storage_available():
    # Store API key securely
    store_api_key("your-api-key-here")

    # Retrieve API key
    api_key = retrieve_api_key()

    # Delete API key
    from crackz_gui.state.secure_storage import delete_api_key
    delete_api_key()
else:
    print("Keyring not available - falling back to plaintext storage")
```

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- 11.



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You: My detection is finding too many false positives. How can I reduce them?

AI: Based on your project, I recommend:

1. Increase confidence_threshold from 0.25 to 0.4
2. Adjust iou_threshold to 0.5 for stricter overlap filtering
3. Consider retraining with more negative examples

Would you like me to explain how to adjust these in your project?

You: How many epochs should I train for?

AI: For your dataset with 150 training images:

- Start with 50 epochs as a baseline
- Monitor validation loss - it should decrease steadily
- If validation loss plateaus before 50 epochs, you can stop early
- If still improving at 50, try 75-100 epochs

Your current batch_size of 8 is good for this dataset size.

You: Analyze my project configuration

AI: Here's what I found in your oslo_project:

- Good splits: 70/20/10 train/val/test distribution
- △ Low image count (150 total) - consider data augmentation
- Reasonable batch size (8) for dataset
- Suggestion: Enable data augmentation in training config
- Suggestion: Consider collecting more diverse training data

You: How do I export detection results to CSV?

AI: To export detection results:

1. Run detection: ``crackz detect run <project>``
2. Results are saved in `project/artifacts/detections/`
3. Use export command: ``crackz export --format csv <project>``

Or in GUI:

1. Click "Run Detection" button
2. Results appear in Detection Results panel
3. Click "Export" → "CSV Format"

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5-20251001 claude-opus-4-1-20250805

claude-sonnet-4-5-20250929 claude-haiku-4-

§ ~ Ctrl+/

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- 6.
- 7.

`chat_window.py`

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`ai_chat.py`

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-

`chat_settings_dialog.py`

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1. `ai_chat.py` `_get_context()`

```
def _get_context(self) -> str:  
    if self.app_state is None or self.app_state.project is None:
```



```
        return "No project loaded"

    project = self.app_state.project

    # Add your new context here
    return f"""
    Current project: {project.name}

    Your new context:
    - Additional info: {project.additional_info}
    """
```

2.

```
uv run task tests
```

- _____
- _____
- _____

Configuration

Configuration Reference

```
crackz init
```

```
CLI flag > ENV var > Project YAML > Code default
```

```
crackz init --name demo-project
```

```
demo-project/_project.yaml
```

```
<name>_project.yaml
```

```
project:
  name: demo-project
  description: Demo project
  created: 2025-01-01T12:00:00+00:00
paths:
  project_path: demo-project
  image_path: demo-project/images
  image_mask_path: demo-project/masks
  artifacts_path: demo-project/artifacts
logging:
  level: INFO
  log_dir: logs
ai:
  model:
    encoder: resnet34
    encoder_weights: imagenet
    device: cpu
    in_channels: 3
    num_classes: 1
    input_size: [512, 512]
    threshold: 0.5
    loss_name: combo
    dice_weight: 0.7
    focal_weight: 0.3
  training:
    batch_size: 8
    num_workers: 4
    epochs: 10
    learning_rate: 0.001
    shuffle: true
  detection:
    threshold: 0.5
    learning_rate: 0.001
    batch_size: 8
    shuffle: true
    output_dir_name: detections
data:
  mask_threshold: 50
schema_version: 2
```

```
ProjectConfig
```

```
from pathlib import Path
from crackz.config.project_config import ProjectConfig

pc = ProjectConfig(
    name="demo-project",
    project_path=Path("demo-project"),
    image_path=Path("demo-project/images"),
    image_mask_path=Path("demo-project/masks"),
    artifacts_path=Path("demo-project/artifacts"),
)
pc.save(Path("project_config.yaml"))

# Later
loaded = ProjectConfig.load(Path("project_config.yaml"))
project = loaded.to_project()
```

```
Project
```

```
from crackz.config.project_config import ProjectConfig
pc = ProjectConfig.from_project(project)
pc.save(Path("backup_config.yaml"))
```

	ModelConfig	
	TrainingConfig	
	DetectionConfig	

--	--	--

segmentation-models-pytorch

efficientnet-b0				
efficientnet-b3				
efficientnet-b7				
resnet18				
resnet34				
resnet50				
resnet101				
densenet121				
mobilenet_v2				
vgg16				
	swsl	imagenet	None	ssl
			null	

Binary Segmentation (Default)

```
ai:
  model:
    encoder: resnet34
    num_classes: 1      # Single output channel
    threshold: 0.5      # Probability threshold for crack/no-crack
```

Multi-Class Segmentation

```
ai:
  model:
    encoder: efficientnet-b0
    num_classes: 3      # E.g., background=0, crack=1, corrosion=2, spalling=3
    threshold: 0.5      # Applied per-class
```

Grayscale Input

```
ai:
  model:
    encoder: mobilenet_v2
    in_channels: 1      # Grayscale images
```

```
normalization:
  mean: [0.5]
  std: [0.5]
```

Training from Scratch (No Pretrained Weights)

```
ai:
  model:
    encoder: resnet34
    encoder_weights: null # Random initialization
```



TrainingConfig

int int | str

```
tensorboard
artifacts/runtime/tensorboard/
  getattr(project.ai.training, "tensorboard", False) False
```

```
# Safe access pattern inside runtime code
use_tb = getattr(project.ai.training, "tensorboard", False)
if use_tb:
  ... # create logger
```

getattr(..., default)

False tests/unit/crackz/ai/test_detection.py::test_tensorboard_logger_disabled_by_default model_fields

TrainingConfig

--	--	--

min_confidence medium_confidence high_confidence

-
-
-

--	--	--

```
ai:
  detection:
    min_confidence: 0.005 # Save detections with ≥0.5% crack area
    medium_confidence: 0.02 # "Probable Crack" at ≥2% crack area
    high_confidence: 0.05 # "Crack Detected" at ≥5% crack area
    min_crack_area_px: 100 # Suppress detections smaller than 100 pixels
```

data

--	--	--

DataConfig

--	--	--

mask_threshold

-
-
-

--	--	--

```
data:
  mask_threshold: 50 # Default - balanced filtering
```

```
crackz import-images --masks_src <path>
scripts/import_masks.py
```

```
project.import_images()
```

```
crackz migrate
```

```
mask_threshold:
```

```
50
```

```
crackz export-config --project demo-project --out tuned_config.yaml
```

```
from crackz.config.project_config import ProjectConfig
pcfg = ProjectConfig.from_project(project)
pcfg.save(Path("tuned_config.yaml"))
```

```
CRACKZ_LOG_DIR=run_logs crackz detect run --project demo-project
```

```
$env:CRACKZ_LOG_DIR='run_logs'; crackz detect run --project demo-project
```

```
ai.model.device
resolve_device_and_accelerator
cuda python -c "import torch; print(torch.cuda.is_available())"
```

```
ai.model.device:
```

```
tensorboard: true
```

```
ai:
  training:
    tensorboard: true
    epochs: 10
    batch_size: 8
```

```
from crackz import Project

project = Project.load(Path("demo-project"))
project.ai.training.tensorboard = True
project.save()
```

```
<artifacts_path>/runtime/tensorboard/
```

```
tensorboard --logdir demo-project/artifacts/runtime/tensorboard
```

```
docker run --rm -v $(pwd)/demo-project:/workspace -p 6006:6006 crackz-cli \
  tensorboard --logdir /workspace/artifacts/runtime/tensorboard
```

```
crackz detect run --project p --cfg cfg.yaml
```

```
training.epochs    batch_size
```

```
detection.batch_size
```

```
<artifacts_path>/models/                best.ckpt                last.ckpt
epoch={N}-val_iou={score}.ckpt          epoch=9-val_iou=0.8234.ckpt
<artifacts_path>/checkpoints/
```

```
ai:
  training:
    save_top_k: 1      # Keep N best checkpoints (by val_iou), default: 1
    save_last: true    # Always save last.ckpt, default: true
    auto_prune_checkpoints: false # Auto-prune after training, default: false
    checkpoint_keep: 3  # Checkpoints to keep when auto-pruning, default: 3
```

1. save_top_k: 1, save_last: true
2. save_top_k: 3, auto_prune_checkpoints: true, checkpoint_keep: 3
3. save_top_k: 1, save_last: false, auto_prune_checkpoints: true, checkpoint_keep: 1

```
crackz detect prune-checkpoints --project demo-project --keep 3
```

```
docker run --gpus all --rm -v %cd%\project:/workspace crackz-cli:gpu \
detect prune-checkpoints --project /workspace/demo-project --keep 2
```

```
from crackz.ai.checkpoints import prune_checkpoints
removed = prune_checkpoints(project, keep=2)
```

0.8234.ckpt last.ckpt best.ckpt 003-

<artifacts_path>/detections/detection_results.json

```
from crackz.ai.results import load_detection_results
from crackz.ai.eval import basic_summary
results = load_detection_results(project)
print(basic_summary(results))
```

crack_severity

seed

```
# project_config.yaml (unified example)
name: demo-project
description: Example project
project_path: demo-project
image_path: demo-project/images
image_mask_path: demo-project/masks
artifacts_path: demo-project/artifacts
logging:
  level: INFO
  log_dir: logs
model:
  encoder: resnet34
  device: cpu
  input_size: [512, 512]
training:
  epochs: 5
  batch_size: 4
  learning_rate: 0.0005
  tensorboard: false # Set to true to enable TensorBoard logging
detection:
  threshold: 0.55
  batch_size: 4
```

schema_version

```
crackz migrate --project ./my_project
```

false schema_version schema_version: 1 tensorboard:


```
crackz migrate --cfg ./my_project/my_project_project.yaml
```

```
crackz migrate --project ./my_project --dry-run
```

```
crackz migrate --project ./my_project --no-backup
```

up-to-date noop

```
for d in projects/*; do
[ -d "$d" ] && crackz migrate --project "$d" --no-backup;
done
```

ProjectConfig

Training Configuration

TrainingConfig Contract

TrainingConfig	crackz.project.config	ai.training
batch_size		
num_workers		
epochs		
learning_rate		
shuffle		
loss		
metrics		
seed		
precision		
devices		
strategy		
tensorboard	artifacts/training/tensorboard	
tests/unit/crackz/project/test_training_config_schema.py		
TrainingConfig		
1.		
2.		
3.		
4.	CURRENT_SCHEMA_VERSION	
1.		
2.		
3.		
4.		
	--epochs	--batch-size
tensorboard=True	<project>/artifacts/training/tensorboard	
seed		

Data Splits & Import

Data Splits and Import Guide

- 1. _____
 - 2. _____
 - 3. _____
 - 4. _____
 - 5. _____
-



images/raw/

raw/

```
crackz import-images --project my-project --src ./photos --type raw --size 512 512
```

images/train/ masks/train/

img001.jpg img001.png

images/val/ masks/val/

`images/test/` `masks/test/`

`--type raw`

```
crackz import-images \  
--project my-project \  
--src ./labeled-data/images \  
--src-masks ./labeled-data/masks \  

```

```
--split-ratios 0.7 0.15 0.15 \
--size 512 512
```

```
crackz import-images \
--project my-project \
--src ./labeled-data/images \
--src-masks ./labeled-data/masks \
--split-ratios 0.7 0.15 0.15 \
--seed 42 \
--size 512 512
```

```
# Small dataset: larger validation set
crackz import-images \
--project my-project \
--src ./small-dataset/images \
--src-masks ./small-dataset/masks \
--split-ratios 0.6 0.25 0.15 \
--size 512 512

# Large dataset: more for training
crackz import-images \
--project my-project \
--src ./large-dataset/images \
--src-masks ./large-dataset/masks \
--split-ratios 0.8 0.1 0.1 \
--size 512 512
```

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.
- 11.
- 12.
- 13.
- 14.
- 15.

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-

- 16.
- 17.
- 18.
- 19.
- 20.
- 21.
- 22.

23.

24.

25.

26.

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```
crackz init --name harbor-inspection
cd harbor-inspection
```

```
crackz import-images \
--project . \
--src ~/datasets/harbor-cracks/images \
--src-masks ~/datasets/harbor-cracks/masks \
--split-ratios 0.7 0.15 0.15 \
--seed 42 \
--size 512 512
```

images/train/
masks/

images/val/

images/test/

```
# Count images
ls images/train/ | wc -l # Should show 140
ls images/val/ | wc -l # Should show 30
ls images/test/ | wc -l # Should show 30

# Count masks (should match images)
ls masks/train/ | wc -l # Should show 140
ls masks/val/ | wc -l # Should show 30
ls masks/test/ | wc -l # Should show 30
```

```
crackz detect train --epochs 50 --batch-size 8
```

images/val/ masks/val/ images/train/ masks/train/ artifacts/metrics.json artifacts/loss_curve.png checkpoints/best.ckpt

```
crackz detect run --split test
```

masks/test/ images/test/ detections/

```
# Import new unlabeled images
crackz import-images \
  --project . \
  --src ~/new-harbor-photos \
  --type raw \
  --size 512 512

# Run detection
crackz detect run --split raw
```

detections/

1.

```
--seed 42 # Same split every time
```

2.

```
cp -r original-data backup/
```

3.

```
# Check if splits are representative
ls images/train/ | wc -l
ls images/val/ | wc -l
```

4.

```
# For 50 images: 0.6, 0.25, 0.15 ensures val gets ~13 images
--split-ratios 0.6 0.25 0.15
```

5.

```
# In project documentation
split_info:
  seed: 42
  ratios: [0.7, 0.15, 0.15]
  date: 2025-10-19
```

1.

2.

3.

4.

5.

6.

- -
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 -
 -
 -
-

```
# Wrong: 0.7 + 0.2 + 0.2 = 1.1 x
# Right: 0.7 + 0.15 + 0.15 = 1.0
--split-ratios 0.7 0.15 0.15
```

images/val/

```
# If you have 10 images and use 0.7/0.15/0.15:
# train=7, val=1, test=2 (works)

# If you have 5 images and use 0.8/0.1/0.1:
# train=4, val=0, test=1 (val is empty!)

# Solution: Use larger val ratio or more images
--split-ratios 0.6 0.25 0.15 # Ensures val gets at least 1
```

```
# Image: images/crack_001.jpg
# Mask must be: masks/crack_001.png (same stem)

# Check your filenames:
ls images/ > image_list.txt
ls masks/ > mask_list.txt
# Compare the stems (filename without extension)
```

```
# Add --seed for reproducible splits
--seed 42
```

```
# Increase validation ratio
--split-ratios 0.65 0.25 0.1 # Gives more to validation
```

```
# Import with smaller size
--size 256 256 # Instead of 512 512

# Or reduce batch size during training
crackz detect train --batch-size 2 # Instead of 8
```



```
# Separate images by class first
mkdir temp/crack temp/no-crack

# Move files to class folders
# ... organize manually or with script ...

# Import each class separately with same ratio
crackz import-images --src temp/crack --split-ratios 0.7 0.15 0.15 --seed 42
crackz import-images --src temp/no-crack --split-ratios 0.7 0.15 0.15 --seed 42
```

```
# Import full dataset to train
crackz import-images --src data --type train --size 512 512

# Manually create folds by moving files between train/val
# Train 5 models with different train/val splits
# Average results for robust performance estimate
```

```
# Import new batch with same seed and ratios
crackz import-images \
  --src new-data \
  --split-ratios 0.7 0.15 0.15 \
  --seed 42 \
  --size 512 512

# WARNING: This adds to existing splits
# Consider fresh re-import of all data for clean splits
```

```
images/raw/
images/train/
images/val/
images/test/
```

```
# Import for training (with split)
crackz import-images --src data/images --src-masks data/masks \
  --split-ratios 0.7 0.15 0.15 --seed 42 --size 512 512

# Import for detection (no split)
crackz import-images --src photos --type raw --size 512 512

# Train on train split, validate on val split
crackz detect train --epochs 50 --batch-size 8

# Test on test split
crackz detect run --split test

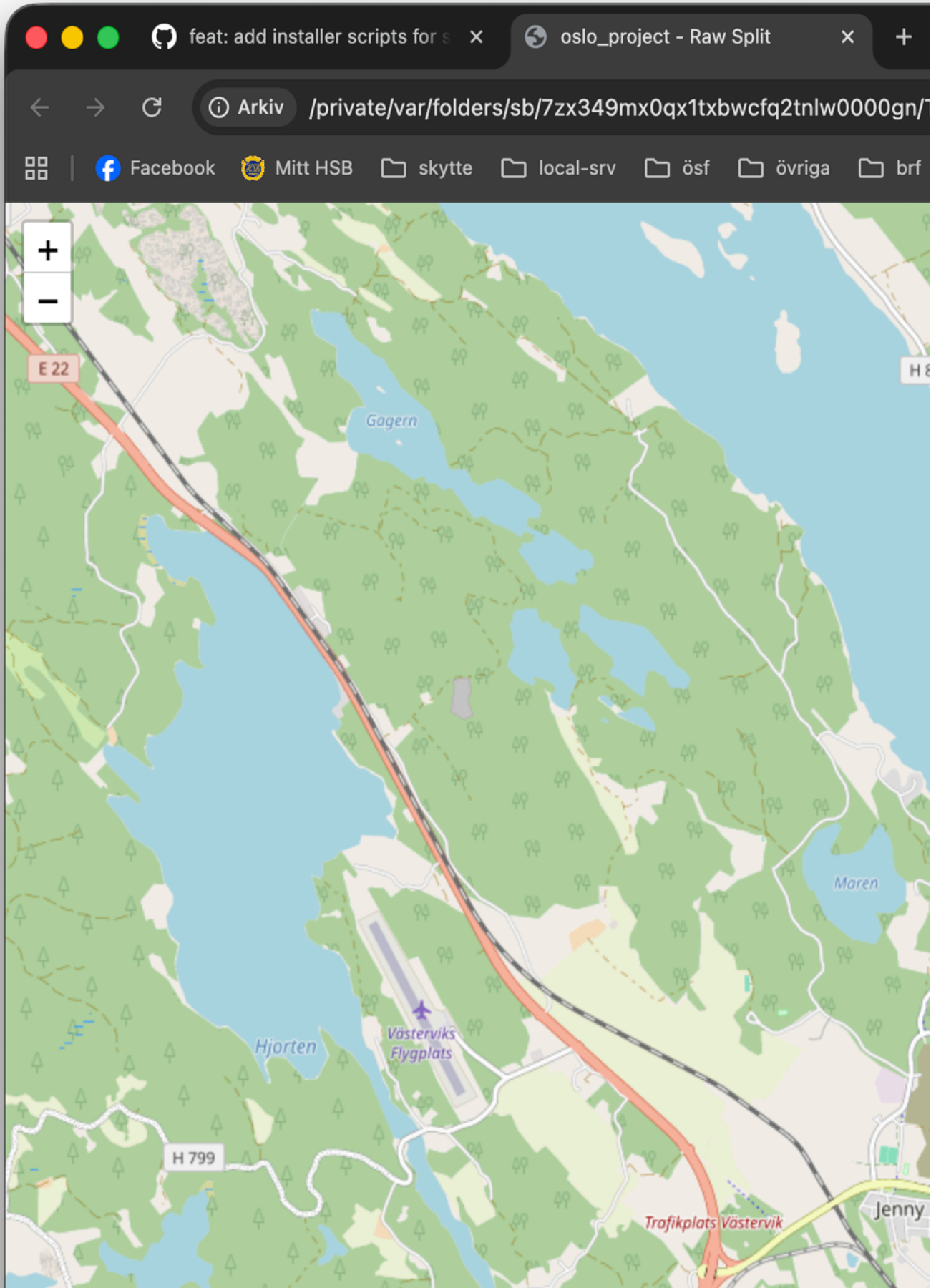
# Detect on raw split
crackz detect run --split raw
```

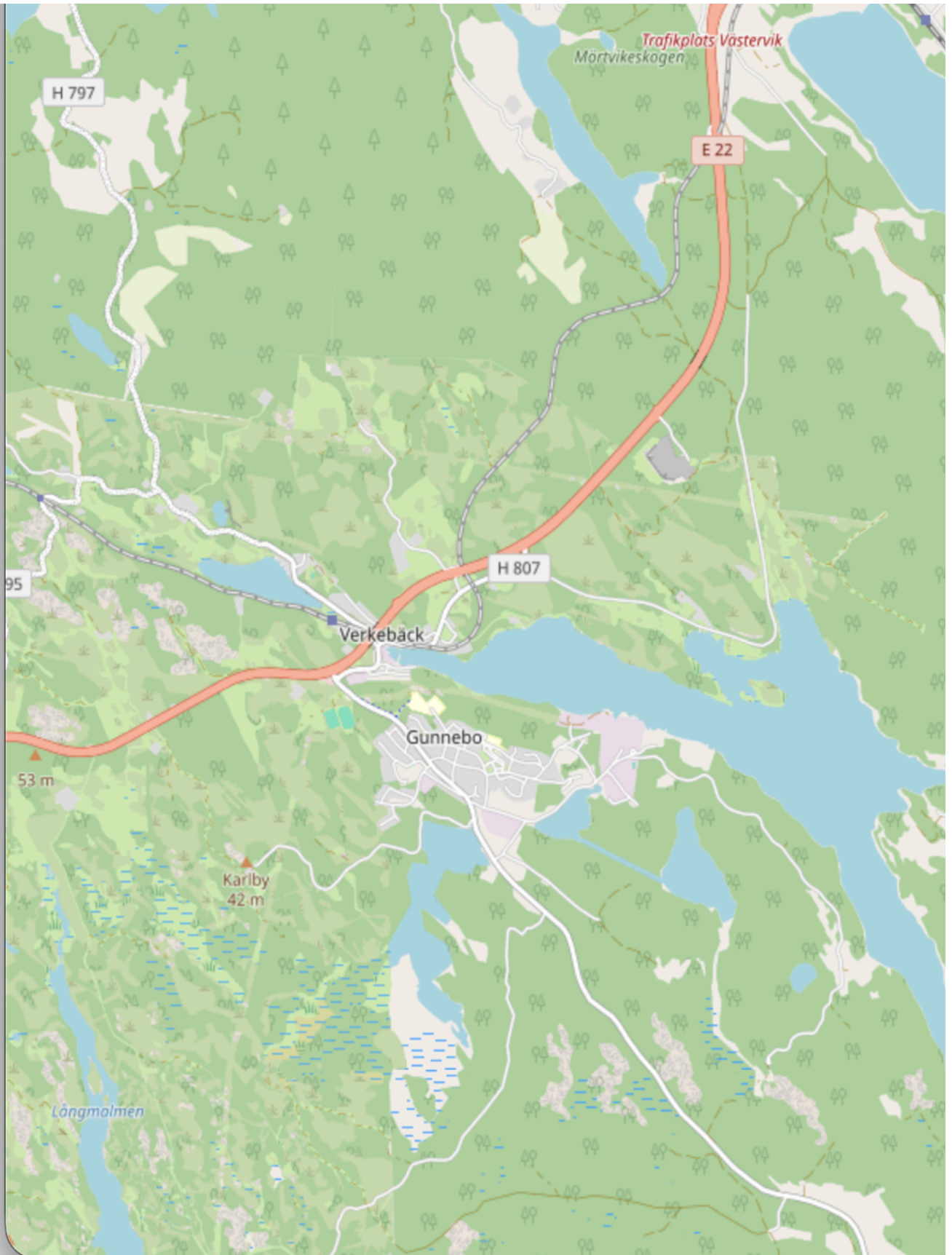
-
-
-

- _____
- _____
- _____
- _____
- _____

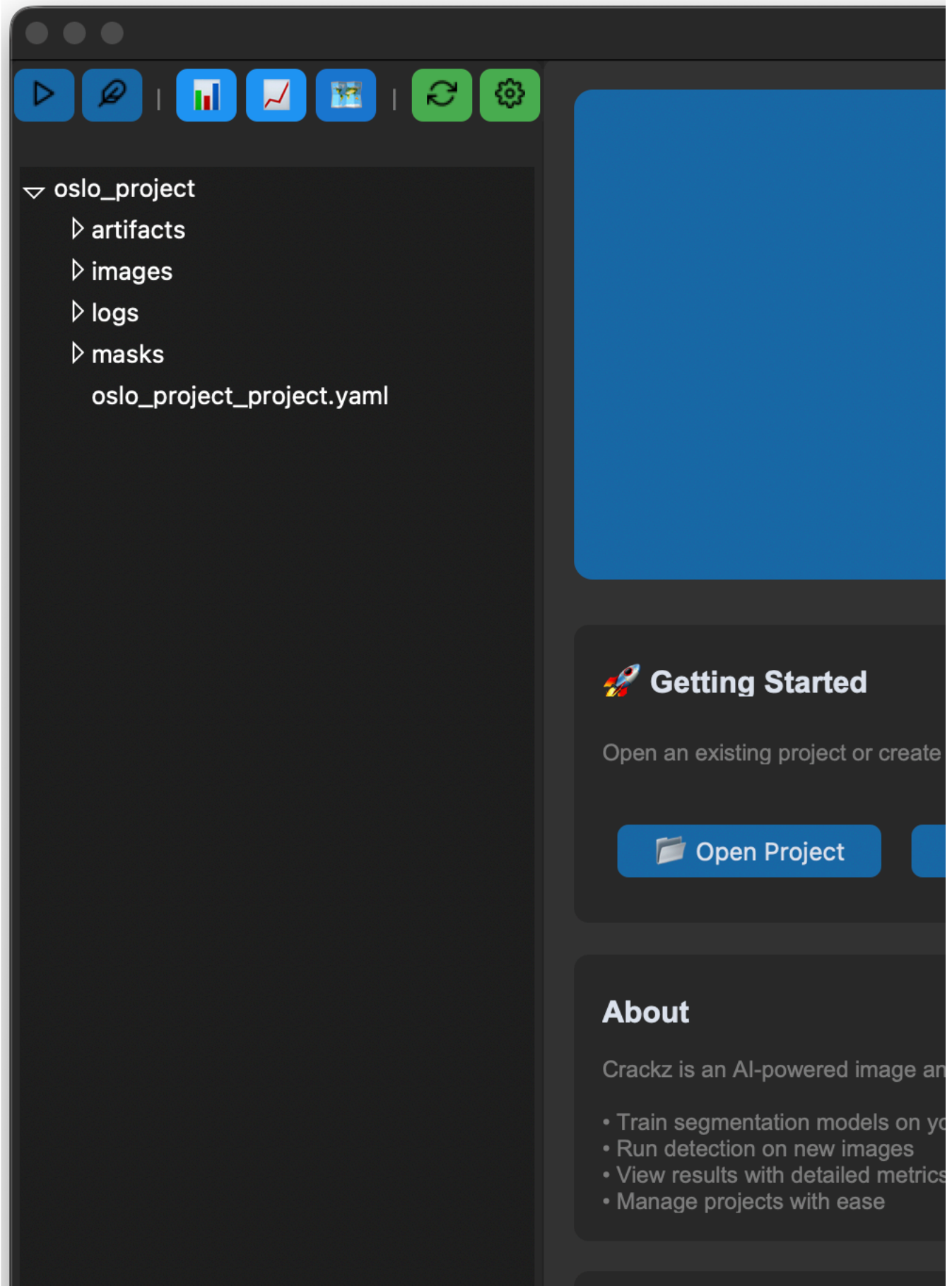
Map Viewer

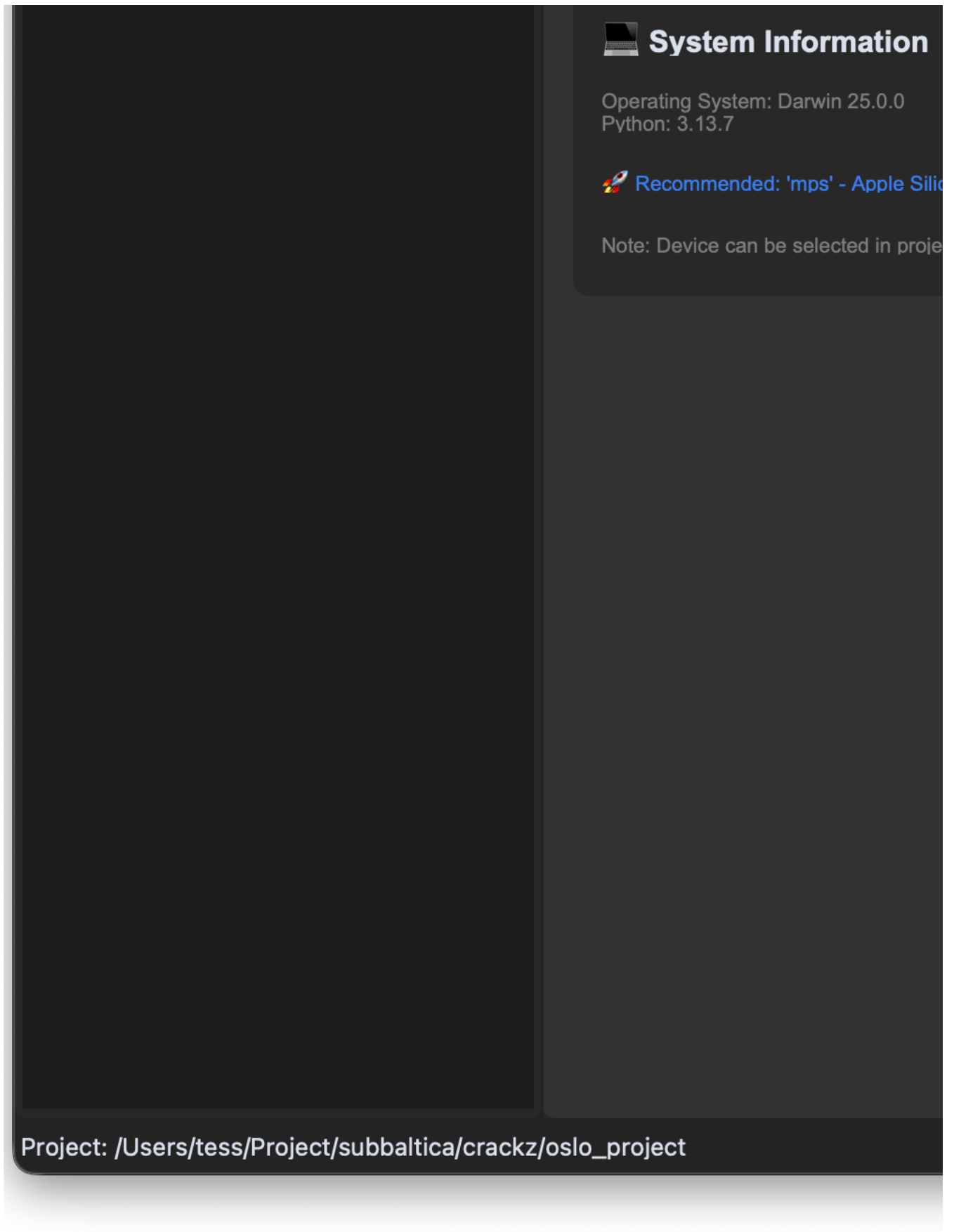
Map Viewer Feature





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•





Interactive map view opened in web browser with geotagged images displayed as markers

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-

- `gui/src/crackz_gui/components/map_viewer.py`
-
-
- `webbrowser.open()`
-

- 1.
- 2.
- 3.
- 4.
- 5.

```
from pathlib import Path
from crackz_gui.qt.views.map_viewer_view import MapViewerDialog
from crackz.project.project import load_project

# Load a project
project = load_project(Path("path/to/project"))

# Note: MapViewerDialog requires a parent GUI app instance and is typically
# used within the Crackz GUI application. For command-line access to GPS data, use:
from crackz.project.geoposition import get_project_geopositions

# Get GPS data for all images in the project
gps_data = get_project_geopositions(project)
for image_path, geo_pos in gps_data.items():
    print(f"{image_path}: {geo_pos.latitude}, {geo_pos.longitude}")
```

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`tests/test_data/` `oslo_project/`

- `tests/test_data/images/`
- `oslo_project/images/`

`scripts/add_test_gps_data.py`

```
python scripts/add_test_gps_data.py
```

- 1.
2. `logs/crackz.log`
- 3.

-
- `crackz.project.geoposition.get_project_geopositions()`
- `python scripts/add_test_gps_data.py`

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- `gui/src/crackz_gui/components/map_viewer.py`
- `gui/src/crackz_gui/components/browser/toolbar.py`

- `core/src/crackz/project/geoposition.py`
- `scripts/add_test_gps_data.py`

Report Generation

Report Generation

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```
crackz report generate --project demo-project
demo-project_report.docx
```

```
crackz report generate --project demo-project --output inspection_report.docx
```

```
crackz report generate --cfg project_config.yaml --output report.docx
```

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-

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-
-

```
ai:
detection:
  min_confidence: 0.005    # 0.5% - Possible Crack threshold
  medium_confidence: 0.02  # 2% - Probable Crack threshold
  high_confidence: 0.05   # 5% - Crack Detected threshold
```

detections/

```
crackz report generate --project harbor_inspection_2024 \
--output "Harbor_Inspection_Report_2024-Q4.docx"
```

```
# Monthly reports
crackz report generate --project monthly_monitoring \
--output "reports/2024-11_monitoring.docx"
```

```
crackz report generate --project compliance_check \
--output "compliance/inspection_$(date +%Y%m%d).docx"
```

```
crackz report generate --project qa_validation \
--output "internal/qa_review.docx"
```

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-
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-

-
-
-

```
crackz detect run
```

```
crackz detect run --project demo-project
```

```
# Check for visualization images (new structure)
ls -la demo-project/artifacts/detections/visualizations/

# Or legacy location (older projects)
ls -la demo-project/artifacts/detections/*.png
```

```
crackz detect run --project demo-project
```

```
artifacts/detections/visualizations/
```

```
data:
size: [512, 512] # Use smaller images
```

```
--output
```

```
crackz report generate --project harbor_main \  
--output "Harbor_Main_Inspection_2024-11-05.docx"
```

```
for project in project1 project2 project3; do  
  crackz report generate --project "$project" \  
  --output "reports/${project}_${date +%Y%m%d}.docx"  
done
```

```
crackz report generate --project demo-project # Creates demo-project/demo-project_report.docx
```

```
# Add to project notes  
project:  
  notes: "Report generated 2024-11-05 with default thresholds"
```

- _____
- _____
- _____
- _____

Docker

Docker Usage

```
# Pull from registry
docker pull ghcr.io/anaxiatech-ab/crackz:latest
docker pull ghcr.io/anaxiatech-ab/crackz:slim
docker pull ghcr.io/anaxiatech-ab/crackz:gpu

# Run from registry
docker run --rm ghcr.io/anaxiatech-ab/crackz:latest --help
```

latest	Dockerfile		
slim	Dockerfile.slim		
gpu	Dockerfile.gpu		

```
# Full
docker build -t crackz-cli .
# Slim
docker build -f Dockerfile.slim -t crackz-cli:slim .
# GPU (CUDA 12.8)
docker build -f Dockerfile.gpu -t crackz-cli:gpu .
```

```
docker build --build-arg PYTHON_VERSION=3.13 -t crackz-cli .
```

1.

ghcr.io/astral-sh/uv:bookworm-slim

uv build --wheel
2.
3.

python:3.13-slim
4.

nvidia/cuda:12.8.0-runtime-ubuntu22.04
5.
6.

--build-arg TORCH_INDEX_URL=...

- uv build
-
-
-

- nvidia/cuda:12.8.0-runtime-ubuntu22.04
-
- git libglib2.0-0 libjpeg-turbo8 libgomp1 ca-certificates
-
-


```
# Default: CUDA 12.8 (recommended, matches PyTorch cu128)
docker build -f Dockerfile.gpu -t crackz-cli:gpu .

# Override for CUDA 12.1
docker build -f Dockerfile.gpu \
  --build-arg TORCH_INDEX_URL=https://download.pytorch.org/whl/cu121 \
  -t crackz-cli:gpu-cu121 .

# Override for CUDA 12.4 (if needed for compatibility)
docker build -f Dockerfile.gpu \
  --build-arg TORCH_INDEX_URL=https://download.pytorch.org/whl/cu124 \
  -t crackz-cli:gpu-cu124 .
```

TORCH_INDEX_URL

```
#18 [runtime 7/9] RUN /app/.venv/bin/python3 -c 'import torch; print("Torch:", torch.__version__, "CUDA available:", torch.cuda.is_available())'
#18 0.699 Torch: 2.8.0+cpu CUDA available: False
```

CUDA available: False

linux/amd64

/home/app

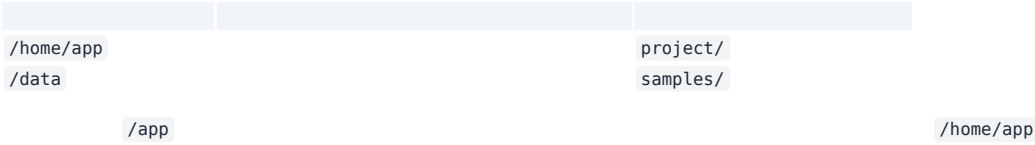
```
# Help
docker run --rm crackz-cli --help

# Init project (creates /home/app/demo-project inside container)
mkdir -p project
docker run --rm -v "$(pwd)/project:/home/app" crackz-cli init --name demo-project

# Import images (mount samples to /data)
mkdir -p samples
# (populate samples/ with images and optional masks/ first)
docker run --rm \
  -v "$(pwd)/project:/home/app" \
  -v "$(pwd)/samples:/data" \
  crackz-cli import-images --project /home/app/demo-project --src /data --type raw --size 512 512

# Detection
docker run --rm -v "$(pwd)/project:/home/app" crackz-cli detect run --project /home/app/demo-project
```

```
docker run --gpus all --rm -v "$(pwd)/project:/home/app" crackz-cli:gpu detect run --project /home/app/demo-project
```



```
CRACKZ_LOG_DIR      logs
PYTHONUNBUFFERED    1

/home/app/central_logs

docker run --rm -e CRACKZ_LOG_DIR=central_logs -v "$(pwd)/project:/home/app" \
  crackz-cli detect run --project /home/app/demo-project
```

/home/app

```
make build      # full image
make build-slim # slim image
make build-gpu  # gpu image (CUDA 12.8)
make init NAME=demo-project # creates project under /home/app
make import PROJECT=demo-project DATA_DIR=./samples
make detect PROJECT=demo-project
make detect-gpu PROJECT=demo-project # GPU detection
make train-gpu PROJECT=demo-project  # GPU training
```

act

```
# Install act (macOS)
brew install act

# Test GPU build with make targets
make act-gpu-dryrun # Dry-run to validate workflow
make act-gpu        # Actually run the GPU build locally
make act-docker-build # Test all Docker builds (full, slim, gpu)
```

	latest
	slim
	gpu
gpu	TORCH_INDEX_URL

--	--	--

CRACKZ_LOG_DIR

--gpus all

TORCH_INDEX_URL

- _____
- _____
- _____

Azure Batch

Azure Batch Deployment

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- 1.
- 2.
- 3.
4. `az login`

```
./scripts/azure/azure_batch.sh \
--batch-account mycrackzaccount \
--storage-account mycrackzstorage
```

mycrackzaccount

mycrackzstorage

```
# Get storage account key
STORAGE_KEY=$(az storage account keys list \
--account-name mycrackzstorage \
--resource-group subbaltica-crackz-rg \
--query "[0].value" -o tsv)
```

```
# Upload project directory
az storage blob upload-batch \
--account-name mycrackzstorage \
--account-key "$STORAGE_KEY" \
--destination crackz-data \
--source ./my-project
```

```
# Create job
az batch job create \
--id crackz-detection-job \
--pool-id crackz-pool

# Add tasks to the job
az batch task create \
--job-id crackz-detection-job \
--task-id detect-task-1 \
--command-line "docker run crackz-cli:latest detect run --project /mnt/batch/tasks/fsmounts/crackz-data/my-project"
```

```

--batch-account
--resource-group          subbaltica-crackz-rg
--location                swedencentral
--pool-id                 crackz-pool
--vm-size                 Standard_D2s_v3
--node-count              2
--image                   crackz-cli:latest
--storage-account
--container                crackz-data

```

```

Standard_D2s_v3
Standard_D4s_v3
Standard_D8s_v3
Standard_NC6s_v3
Standard_NC4as_T4_v3

```

```

./scripts/azure/azure_batch.sh \
--batch-account crackz-batch \
--pool-id high-performance-pool \
--vm-size Standard_D8s_v3 \
--node-count 10 \
--storage-account crackzstore

```

```

./scripts/azure/azure_batch.sh \
--batch-account crackz-batch-gpu \
--pool-id gpu-pool \
--vm-size Standard_NC4as_T4_v3 \
--node-count 2 \
--image crackz-cli:gpu

```

```

# Enable auto-scaling
az batch pool autoscale enable \
--pool-id crackz-pool \
--account-name mycrackzaccount \
--auto-scale-formula '
    startingNumberOfVMs = 2;
    maxNumberOfVMs = 10;
    pendingTaskSamplePercent = $PendingTasks.GetSamplePercent(180 * TimeInterval_Second);
    pendingTaskSamples = pendingTaskSamplePercent < 70 ? startingNumberOfVMs : avg($PendingTasks.GetSample(180 * TimeInterval_Second));
    $TargetDedicatedNodes = min(maxNumberOfVMs, pendingTaskSamples);
'

```

```
#!/bin/bash
# Create job
az batch job create --id multi-project-job --pool-id crackz-pool

# Submit tasks for each project
for project in project1 project2 project3; do
  az batch task create \
    --job-id multi-project-job \
    --task-id "detect-$project" \
    --command-line "docker run crackz-cli:latest detect run --project /data/$project"
done
```

```
# Training task
az batch task create \
  --job-id pipeline-job \
  --task-id train \
  --command-line "docker run crackz-cli:latest detect train --project /data/my-project --epochs 50"

# Detection task (depends on training)
az batch task create \
  --job-id pipeline-job \
  --task-id detect \
  --depends-on train \
  --command-line "docker run crackz-cli:latest detect run --project /data/my-project"
```

```
az batch task create \
  --job-id output-job \
  --task-id detect-with-output \
  --command-line "docker run -v /mnt/output:/output crackz-cli:latest detect run --project /data/project" \
  --resource-files '["httpUrl":"https://mystorage.blob.core.windows.net/input/data.zip","filePath":"data.zip"]' \
  --output-files '["filePattern":"/mnt/output/**/*","destination":{"container":{"containerUrl":"https://mystorage.blob.core.windows.net/results"}},{"uploadOptions":{"uploadC
```

```
az batch pool show \
  --pool-id crackz-pool \
  --account-name mycrackzaccount \
  --query "{State:allocationState, DedicatedNodes:currentDedicatedNodes}"
```

```
az batch job list \
  --account-name mycrackzaccount \
  -o table
```

```
az batch task list \
  --job-id crackz-detection-job \
  --account-name mycrackzaccount \
  -o table
```

```
az batch task file download \
  --job-id crackz-detection-job \
  --task-id detect-task-1 \
  --file-path stdout.txt \
  --destination ./task-stdout.txt \
  --account-name mycrackzaccount
```

```
# After pool creation, modify to use low-priority nodes
az batch pool resize \
```

```
--pool-id crackz-pool \
--account-name mycrackzaccount \
--target-dedicated-nodes 0 \
--target-low-priority-nodes 4
```

```
./scripts/azure/azure_batch.sh \
--batch-account mycrackzaccount \
--pool-id crackz-pool \
--destroy
```

```
az batch location quotas show --location <region>
az batch task file download
```

```
az batch account login
```

```
# Get task details
az batch task show \
--job-id crackz-detection-job \
--task-id detect-task-1 \
--account-name mycrackzaccount

# Download stderr
az batch task file download \
--job-id crackz-detection-job \
--task-id detect-task-1 \
--file-path stderr.txt \
--destination ./task-stderr.txt \
--account-name mycrackzaccount
```

```
# Check current quotas
az batch location quotas show --location swedencentral

# Submit quota increase request through Azure Portal
# Support > New support request > Issue type: Service and subscription limits
```

```
name: Azure Batch Processing
on:
  push:
    branches: [main]
jobs:
  batch-process:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v5

      - name: Azure Login
        uses: azure/login@v1
        with:
          creds: ${ secrets.AZURE_CREDENTIALS }

      - name: Create Batch Pool
        run: |
          ./scripts/azure/azure_batch.sh \
            --batch-account ${ secrets.BATCH_ACCOUNT } \
            --storage-account ${ secrets.STORAGE_ACCOUNT }

      - name: Submit Batch Job
        run: |
```

```
az batch job create --id ci-job-{{ github.run_number }} --pool-id crackz-pool
az batch task create \
  --job-id ci-job-{{ github.run_number }} \
  --task-id detect \
  --command-line "docker run crackz-cli:latest detect run --project /data/project"
```

- 1.
- 2.
- 3.
- 4.
- 5.

taskSlotsPerNode

```
# Modify pool to allow multiple tasks per node
az batch pool set \
  --pool-id crackz-pool \
  --account-name mycrackzaccount \
  --json-file pool-config.json
```

pool-config.json

```
{
  "taskSlotsPerNode": 4
}
```

```
ai:
  detection:
    batch_size: 16 # Adjust based on VM size
```

- _____
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- _____

Azure VM with CUDA

Azure Windows VM with CUDA for Crackz

```
azure_vm_cuda.sh
```

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-
-
-

```
# Deploy default GPU VM
./scripts/azure/azure_vm_cuda.sh

# Deploy with custom configuration
./scripts/azure/azure_vm_cuda.sh \
  --name ml-workstation \
  --size Standard_NC12s_v3 \
  --location westus2

# Connect via RDP and verify
# RDP to the public IP, then run:
nvidia-smi
cd C:\Crackz && .venv\Scripts\Activate.ps1
python -m crackz --version
```

```
Standard_NC6s_v3
Standard_NC12s_v3
Standard_NC24s_v3
```

```
Standard_NC6s_v2
Standard_NV6
```

```
# Default deployment (V100 GPU, East US)
./scripts/azure/azure_vm_cuda.sh

# Custom name and location
./scripts/azure/azure_vm_cuda.sh --name gpu-crackz --location westeurope

# Specific GPU configuration
./scripts/azure/azure_vm_cuda.sh --size Standard_NC12s_v3 --location northeurope
```


Option	Value
--name NAME	crackz-gpu-vm
--location REGION	eastus
--size SIZE	Standard_NC6s_v3
--user USERNAME	crackzuser
--password PASSWORD	CrackzGpu2024!
--mode MODE	standard
--data-disk-size SIZE	1024
--storage-type TYPE	Premium_LRS
--no-data-disk	
--no-cuda	
--no-crackz	
--enable-nested-virt	
--destroy	
--help	

```
# Production inference server
./scripts/azure/azure_vm_cuda.sh \
--name crackz-prod \
--size Standard_NC12s_v3 \
--location westus2 \
--mode production

# Development environment with large storage
./scripts/azure/azure_vm_cuda.sh \
--name crackz-dev \
--size Standard_NV6 \
--mode development \
--data-disk-size 2048 \
--storage-type Premium_LRS

# Budget setup without data disk
./scripts/azure/azure_vm_cuda.sh \
--name crackz-simple \
--size Standard_NV6 \
--no-data-disk

# Manual setup (no auto-install)
./scripts/azure/azure_vm_cuda.sh \
--name crackz-manual \
--size Standard_NC6s_v3 \
--no-cuda \
--no-crackz

# Docker development (with nested virtualization)
./scripts/azure/azure_vm_cuda.sh \
--name crackz-docker \
--size Standard_NC6s_v3 \
--enable-nested-virt

# Clean up resources
./scripts/azure/azure_vm_cuda.sh --name crackz-gpu-vm --destroy
```

```
--data-disk-size SIZE
--storage-type TYPE
--no-data-disk
```

```
C:\ (OS Disk - 127 GB)
├─ Windows/
├─ Program Files/
├─ Users/crackzuser/Desktop/Crackz/ # Crackz installation
└─ ...

D:\ (Data Disk - configurable size)
├─ Projects/ # ML project workspace
│   ├── datasets/ # Training datasets
│   ├── models/ # Model checkpoints
│   └─ outputs/ # Detection results
```

	└─ exports/	# Exported models
	└─ temp/	# Temporary processing

--	--	--	--

```
# Production: High-performance storage
./scripts/azure/azure_vm_cuda.sh \
--name ml-prod \
--data-disk-size 2048 \
--storage-type Premium_LRS

# Development: Cost-effective storage
./scripts/azure/azure_vm_cuda.sh \
--name ml-dev \
--data-disk-size 512 \
--storage-type Standard_LRS

# Simple setup: No separate disk
./scripts/azure/azure_vm_cuda.sh \
--name ml-simple \
--no-data-disk
```

- 1.
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 - 7.
 8. C:\Crackz\.venv
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 - 14.
 - 15.
 - 16.
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```
# 1. Connect via RDP to the VM
# 2. Double-click "File Upload Server" desktop shortcut
# 3. Open browser and go to: http://localhost:8080
# 4. Or access remotely: http://<VM-PUBLIC-IP>:8080
```

```
# From your local computer:
# Windows: \\<VM-PUBLIC-IP>\CrackzUploads
# Mac/Linux: smb://<VM-PUBLIC-IP>/CrackzUploads

# Credentials: crackzuser / CrackzGpu2024!
```

```
# Using any SFTP client (WinSCP, FileZilla, etc.)
sftp crackzuser@<VM-PUBLIC-IP>
# Upload to: /d/Projects/uploads/images/
```

```
D:\Projects\uploads\
├─ images/          # Individual images for analysis
└─ batch/           # Batch processing folders
```

```
# Use Windows built-in Remote Desktop
mstsc /v:<PUBLIC_IP>

# Or download RDP file
az vm show -d -g crackz-gpu-rg -n crackz-gpu-vm --query publicIps -o tsv
```

```
# Check GPU status
nvidia-smi

# Test CUDA availability
python -c "import torch; print(f'CUDA available: {torch.cuda.is_available()}')"
python -c "import torch; print(f'GPU count: {torch.cuda.device_count()}')"
```

```
# Activate virtual environment
cd C:\Crackz
.\.venv\Scripts\Activate.ps1

# Check Crackz version
python -m crackz --version

# Test GPU detection
python -m crackz gui # Should detect CUDA device
```

```
# Activate virtual environment
cd C:\Crackz
.\.venv\Scripts\Activate.ps1

# Option 1: Direct analysis on uploaded images
python -m crackz detect run --images "D:\Projects\uploads\images" --device cuda

# Option 2: Create project and import uploaded images
python -m crackz init harbor-inspection
python -m crackz import-images "D:\Projects\uploads\images" harbor-inspection/images/raw
python -m crackz detect run harbor-inspection --device cuda

# Option 3: Batch processing multiple folders
python -m crackz detect run --images "D:\Projects\uploads\batch\*" --device cuda

# View results
explorer "harbor-inspection\artifacts\detection_results"
```

```
# Create test project
python -m crackz init test-project

# Import sample images
python -m crackz import-images C:\path\to\images test-project/images/raw

# Run GPU-accelerated detection
python -m crackz detect run test-project --device cuda
```

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```
az vm deallocate
```

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```
CrackzGpu2024!
```

```
# Manually install CUDA
# Download from: https://developer.nvidia.com/cuda-11-8-0-download-archive
# Choose Windows x86_64 local installer
```

```
# Manual installation
cd C:\Crackz
.\.venv\Scripts\Activate.ps1
uv pip install --find-links https://download.pytorch.org/whl/cu118 torch torchvision
# Download wheel from GitHub releases and install
```

```
# Check drivers
nvidia-smi

# Verify CUDA installation
nvcc --version

# Check PyTorch CUDA
python -c "import torch; print(torch.version.cuda)"
```

- `nvidia-smi -l 1`
-
-
-

```
# Windows Event Logs
eventvwr.msc

# CUDA installation logs
Get-Content C:\ProgramData\NVIDIA Corporation\CUDA Installer\log.txt

# Crackz logs
Get-Content $env:TEMP\crackz-*.log
```

```
az vm list-sizes --location eastus --query "[?contains(name, 'NC')]"
```

```
# Use spot instances (60-90% discount)
az vm create --priority Spot --max-price -1 ...

# Auto-shutdown schedule
az vm auto-shutdown --name crackz-gpu-vm --resource-group crackz-gpu-rg --time 1800

# Deallocate when not in use
az vm deallocate --name crackz-gpu-vm --resource-group crackz-gpu-rg
```

-

```
%TEMP%\crackz-*.log
```

VM vs Batch Comparison

Azure Deployment Comparison

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```
# Create Windows 11 VM
./scripts/azure/azure_vm.sh \
  --name crackz-dev \
  --size Standard_D4s_v3

# Connect via RDP and use GUI
# IP address displayed after creation
```

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```
# Create batch pool with 10 nodes
./scripts/azure/azure_batch.sh \
--batch-account mybatchaccount \
--storage-account mystorage \
--node-count 10 \
--vm-size Standard_D4s_v3

# Submit 100 detection jobs
for i in {1..100}; do
  az batch task create \
    --job-id detection-job \
    --task-id "task-$i" \
    --command-line "docker run crackz-cli:latest detect run --project /data/project-$i"
done
```

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- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.

```
# Phase 1: Development on VM
./scripts/azure/azure_vm.sh --name crackz-dev --size Standard_D4s_v3
# RDP in, use GUI to develop and test
# Export config: crackz export-config --project demo --out optimized.yaml

# Phase 2: Production on Batch
./scripts/azure/azure_batch.sh \
--batch-account prod-batch \
--pool-id production-pool \
```



```
--node-count 20 \  
--vm-size Standard_D8s_v3  
  
# Upload optimized config and data to storage  
# Submit batch jobs with optimized parameters
```

	azure_vm.sh	azure_batch.sh

- 1.
- 2.
- 3.
- 4.

- 1.
- 2.
- 3.
- 4.
- 5.

- 1.
- 2.
- 3.
- 4.

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-

```
# 1. Develop on VM
./scripts/azure/azure_vm.sh --name dev-vm

# 2. Test configuration
crackz detect train --project my-project --epochs 50
crackz detect run --project my-project

# 3. Export configuration
crackz export-config --project my-project --out batch-config.yaml

# 4. Upload to Azure Storage
az storage blob upload-batch \
  --account-name mystorage \
  --destination crackz-data \
  --source my-project

# 5. Create Batch pool
./scripts/azure/azure_batch.sh \
  --batch-account mybatch \
  --storage-account mystorage \
  --node-count 10

# 6. Submit batch jobs
az batch job create --id production-job --pool-id crackz-pool
# ... add tasks
```

- `scripts/azure/azure_vm.sh --help`
- _____
- _____
- _____

`--size Standard_NC*`

`az batch task file download`

- _____
- _____
- _____
- _____

Advanced Performance

Advanced Performance & Reproducibility

	ai.training	ai.detection	ProjectConfig
precision	ai.training		
devices	ai.training		
strategy	ai.training		ddp
seed	ai.training	ai.detection	

321616-mixedbf16-mixed

```
ai:
  training:
    precision: 16-mixed
    batch_size: 16
```

```
crackz detect train --project demo-project
```

```
ai:
  training:
    devices: 2
    strategy: ddp
    batch_size: 8
    precision: bf16-mixed
```

```
crackz detect train --project demo-project
```

```
docker run --gpus all --rm -v %cd%\project:/home/app crackz-cli:gpu \
detect train --project /home/app/demo-project
```

```
ddp
ddp_spawn
auto
```

seed

```
ai:
  training:
```

```
seed: 1234
detection:
seed: 1234
```

PYTHONHASHSEED

random

```
import torch
torch.use_deterministic_algorithms(True)
```

-
-

```
project:
  name: demo-project
  description: Perf tuned
paths:
  project_path: demo-project
  image_path: demo-project/images
  image_mask_path: demo-project/masks
  artifacts_path: demo-project/artifacts
logging:
  level: INFO
  log_dir: logs
ai:
  model:
    encoder: resnet34
    device: cuda
    input_size: [512, 512]
    threshold: 0.5
  training:
    batch_size: 16
    epochs: 20
    learning_rate: 0.0005
    precision: 16-mixed
    devices: 2
    strategy: ddp
    seed: 4242
  detection:
    batch_size: 8
    seed: 4242
```

crackz export-config

```
crackz export-config --project demo-project --out tuned_config.yaml
```

tuned_config.yaml

ai.detection.seed

ddp_spawn

```
torch.use_deterministic_algorithms(True)
```

⋮

Migration Guide

Project Schema Migration

```
ai.training.tensorboard schema_version
```

```
CURRENT_SCHEMA_VERSION schema_version crackz.project.migration schema_version
```

```
schema_version ai.training.tensorboard
schema_version: 1 ai.training.tensorboard: false
data mask_threshold: 1
project.crackz_version
```

```
schema_version: 3
project:
  name: my project
  description: My crack detection project
  created: '2025-01-15T10:30:00Z'
  crackz_version: '0.50.0' # NEW: Tracks Crackz app version
```

```
project.crackz_version
checkpoints/ models/ runtime/
detections/visualizations/
checkpoints_path checkpoints/
models/ runtime/ crackz_version
```

```
data
data:
  mask_threshold: 1
  schema_version: 2
```

```
data.mask_threshold
```

```
detect train detect run
```

- 1.
- 2. Project schema v0 < v1. Migrate now? [y/N]:
- 3. .bak
- 4.
- 5. CRACKZ_AUTO_MIGRATE=1
- 6.

```
crackz migrate --project <path-to-project-dir>
# or
crackz migrate --cfg <path-to-project-yaml>
```

```
-migrate-files      --dry-run      --no-backup      <file>.bak      -
```

```
# Using project directory
crackz migrate --project my_project --dry-run
crackz migrate --project my_project

# Using direct YAML file path
crackz migrate --cfg my_project/my_project_project.yaml
```

```
# Preview what files would be moved
crackz migrate --project my_project --migrate-files --dry-run

# Migrate both schema AND files
crackz migrate --project my_project --migrate-files
```

```
runtime/lightning_logs/ checkpoints/ models/ training/tensorboard/ runtime/tensorboard/ training/lightning_logs/
detections/*.png detections/visualizations/
```

```
{filename}.v{version}.{timestamp}.bak
```

```
myproject_project.v2.20250115_143022.bak
```

```
v2
```

```
20250115_143022
```

```
myproject_project.v2.20250115_143022.bak
.v2.20250115_143022.bak
```

```
CRACKZ_AUTO_MIGRATE=1
```

```
from pathlib import Path
from crackz.project.migration import migrate_project_file

result = migrate_project_file(Path("my_project")) # or path/to_yaml
print(result)
# Example result:
# {
#   'status': 'migrated',
#   'from_version': 2,
#   'to_version': 3,
#   'steps': ['2->3: updated to artifacts structure v3 + added crackz_version tracking'],
#   'path': 'my_project/my_project_project.yaml',
#   'backup_path': 'my_project/my_project_project.v2.20250115_143022.bak',
```

```
# 'dry_run': False
# }
```

dry_run steps status path from_version backup_path to_version

- 1.
- 2.

--	--	--

.bak

.bak

Shell Completion

Shell Completion

crackz

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-

```
crackz --show-completion bash
crackz --show-completion zsh
crackz --show-completion fish
crackz --show-completion powershell
```

source Invoke-Expression

```
crackz --install-completion SHELL
```

SHELL bash zsh fish powershell

```
crackz det<TAB>
```

```
crackz detect
```

```
crackz --install-completion bash
# New shell OR
source ~/.bash_completion
```

bash-completion

--show-completion

```
crackz --install-completion zsh
# Then restart shell or:
autoload -U compinit && compinit
```

\$fpath

fpath

```
crackz --install-completion fish
# Typically installs into ~/.config/fish/completions/
# Reload:
exec fish
```

```
pwsh -File scripts/setup_completion.ps1
# Or force re-install:
pwsh -File scripts/setup_completion.ps1 -Force
```

crackz

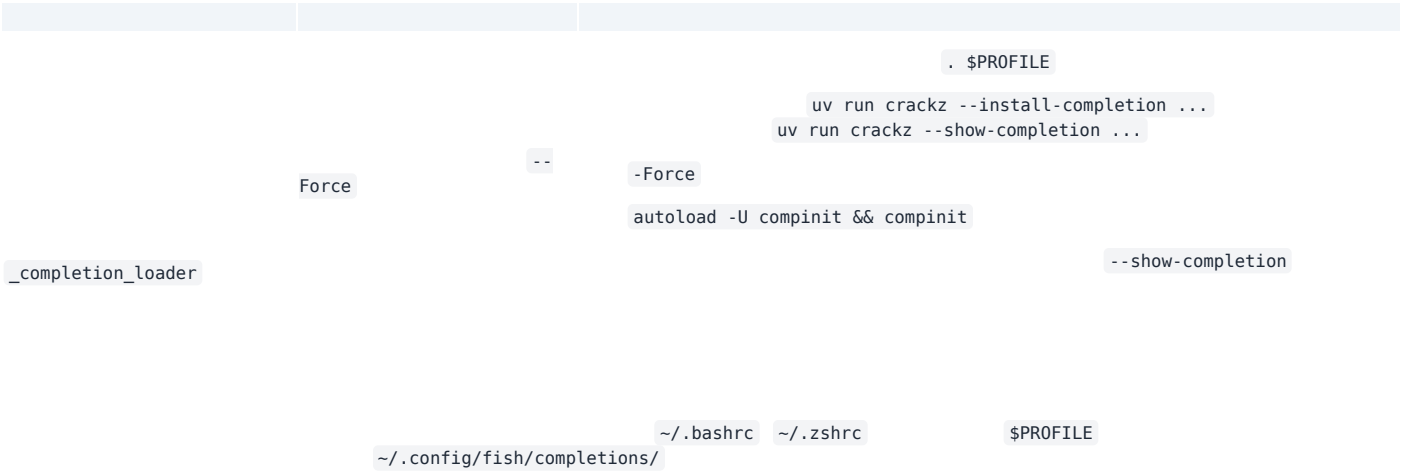
crackz --install-completion powershell

```
crackz --show-completion powershell | Out-String | Invoke-Expression
```

. \$PROFILE

```
crackz --install-completion powershell
# If the profile wasn't updated, add manually:
Add-Content $PROFILE "crackz --show-completion powershell | Out-String | Invoke-Expression"
. $PROFILE
```

```
crackz --show-completion bash > /tmp/crackz.sh && source /tmp/crackz.sh
source <(crackz --show-completion zsh)
crackz --show-completion fish | source
crackz --show-completion powershell | Out-String | Invoke-Expression
```



```
crackz --show-completion bash > .crackz_comp.sh
source .crackz_comp.sh
# run tests that exercise completion
```

: _____

Project Scope

Project Scope

- 1.
- 2.
- 3.
- 4.
- 5.

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- 1.
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Current Tasks

Current Tasks & Backlog

- - [x] Task description
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- [illegible]

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crackz-gui

crackz.app

crackz-gui

crackz.app.active_learning

crackz-core

crackz

crackz-gui

crackz-gui

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docs/superpowers/plans/2026-04-21-multiclass-defect-segmentation.md

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crackz report export --format docx

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update task list

docs:

Architecture

System Architecture

```
crackz/
├── pyproject.toml      # Workspace root
├── core/               # Workspace config, dev deps, tool settings (no [project])
│   ├── pyproject.toml # crackz-core package
│   ├── src/crackz/    # Package metadata and dependencies
│   ├── tests/         # Core library (AI, CLI, config, logging, project)
│   └── tests/         # Core unit + integration tests
├── gui/               # crackz-gui package (depends on crackz-core)
│   ├── pyproject.toml # Package metadata, dependency on crackz-core
│   ├── src/crackz gui/ # PySide6 Qt GUI application
│   └── tests/         # GUI unit + integration tests
└── tests/             # Root-level tests (cross-package, legacy)
```

```
core/
crackz_gui.qt import ...

uv sync --dev

gui/
from crackz.ai import ...

core/
from
```



```
graph TD
    subgraph L4["Layer 4 – Presentation"]
        CLI["crackz.cli"]
        GUI["crackz_gui"]
        DASH["crackz.dashboard"]
    end
    subgraph L3["Layer 3 – Application Façade"]
        APP["crackz.app"]
    end
    subgraph L2["Layer 2 – Domain"]
        AI["crackz.ai"]
        PROJECT["crackz.project"]
        REPORT["crackz.report"]
    end
    subgraph L1["Layer 1 – Infrastructure"]
        CONFIG["crackz.config"]
        LOG["crackz.log"]
        TRACING["crackz.tracing"]
        UTIL["crackz.util"]
    end

    CLI --> APP
    GUI --> APP
    DASH --> APP
    APP --> AI
    APP --> PROJECT
    APP --> REPORT
    AI --> CONFIG
    AI --> LOG
    AI --> UTIL
    PROJECT --> CONFIG
    PROJECT --> LOG
    PROJECT --> UTIL
    REPORT --> CONFIG
    REPORT --> LOG
    CONFIG --> LOG
    CONFIG --> UTIL
```

[illegible]

```
graph LR
    Dataset[Dataset] --> raw_train_val_test[raw/ train/ val/ test/]
    raw_train_val_test --> pydantic_settings[pydantic-settings]
    pydantic_settings --> CrackzDataset[CrackzDataset]
    loguru[loguru] --> artifacts_models[artifacts/models/]
    artifacts_models --> artifacts_runtime[artifacts/runtime/]
```

- 1.
- 2.
- 3.
4. `crackz detect train`
- 5.
- 6.
- 7.
8. `artifacts/models/`
9. `crackz detect run`
10. `segmentation-models-pytorch`

```
project.yaml
```

```
ai:
  model:
    encoder: resnet34          # or efficientnet-b0, mobilenet_v2, etc.
    encoder weights: imagenet # or None for random initialization
```

```

C4Container
title Crackz – Container View (Implemented)
Person(user, "User")
System Boundary(crackz, "Crackz Local System") {
    Container(cli, "CLI", "Typer", "Command parsing & dispatch")
    Container(gui, "GUI", "PySide6 (Qt 6)", "Desktop inspection & invocation")
    Container(core, "Core Library", "Python Packages", "project, ai, config, logging")
    Container(fs, "Local Filesystem", "images/ + artifacts/", "Images, masks, checkpoints, metrics")
}
Rel(user, cli, "Runs commands")
Rel(user, gui, "Interacts (point & click)")
Rel(cli, core, "Invokes APIs")
Rel(gui, core, "Invokes APIs")
Rel(core, fs, "read/write images, masks, checkpoints, metrics")

```

```

C4Context
title Crackz – System Context (Local)
Person(user, "User", "Engineer / Operator")
System(local, "Crackz Application", "CLI + GUI + Core Library")
System_Ext(osfs, "Host Filesystem", "Images + Artifacts")
Rel(user, local, "Invoke CLI / Launch GUI")
Rel(local, osfs, "read / write")

```

```

C4Component
title Core Library Components (Implemented)
Container(core, "core", "Python")
Component(project, "Project & Paths", "pydantic", "Project metadata & path helpers")
Component(config, "Configuration", "pydantic-settings", "Merge defaults / env / YAML")
Component(importer, "Image Importer", "Pillow", "Resize, quality, organize splits & masks")
Component(dataset, "Dataset Layer", "Torch Dataset", "CrackzDataset + DataLoader factory")
Component(train, "Training Orchestrator", "PyTorch Lightning", "Fit/validate/test + callbacks")
Component(detect, "Detection Pipeline", "PyTorch", "Batch inference over dataset splits (raw/test)")
Component(metrics, "Metrics & Checkpoints", "Lightning + JSON", "Save val/test metrics & best.ckpt/last.ckpt")
Component(perf, "Performance Utilities", "Seeding", "Seed + trainer overrides")
Component(logging, "Logging", "loguru", "Structured log sink & setup")
Component(evaluation, "Evaluation Framework", "Metrics", "IoU, precision, recall, F1, accuracy")
Component(reporter, "Report Generator", "python-docx", "Branded Word reports")
Component(viz, "Visualization", "Matplotlib", "Detection overlays (3-panel)")
Component(pipeline_orch, "Pipeline Orchestrator", "Workflow", "Train-detect + cancellation")
Component(device_mgmt, "Device Management", "Abstraction", "CPU/GPU/MPS selection")
Component(severity, "Severity Classifier", "Categorization", "3-tier crack severity")
Component(suggester, "Mask Suggester", "Semi-automation", "Model-assisted annotation")
Component(perf_est, "Performance Estimator", "RuntimeTracker", "Time estimates")
Component(async_cb, "Async Callbacks", "Janus", "Sync/async bridging")
Component(tracing_mod, "Tracing", "OpenTelemetry", "AI API observability")
Component(ai_chat_svc, "AI Chat", "Anthropic", "User assistance")
Rel(importer, project, "Uses path layout")
Rel(dataset, project, "Resolves split directories")
Rel(train, dataset, "Consumes (train/val/test) loaders")
Rel(detect, dataset, "Consumes detection loader (raw or test)")

```

```

Rel(train, metrics, "Saves checkpoints + metrics")
Rel(detect, metrics, "Writes detection results JSON/PNGs")
Rel(project, config, "Initializes from effective settings")
Rel(train, perf, "Reads overrides / seed")
Rel(detect, perf, "Sets seed (optional)")
Rel(dataset, logging, "Reports warnings (missing masks / size mismatch)")
Rel(detect, evaluation, "Evaluates performance")
Rel(evaluation, metrics, "Calculates metrics")
Rel(detect, viz, "Visualizes results")
Rel(detect, reporter, "Generates reports")
Rel(train, pipeline_orch, "Orchestrated by")
Rel(detect, pipeline_orch, "Orchestrated by")
Rel(train, device_mgmt, "Selects device")
Rel(detect, device_mgmt, "Selects device")
Rel(detect, severity, "Categorizes cracks")
Rel(suggester, detect, "Uses model")
Rel(train, perf_est, "Tracks runtime")
Rel(detect, perf_est, "Tracks runtime")
Rel(train, async_cb, "Reports progress")
Rel(detect, async_cb, "Reports progress")
Rel(ai_chat_svc, tracing_mod, "Optionally traced")

```

flowchart LR

```

A["Test Dataset"] --> B["Evaluator"]
B --> C["Metrics Calculator"]
C --> D["Evaluation Results"]
D --> E["JSON/CSV Reporter"]
E --> F["Report Generator"]
F --> G["Branded Word Document"]
I["Detection Results"] --> J["Visualization"]
J --> K["Detection Overlays"]
I --> F

```

flowchart TB

```

A["Training Thread (sync)"] --> B["CallbackDispatcher.sync_q"]
B --> C["Janus Queue"]
C --> D["CallbackDispatcher.async_q"]
D --> E["GUI Event Loop (async)"]
E --> F["UI Update"]

```

flowchart LR

```

A["Import Command / GUI Action"] --> B["Image Importer"]
B --> C["Split Directories (images/*, masks/*)"]
C --> D["Train Command"]
D --> E["Training Orchestrator<br/>(Trainer.fit)"]
E --> F["Checkpoints & Metrics<br/>(best.ckpt / last.ckpt / JSON)"]
F --> G["Detect Command"]
G --> H["Inference Loop<br/>(sigmoid + threshold)"]
H --> I["Annotated Output<br/>(PNGs + detection_results.json)"]

```

sequenceDiagram

```

participant User
participant CLI as CLI / GUI
participant Project
participant Import as ImageImporter
participant FS as Filesystem
User->>CLI: import-images --src ./samples --type train
CLI->>Project: load_project(path|yaml)
CLI->>Import: iterate source images
Import->>FS: resize + normalize + write split file
Import->>FS: write mask (if provided)
Import-->>CLI: progress callback
CLI-->>User: summary (count, skipped)

```

```
sequenceDiagram
    participant User
    participant CLI as CLI / GUI
    participant Project
    participant DS as CrackzDataset
    participant Trainer as Lightning Trainer
    participant Model
    participant FS as Filesystem
    User->>CLI: detect train --project P
    CLI->>Project: load_project()
    CLI->>DS: build train/val/test datasets
    DS->>CLI: DataLoaders
    CLI->>Trainer: fit(model, loaders)
    Trainer->>FS: save best.ckpt / last.ckpt
    Trainer->>CLI: metrics (val/test)
    CLI->>User: training summary
    User->>CLI: detect run --project P
    CLI->>Project: load_project()
    CLI->>Model: load checkpoint (best or last)
    CLI->>DS: build raw dataset (or test)
    DS->>CLI: DataLoader
    CLI->>Model: forward (batched)
    Model->>CLI: probabilities
    CLI->>CLI: threshold -> mask
    CLI->>FS: write annotated PNG + JSON metrics
    CLI->>User: detection summary
```

```
flowchart TD
    Defaults[Built-in Defaults] --> Merge
    YAML[Project YAML] --> Merge
    Env[Environment] --> Merge
    Merge --> Effective[Effective Settings]
```

```
flowchart LR
    User([User]) --> CLI["CLI (Typer)"]
    User --> GUI["GUI (PySide6)"]
    CLI --> CORE["Core Library"]
    GUI --> CORE
    CORE --> FS["Filesystem<br/>images / artifacts/models / artifacts/runtime"]
    CORE -.optional.-> GPU["GPU"]

    subgraph Optional_Docker [Optional Docker]
        IMG["Container Image<br/>(Python 3.13 + deps)"]
    end

    %% Show that container can host CLI/GUI
    IMG --> CLI
    IMG --> GUI
```

```
crackz.project.project
pydantic-settings
project.import_images
crackz.ai.data
crackz.ai.detection.train
crackz.ai.detection.detect
crackz.ai.model
crackz.ai.checkpoints
crackz.ai.device
crackz.util
crackz.ai.performance
crackz.log.crackz_logger
crackz_gui.*
crackz.ai.evaluation
crackz.report.generator
crackz.ai.visualization
```


crackz.ai.pipeline
crackz.ai.pipeline.cancellation
crackz.ai.detection.severity
crackz.ai.mask_suggester
crackz.ai.performance.runtime_estimation
crackz.ai.core.async_callbacks
crackz.tracing
crackz_gui.services.ai_chat

test/

raw/

train/

val/

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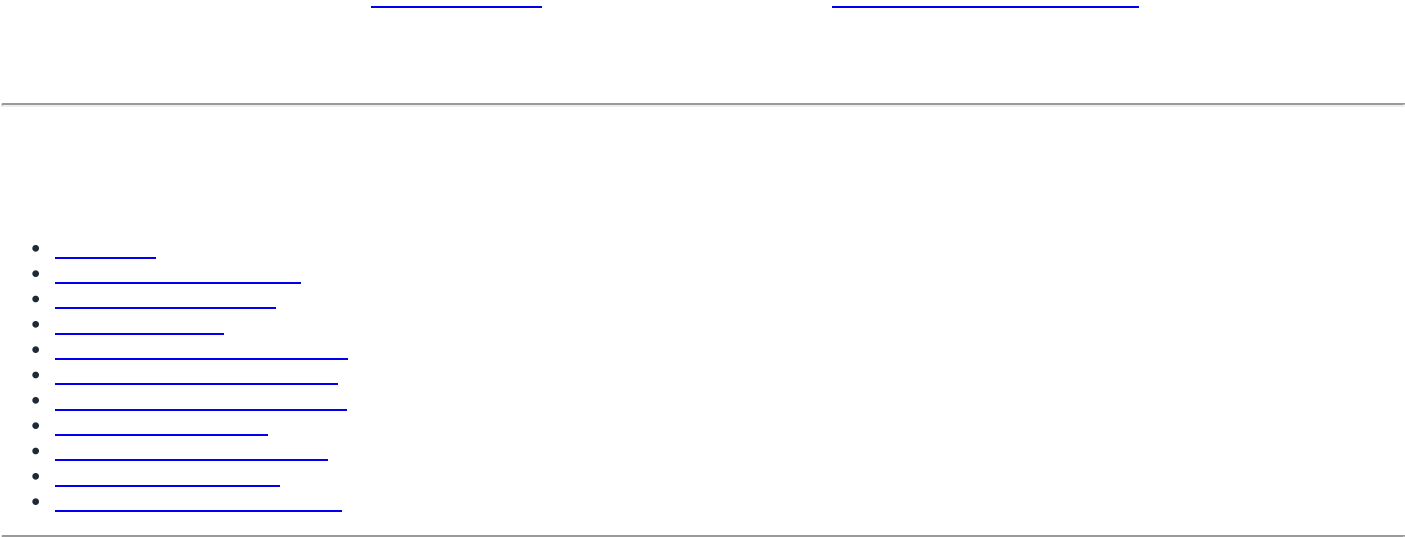
core/

gui/

Last updated: synchronized with repository implementation on commit timeline of current editing session.

Architecture Guidelines

Architecture Guidelines



User Interfaces (CLI, GUI)	↳ Entry points
Business Logic (Commands, Operations)	↳ Use cases
Core Domain (Project, AI, Config)	↳ Domain models
Infrastructure (Logging, Storage)	↳ Technical concerns

```
# [] GOOD: Clear separation
# core/src/crackz/cli/detection_cli.py - CLI commands only
# core/src/crackz/ai/detection.py - Detection logic only
# core/src/crackz/project/structure.py - Project layout only

# x BAD: Mixed concerns
# core/src/crackz/everything.py - CLI, detection, project management all in one file
```

```
# [] GOOD: Depend on protocol
from typing import Protocol

class ModelProtocol(Protocol):
    def predict(self, image: torch.Tensor) -> torch.Tensor: ...

def run_detection(model: ModelProtocol, image: torch.Tensor) -> dict:
    """Works with any model implementing the protocol."""
    return {"mask": model.predict(image)}

# x BAD: Depend on concrete class
def run_detection(model: SegmentationModel, image: torch.Tensor) -> dict:
    """Tightly coupled to SegmentationModel."""
    return {"mask": model.predict(image)}
```

```
# [] GOOD: Single responsibility
class ProjectLoader:
    """Loads project configuration from disk."""
    def load(self, path: Path) -> ProjectConfig: ...

class ProjectValidator:
    """Validates project structure and configuration."""
    def validate(self, project: ProjectConfig) -> list[str]: ...

# x BAD: Multiple responsibilities
class ProjectManager:
    """Loads, validates, trains models, exports results."""
    def load(self, path: Path) -> ProjectConfig: ...
    def validate(self, project: ProjectConfig) -> list[str]: ...
    def train_model(self, project: ProjectConfig) -> None: ...
    def export_results(self, project: ProjectConfig) -> None: ...
```

```
# [] GOOD: Extensible via configuration
def create_model(config: ModelConfig) -> nn.Module:
    """Factory method - add new encoders via config."""
    if config.encoder in SUPPORTED_ENCODERS:
        return create_segmentation_model(config)
    raise ValueError(f"Unsupported encoder: {config.encoder}")

# x BAD: Requires code changes for new models
def create_model(encoder_name: str) -> nn.Module:
    if encoder_name == "resnet34":
        return ResNet34Model()
    elif encoder_name == "efficientnet":
        return EfficientNetModel()
    # Must modify this function for each new encoder!
```

```
# [] GOOD: Explicit parameters
def train_model(
    project_path: Path,
    epochs: int,
    batch_size: int,
    learning_rate: float,
) -> dict[str, float]:
    """All parameters explicit."""
    pass
```

```
# x BAD: Implicit global state
GLOBAL_CONFIG = {} # Modified elsewhere

def train_model():
    """What parameters? Where do they come from?"""
    epochs = GLOBAL_CONFIG["epochs"] # Implicit dependency
```

```
core/src/crackz/      # Core library (crackz-core package)
├─ ai/                # AI/ML pipeline (domain logic)
│   ├── data.py       # Dataset, data loading
│   ├── models.py     # Model definitions
│   ├── training.py   # Training pipeline
│   └── detection.py  # Inference pipeline
├─ cli/              # CLI interface (entry points)
│   ├── options.py    # Shared CLI options
│   ├── exceptions.py # CLI error handling
│   ├── project_cli.py
│   └── detection_cli.py
├─ config/           # Configuration (domain models)
│   ├── project.py    # ProjectConfig
│   ├── ai.py         # AIConfig
│   └── logging.py    # LogConfig
├─ log/              # Logging (infrastructure)
│   ├── setup.py      # Loguru configuration
├─ project/          # Project management (domain logic)
│   └── structure.py  # Directory layout

gui/src/crackz_gui/   # GUI package (crackz-gui, depends on crackz-core)
├─ qt/               # PySide6 Qt application
│   ├── app_state.py  # QtAppState (reactive state)
│   └── async_bridge.py # TaskSignals, run_in_background
├─ services/         # External service integrations
├─ state/             # State management
└─ util/              # GUI utilities
```

```
CLI (core)  → Business Logic → Domain Models → Infrastructure
GUI (gui)   → (ai/, project/) (config/) (log/)
```

Examples:

```
- core: cli/detection_cli.py → ai/detection.py → config/ai.py → log/setup.py □
- gui: crackz_gui/qt/panels/train.py → crackz.ai.training → crackz.config □
- gui: crackz-gui depends on crackz-core (declared in gui/pyproject.toml) □
```

```
Domain →X- GUI/CLI      # Domain shouldn't know about UI
Infrastructure →X- Domain # Infrastructure shouldn't depend on business logic
GUI →X- CLI            # GUI shouldn't import CLI modules
```

Examples:

```
- ai/detection.py → cli/options.py x # Domain shouldn't import from CLI
- log/setup.py → config/project.py x # Infrastructure shouldn't import domain
- crackz_gui → crackz.cli x # GUI should use core APIs, not CLI
```

```
# x BAD: Domain imports from CLI
from crackz.cli.options import PROJECT_PATH_OPTION
def train_model(project_path: Path = PROJECT_PATH_OPTION): ...

# □ GOOD: CLI calls domain with explicit parameters
from crackz.cli.options import PROJECT_PATH_OPTION
from crackz.ai.training import train_model

def train_cli(project_path: Path = PROJECT_PATH_OPTION):
    train_model(project_path) # Domain function has no CLI dependency
```

core/src/crackz/ai/models.py

```
def create_segmentation_model(config: ModelConfig) -> nn.Module:
    """Factory for creating segmentation models."""
    import segmentation_models_pytorch as smp

    return smp.Unet(
        encoder_name=config.encoder,
        encoder_weights=config.encoder_weights,
        in_channels=config.in_channels,
        classes=config.num_classes,
    )
```

core/src/crackz/ai/detection.py

```
class DetectionStrategy(Protocol):
    def detect(self, images: list[Path]) -> dict[Path, np.ndarray]: ...

class SingleImageDetection:
    def detect(self, images: list[Path]) -> dict[Path, np.ndarray]:
        """Process one at a time."""
        return {img: self._detect_one(img) for img in images}

class BatchDetection:
    def detect(self, images: list[Path]) -> dict[Path, np.ndarray]:
        """Process in batches for efficiency."""
        results = {}
        for batch in chunk_list(images, self.batch_size):
            results.update(self._detect_batch(batch))
        return results
```

core/src/crackz/config/project.py

```
class ProjectConfig(BaseModel):
    """Pydantic acts as builder with validation."""
    name: str
    project_path: Path
    training: TrainingConfig = Field(default_factory=TrainingConfig)
    logging: LogConfig = Field(default_factory=LogConfig)

    @classmethod
    def load(cls, path: Path) -> ProjectConfig:
        """Build from YAML with defaults."""
        with open(path) as f:
            data = yaml.safe_load(f)
        return cls(**data) # Validation + defaults applied
```

core/src/crackz/cli/*.py

```
@app.command(name="train")
def train_command(
    project_path: Path = PROJECT_PATH_OPTION,
    epochs: int | None = EPOCHS_OPTION,
) -> None:
    """Each CLI command is a self-contained operation."""
    try:
        # Load configuration
        config = ProjectConfig.load(project_path / f"{project_path.name}_project.yaml")

        # Override with CLI args
        if epochs:
            config.training.epochs = epochs

        # Execute
        train_model(config)
```

```
typer.secho("[ ] Training completed", fg=typer.colors.GREEN)
except Exception as e:
    cli_failure(e, "Training")
```

core/src/crackz/project/structure.py

```
class ProjectRepository:
    """Abstracts project storage."""

    def load(self, path: Path) -> ProjectConfig:
        """Load project from filesystem."""
        return ProjectConfig.load(path / f"{path.name}_project.yaml")

    def save(self, project: ProjectConfig) -> None:
        """Save project to filesystem."""
        project.save(project.project_path / f"{project.name}_project.yaml")

    def exists(self, path: Path) -> bool:
        """Check if project exists."""
        return (path / f"{path.name}_project.yaml").exists()
```

core/src/crackz/ai/core/async_callbacks.py

RuntimeError: no running event loop

sync_q

async_q

```
from crackz.ai.core.async_callbacks import CallbackDispatcher

# In training setup (main thread)
dispatcher = CallbackDispatcher(maxsize=100)

# In PyTorch Lightning callback (training thread)
sync_reporter = dispatcher.create_sync_progress_reporter()
sync_reporter(0.5, "Training epoch 5/10") # Thread-safe put

# In GUI (async event loop)
async for progress, message in dispatcher.iter_progress():
    update_progress_bar(progress, message) # Async get
```

AsyncProgressReporter AsyncMetricsReporter

core/src/crackz/tracing/

--extra tracing

```
# In core/src/crackz/tracing/setup.py
from crackz.tracing import setup_tracing, is_tracing_enabled

# Automatic instrumentation on setup
if is_tracing_enabled():
    setup_tracing() # Instruments Anthropic client automatically

# The instrumentation happens automatically via AnthropicInstrumentor
# No manual client instrumentation needed
```

```
# Install tracing dependencies
uv sync --extra tracing

# Enable at runtime
export CRACKZ_TRACING_ENABLED=1
```

```
core/src/crackz/ai/evaluation/reporter.py core/src/crackz/report/generator.py
```

```
from crackz.ai.evaluation.reporter import export_json, export_csv
from crackz.report.generator import ReportGenerator

# Aggregation (format-agnostic)
results = {
    "accuracy": 0.95,
    "precision": 0.92,
    "recall": 0.89,
    "f1_score": 0.90,
    "iou": 0.85
}

# Export to JSON (lightweight)
export_json(
    results,
    output_path=project_path / "evaluation_results.json"
)

# Export to CSV (analysis-friendly)
export_csv(
    results,
    output_path=project_path / "evaluation_results.csv"
)

# Export to DOCX (rich formatting)
generator = ReportGenerator(
    project_path=project_path,
    template_path=template_path
)
generator.generate_report(
    detection_results=results,
    output_path=project_path / "report.docx"
)
```

```
core/src/crackz/ai/device/device.py
```

```
torch.device("cuda")
```

```
from crackz.ai.device.device import AIDevice

# Auto-select best available device
device = AIDevice.default() # Returns CUDA if available, else MPS, else CPU

# Check availability
available = AIDevice.available_devices() # [AIDevice.CPU, AIDevice.CUDA]

# Test device works
success, message, details = AIDevice.test_device("cuda")
if success:
    print(f"CUDA available: {details['device_name']}")

# Use in model
model = SegmentationModel()
model.to(device.value) # device.value is "cuda", "mps", "cpu", etc.
```

core/src/crackz/ai/detection/severity.py

```
from crackz.ai.detection.severity import categorize_crack_severity

# Default thresholds: 5% (high), 2% (medium)
severity = categorize_crack_severity(confidence=0.08) # "Crack Detected"
severity = categorize_crack_severity(confidence=0.03) # "Suspected Crack"
severity = categorize_crack_severity(confidence=0.01) # "No Crack"

# Custom thresholds via config
severity = categorize_crack_severity(
    confidence=0.04,
    high_threshold=0.06, # Raise "Crack Detected" threshold
    medium_threshold=0.03 # Raise "Suspected" threshold
) # "Suspected Crack" (was "Crack Detected" with defaults)

# Override with has_crack flag (confidence + area thresholds met)
severity = categorize_crack_severity(confidence=0.01, has_crack=True)
# "Crack Detected" (flag indicates combined criteria met)
```

```
# project.yaml
ai:
  detection:
    high_confidence_threshold: 0.05 # 5% crack area
    medium_confidence_threshold: 0.02 # 2% crack area
    min_confidence: 0.005 # 0.5% minimum to save
```

```
# core/src/crackz/cli/options.py
PROJECT_PATH_OPTION = typer.Option(
    None,
    "--project",
    help="Path to project directory",
)

FORCE_OPTION = typer.Option(
    False,
    "--force",
    help="Force operation without confirmation",
)

# core/src/crackz/cli/some_command.py
from crackz.cli.options import PROJECT_PATH_OPTION, FORCE_OPTION

def my_command(
    project_path: Path | None = PROJECT_PATH_OPTION,
    force: bool = FORCE_OPTION,
) -> None:
    pass
```



```
print() logger typer.echo()
```

```
from crackz.log import logger
import typer

# User-facing success/error messages
typer.secho("[] Detection completed", fg=typer.colors.GREEN)
typer.secho("x Training failed", fg=typer.colors.RED)

# Technical/operational logs
logger.info("Loading {} images from {}", len(images), path)
logger.debug("Model config: {}", config.model_dump())
logger.error("Failed to load checkpoint: {}", error, exc_info=True)
```

```
typer.secho()
```

```
logger.*
```

```
print() typer.echo()
```

```
from crackz.cli.exceptions import cli_failure

def train_command():
    try:
        # Business logic
        result = train_model(project_path)
        typer.secho("[] Training completed", fg=typer.colors.GREEN)
    except FileNotFoundError as e:
        cli_failure(e, "File access")
    except RuntimeError as e:
        cli_failure(e, "Training execution")
    except Exception as e:
        cli_failure(e, "Unexpected error")
```

```
cli_failure
```

```
logger.error()
```

```
typer.secho()
```

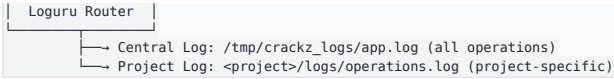
```
class AIConfig(BaseModel):
    model: ModelConfig = Field(default_factory=ModelConfig)
    training: TrainingConfig = Field(default_factory=TrainingConfig)

# Usage
config = AIConfig() # Loads from defaults only
config_from_yaml = AIConfig(**yaml_data) # Loads from YAML + defaults

# CLI override (doesn't persist to YAML)
if cli_epochs:
    config.training.epochs = cli_epochs # Runtime only
```

```
logger.info("message")
```

```
↓
```



```

from loguru import logger

# Configure at startup
logger.add(
    central_log_path,
    format="{time} {level} {message}",
    level="INFO",
)

# Add project-specific handler when project opens
logger.add(
    project_log_path,
    format="{time} {level} {message}",
    level="DEBUG",
    filter=lambda record: record["extra"].get("project") == project_name,
)

# Usage
logger.bind(project=project.name).info("Training started") # Goes to both logs

```

frozen=False

schema_version

Field(description=...)

```

class ProjectConfig(BaseModel):
    """Project configuration with validation and versioning."""

    schema_version: int = Field(
        default=2,
        ge=1,
        description="Config schema version for migration support",
    )

    name: str = Field(
        ..., # Required
        min_length=1,
        max_length=100,
        pattern=r"^[a-zA-Z0-9_-]+$",
        description="Project name (alphanumeric, underscore, hyphen)",
    )

    root: Path = Field(
        ..., # Required
        description="Project root directory",
    )

    ai: AIConfig = Field(
        default_factory=AIConfig,
        description="AI/ML configuration",
    )

    @field_validator("root")
    @classmethod
    def root_must_exist(cls, v: Path) -> Path:
        """Validate project root exists."""
        if not v.exists():
            raise ValueError(f"Project root does not exist: {v}")
        return v

    @model_validator(mode="after")
    def name_matches_root(self) -> Self:
        """Validate name matches root directory."""
        if self.root.name != self.name:
            raise ValueError(f"Name '{self.name}' doesn't match directory '{self.root.name}'")
        return self

```

schema_version: int

crackz migrate <project>

```
def migrate_v1_to_v2(config_data: dict) -> dict:
    """Migrate project config from v1 to v2."""
    if config_data.get("schema_version", 1) == 1:
        # v2 added 'ai.detection.batch_size'
        if "ai" not in config_data:
            config_data["ai"] = {}
        if "detection" not in config_data["ai"]:
            config_data["ai"]["detection"] = {}
        if "batch_size" not in config_data["ai"]["detection"]:
            config_data["ai"]["detection"]["batch_size"] = 8

        config_data["schema_version"] = 2

    return config_data
```

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.
- 11.
- 12.

```
class CrackzError(Exception):
    """Base exception for all Crackz errors."""
    pass

class ConfigurationError(CrackzError):
    """Invalid configuration."""
    pass

class ProjectError(CrackzError):
    """Project-related error."""
    pass

class ModelError(CrackzError):
    """Model loading/execution error."""
    pass
```

```
def command():
    try:
        # Business logic
        result = operation()
        typer.secho("✅ Success", fg=typer.colors.GREEN)

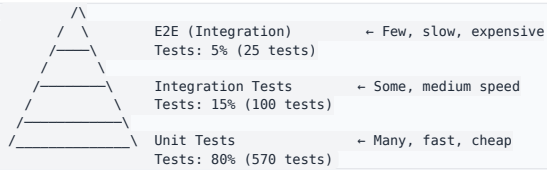
    except FileNotFoundError as e:
        # User error: file doesn't exist
        logger.error("File not found: {}", e)
        typer.secho(f"❌ File not found: {e}", fg=typer.colors.RED)
        raise typer.Exit(1)

    except ConfigurationError as e:
        # User error: invalid config
        logger.error("Configuration error: {}", e)
        typer.secho(f"❌ Configuration error: {e}", fg=typer.colors.RED)
        raise typer.Exit(1)

    except MemoryError:
        # System error: out of memory
        logger.error("Out of memory during operation")
        typer.secho("❌ Out of memory. Try reducing batch size.", fg=typer.colors.RED)
        raise typer.Exit(1)

    except Exception as e:
        # Programmer error: unexpected
        logger.exception("Unexpected error: {}", e)
```

```
typer.secho("x Unexpected error. Check logs for details.", fg=typer.colors.RED)
raise typer.Exit(1)
```



```
core/tests/
├── unit/                                # Fast, isolated, no I/O
│   ├── crackz/
│   │   ├── ai/
│   │   │   ├── test_models.py        # Model creation, logic
│   │   │   ├── test_training.py      # Training loop logic
│   │   │   └── test_detection.py     # Detection logic
│   │   ├── cli/
│   │   │   ├── test_options.py       # CLI option validation
│   │   │   └── test_commands.py      # Command parsing
│   │   └── config/
│   │       └── test_project.py       # Config validation
│   └── integration/                  # Slower, system-level, I/O allowed
│       ├── crackz/
│       │   ├── test_cli_workflows.py # End-to-end CLI
│       │   └── test_training_pipeline.py # Full training
│       └── docs/
│           └── test_docs_build.py    # Documentation build
└── gui/tests/
    ├── unit/                        # GUI unit tests
    └── integration/                 # GUI integration tests
```

```
def test_model_creation():
    """Test model factory with different configs."""
    # Arrange
    config = ModelConfig(encoder="resnet34", num_classes=1)

    # Act
    model = create_segmentation_model(config)

    # Assert
    assert isinstance(model, nn.Module)
    assert hasattr(model, "encoder")
    assert hasattr(model, "decoder")
```

```
@pytest.mark.integration
def test_full_training_workflow(tmp_path):
    """Test complete training workflow."""
    # Arrange
    project = create_test_project(tmp_path, num_images=10)

    # Act
    result = train_model(project, epochs=1)

    # Assert
    assert result["final_loss"] < result["initial_loss"]
    assert (project.project_path / "artifacts" / "models" / "checkpoint.ckpt").exists()
```

```
@pytest.mark.benchmark
def test_detection_performance(benchmark, sample_images):
    """Ensure detection meets performance targets."""
    model = load_model(TEST_CHECKPOINT)

    result = benchmark(detect_cracks, model, sample_images)

    # Verify performance target: <5s per image on CPU
    assert result.mean < 5.0
```

core/tests/conftest.py

```
@pytest.fixture
def tiny_model():
    """Fast model for unit tests."""
    return create_segmentation_model(ModelConfig(
        encoder="resnet18",
        encoder_weights=None, # No pre-trained weights
        num_classes=1,
    ))

@pytest.fixture
def sample_project(tmp_path):
    """Sample project with test data."""
    project_dir = tmp_path / "test_project"
    initialize_project(project_dir, name="test_project")

    # Add minimal test data
    (project_dir / "images" / "raw").mkdir(parents=True)
    create_test_image(project_dir / "images" / "raw" / "test.png")

    return project_dir

@pytest.fixture
def mock_model(mocker):
    """Mock model for testing without actual inference."""
    mock = mocker.Mock(spec=nn.Module)
    mock.predict.return_value = torch.zeros(1, 1, 512, 512)
    return mock
```

```
# x BAD: Load everything into memory
all_images = [Image.open(p) for p in image_paths] # OOM for large datasets
for img in all_images:
    process(img)

# [] GOOD: Process in batches
for batch_paths in chunk_list(image_paths, batch_size=8):
    batch = [Image.open(p) for p in batch_paths]
    process_batch(batch)
    # batch garbage collected after each iteration
```

```
def train_with_cleanup():
    try:
        # Training code
        model.train()
        for batch in dataloader:
            loss = model.training_step(batch)
            loss.backward()
            optimizer.step()
    finally:
        # Always clean up GPU memory
        if torch.cuda.is_available():
            torch.cuda.empty_cache()
```

```
class Project:
    def __init__(self, root: Path):
        self.root = root
        self.config: ProjectConfig | None = None # Not loaded yet
        self.images: list[Path] | None = None # Not loaded yet

    @property
    def config(self) -> ProjectConfig:
        """Lazy load configuration."""
        if self._config is None:
            self._config = ProjectConfig.load()
```

```

        self.root / f"{self.root.name}_project.yaml"
    )
    return self._config

@property
def images(self) -> list[Path]:
    """Lazy load image list."""
    if self._images is None:
        self._images = list((self.root / "images").rglob("*.png"))
    return self._images

```

- 1.
- 2.
- 3.
- 4.
- 5.

```

from pydantic import BaseModel, Field, validator

class ProjectConfig(BaseModel):
    name: str = Field(
        ...,
        min_length=1,
        max_length=100,
        pattern=r"^[a-zA-Z0-9_-]+$", # Prevent path traversal
    )

    @validator("name")
    def no_path_traversal(cls, v: str) -> str:
        """Prevent directory traversal attacks."""
        if "." in v or "/" in v or "\\" in v:
            raise ValueError("Invalid project name: contains path separators")
        return v

```

```

def safe_path_join(base: Path, *parts: str) -> Path:
    """Join paths safely, preventing traversal attacks."""
    result = base
    for part in parts:
        if "." in part or part.startswith("/") or part.startswith("\\"):
            raise ValueError(f"Unsafe path component: {part}")
        result = result / part

    # Verify result is still under base
    if not result.resolve().is_relative_to(base.resolve()):
        raise ValueError("Path traversal detected")

    return result

```

1. `core/src/crackz/cli/my_command.py`
- 2.

```

from __future__ import annotations

import typer
from crackz.cli.options import PROJECT_PATH_OPTION
from crackz.cli.exceptions import cli_failure
from crackz.log import logger

app = typer.Typer(no_args_is_help=True)

@app.command(name="my-command")
def my_command(
    project_path: Path = PROJECT_PATH_OPTION,
) -> None:
    """Description shown in --help."""

```

```

try:
    # Business logic
    logger.info("Starting operation")
    result = do_operation(project_path)
    typer.secho("[ ] Operation completed", fg=typer.colors.GREEN)
except Exception as e:
    cli_failure(e, "Operation name")

```

3. `core/src/crackz/cli/main.py`
4. `core/tests/unit/crackz/cli/test_my_command.py`
- 5.

1. `core/src/crackz/config/my_feature.py`

```

from pydantic import BaseModel, Field

class MyFeatureConfig(BaseModel):
    """Configuration for my feature."""

    enabled: bool = Field(
        default=False,
        description="Enable my feature",
    )

    option: str = Field(
        default="default_value",
        description="Feature option",
    )

```

2. `core/src/crackz/config/project.py`

```

class ProjectConfig(BaseModel):
    # ... existing fields ...
    my_feature: MyFeatureConfig = Field(default_factory=MyFeatureConfig)

```

3. `core/src/crackz/project/migrations.py`
- 4.
- 5.

1. `core/src/crackz/ai/models/my_model.py`
- 2.

```

class MyCustomModel(nn.Module):
    def __init__(self, config: ModelConfig):
        super().__init__()
        # Model setup

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        # Forward pass
        return x

    def predict(self, x: torch.Tensor) -> torch.Tensor:
        """Inference interface."""
        self.eval()
        with torch.no_grad():
            return self(x)

```

3. `core/src/crackz/ai/models/__init__.py`

```

def create_model(config: ModelConfig) -> nn.Module:
    if config.type == "segmentation":
        return create_segmentation_model(config)
    elif config.type == "my_custom":
        return MyCustomModel(config)
    else:
        raise ValueError(f"Unknown model type: {config.type}")

```

4. `core/src/crackz/config/ai.py`
- 5.

1. `core/src/crackz/ai/detection/my_strategy.py`
- 2.

```

class MyDetectionStrategy:
    def __init__(self, model: nn.Module, config: DetectionConfig):
        self.model = model
        self.config = config

    def detect(self, images: list[Path]) -> dict[Path, np.ndarray]:
        """Run detection with my strategy."""
        results = {}
        for image_path in images:
            mask = self._detect_single(image_path)
            results[image_path] = mask
        return results

```

3. `core/src/crackz/ai/detection/__init__.py`

4. `core/src/crackz/config/ai.py`
5.

- [_____](#)
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API Reference

API Reference

```
from pathlib import Path
from crackz.project.project import load_project_from_path
from crackz.ai.detection import detect

project = load_project_from_path(Path("demo-project"))
# Optional: custom progress reporter
progress = lambda p, msg=None: print(f"{p:.0%}", msg or "")

detect(project, progress_reporter=progress)
```

```
from crackz.project.project import load_project_from_path
from crackz.ai.training import train

project = load_project_from_path("demo-project")
train(project) # writes checkpoints & metrics under project.artifacts/checkpoints
```

```
from pathlib import Path
from crackz.project.project import (
    ProjectMetadata, ProjectPaths, create_project, save_project
)

project = create_project(
    metadata=ProjectMetadata(name="demo-project", description="Example"),
    paths=ProjectPaths(
        project_path=Path("demo-project"),
        image_path=Path("demo-project/images"),
        image_mask_path=Path("demo-project/masks"),
        artifacts_path=Path("demo-project/artifacts"),
    ),
    force=False,
)
save_project(project)
```

```
from crackz.project.project import load_project_from_path
project = load_project_from_path("demo-project")
# Uses same logic as CLI import-images (see project.import_images signature)
project.import_images(
    source_path="./samples", # directory with raw images
    masks_source_path="./samples/masks", # optional
    image_type=project.ImagesType.RAW if hasattr(project, 'ImagesType') else None, # or enum from crackz.ImagesType
    size=(512, 512),
    quality=90,
    overwrite=False,
)
```

CrackzSettings

CRACKZ_

CRACKZ_LOG_LEVEL=DEBUG

BaseSettings

► core/src/crackz/config/crackz_settings.py

classmethod

from_conf(conf_file)

► core/src/crackz/config/crackz_settings.py

CrackzError

► core/src/crackz/config/exceptions.py

[ConfigError](#)

► core/src/crackz/config/exceptions.py

[ConfigError](#)

► core/src/crackz/config/exceptions.py

project.project

Project
Project

BaseModel

► `core/src/crackz/config/project_config.py`

`classmethod`

`load(path)`

► `core/src/crackz/config/project_config.py`

`save(path)`

► `core/src/crackz/config/project_config.py`

`to_project()`

`Project`

`ProjectConfig`

► `core/src/crackz/config/project_config.py`

`classmethod`

`from_project(project)`

`ProjectConfig`

`Project`

► `core/src/crackz/config/project_config.py`

`Project`

`BaseModel`

► `core/src/crackz/project/project.py`

`property`

`name`

`property`

root

property

config_file

property

detections_path

Path Path

property

models_path

Path Path

property

checkpoints_path

Path Path

property

runtime_path

Path Path

property

training_path

property

detections_visualizations_path

Path

Path

property

detection_results_path

property

training_metrics_path

property

tensorboard_logs_path

property

registry_db_path

property

lightning_logs_path

property

val_path

Path

Path

property

test_path

Path

Path

property

train_path

Path Path

property

train_mask_path

Path Path

property

val_mask_path

Path Path

property

test_mask_path

Path Path

property

raw_path

Path Path

property

raw_mask_path

Path Path

property

project_file_path

map_type_to_dir(image_type)

test val raw train

Path

None

► core/src/crackz/project/project.py

map_type_to_masks_dir(image_type)

test val raw train

Path

None

► core/src/crackz/project/project.py

save_project()

model_dump

model_dump

► core/src/crackz/project/project.py

```
import_image_files(  
    files,  
    dest_dir,  
    size=(512, 512),  
    quality=90,  
    *,  
    overwrite=False,  
    progress_cb=None,  
)
```

•
•
•

► core/src/crackz/project/project.py

```
import_images(  
    source_path,  
    masks_source_path=None,  
    image_type=ImagesType.RAW,
```

```

size=(512, 512),
quality=90,
*,
overwrite=False,
progress_cb=None,
split=False,
train_ratio=0.7,
val_ratio=0.15,
test_ratio=0.15,
random_seed=None,
)

```

-
-
-
-
-

► `core/src/crackz/project/project.py`

```
create_project_dir(*, force)
```

`force`

`project_path`

`artifacts_path`

► `core/src/crackz/project/project.py`

```
clean(*, logs_only=False, keep_best_model=True)
```

► `core/src/crackz/project/project.py`

```

create_project_and_save(
    metadata,
    paths,
    ai=default_ai_config,
    logging=default_logging_config,
    *,
    force=False,
)

```

► `core/src/crackz/project/project.py`

```

create_project(
    metadata, paths, ai=None, logging=None, *, force=False
)

```


► core/src/crackz/project/project.py

save_project(project)

Project.save_project

► core/src/crackz/project/project.py

load_project(project_file_path)

Project

project_file_path	Path		required

Project	Project
-------------------------	---------

Project	None
-------------------------	------

► core/src/crackz/project/project.py

load_project_from_path(project_path)

► core/src/crackz/project/project.py

load_from_project_path_or_cfg(cfg, project_path)

► core/src/crackz/project/project.py

StrEnum

RAW

TRAIN

TEST

VAL

► `core/src/crackz/project/types.py`

CrackzError

► `core/src/crackz/project/exceptions.py`

[ProjectError](#)

► `core/src/crackz/project/exceptions.py`

[ProjectError](#)

► `core/src/crackz/project/exceptions.py`

[ProjectError](#)

► `core/src/crackz/project/exceptions.py`

[ProjectError](#)

► core/src/crackz/project/exceptions.py

[ProjectError](#)

► core/src/crackz/project/exceptions.py

BaseModel

► core/src/crackz/ai/detection/results.py

```
def categorize_crack_severity(
    confidence,
    *,
    high_threshold=DEFAULT_HIGH_CONFIDENCE,
    medium_threshold=DEFAULT_MEDIUM_CONFIDENCE,
    has_crack=False,
)
```

confidence	float		required
high_threshold	float		DEFAULT_HIGH_CONFIDENCE
medium_threshold	float		DEFAULT_MEDIUM_CONFIDENCE
has_crack	bool		False

str

```
>>> categorize_crack_severity(0.08) # 8% crack area
'Crack Detected'
>>> categorize_crack_severity(0.03) # 3% crack area
'Suspected Crack'
>>> categorize_crack_severity(0.01) # 1% crack area
'No Crack'
>>> categorize_crack_severity(0.01, has_crack=True)
'Crack Detected' # has_crack flag indicates both confidence and area thresholds met
```

► core/src/crackz/ai/detection/severity.py

```
create_tensorboard_logger(project)
```

project [Project](#) required

TensorBoardLogger | None

► core/src/crackz/ai/training/tensorboard.py

```
train(
    project,
    progress_reporter=None,
    cancellation_token=None,
    metrics_callback=None,
    ckpt_path=None,
)
```

project	Project	required
progress_reporter	ProgressReporter None	None
cancellation_token	CancellationToken None	None
metrics_callback	MetricsReporter None	None
ckpt_path	Path None	None

CancellationError

core/src/crackz/ai/training/training_runner.py

```
detect(
    project,
    progress_reporter=None,
    cancellation_token=None,
    checkpoint_name=None,
    result_callback=None,
    *,
    on_incompatible="original",
)
```

project	Project	required
progress_reporter	ProgressReporter None	None
cancellation_token	CancellationToken None	None
checkpoint_name	str None	None
result_callback	Callable[[dict[str, Any]], None] None	None
on_incompatible	IncompatibleMode	"untrained" "original" "raise" 'original'

None

None detection_results.json

CancellationError

CheckpointIncompatibleError on_incompatible "raise"

core/src/crackz/ai/detection/detection_runner.py

__getattr__(name)

► `core/src/crackz/ai/detection/__init__.py`

► `core/src/crackz/ai/data/dataset.py`

`apply(im)`

► `core/src/crackz/ai/data/dataset.py`

`Dataset[Sample]`

► `core/src/crackz/ai/data/dataset.py`

`load_image_with_crackz_image(image_path)`

`image_path` `Path` *required*

`Image`

[ImageLoadingError](#)

► `core/src/crackz/ai/data/dataset.py`

```
CrackzDetectionDataset(
    image_dir, transform=None, normalization_config=None
)
```

image_dir	Path str		required
transform	ImageTransform None		None
normalization_config	NormalizationConfig None		None

--	--

[CrackzDataset](#)

► core/src/crackz/ai/data/dataset.py

```
create_training_datasets(  
    train_path,  
    train_mask_path,  
    val_path,  
    val_mask_path,  
    test_path,  
    test_mask_path,  
)
```

train_path	Path		required
train_mask_path	Path		required
val_path	Path		required
val_mask_path	Path		required
test_path	Path		required
test_mask_path	Path		required

--	--

tuple[[CrackzDataset](#), [CrackzDataset](#), [CrackzDataset](#)]



► core/src/crackz/ai/data/dataset.py

```
create_dataset(  
    project,  
    transforms,  
    mask_transforms,  
    image_dir=None,  
    image_masks_dir=None,  
)
```

project	Project		required
transforms	ImageTransform None		required
mask_transforms	MaskTransform None		required
image_dir	Path None		None
image_masks_dir	Path None		None

[CrackzDataset](#)

► core/src/crackz/ai/data/dataset.py

```
collate(batch)
```

► core/src/crackz/ai/data/dataloader.py

```
make_crackz_loader(  
    dataset, batch_size=32, num_workers=8, *, shuffle=True  
)
```

► core/src/crackz/ai/data/dataloader.py

```
create_training_data loaders(  
    train_ds,  
    val_ds,  
    test_ds,  
    batch_size,  
    num_workers,  
    *,  
    shuffle_train=True,  
)
```


train_ds	CrackzDataset		required
val_ds	CrackzDataset		required
test_ds	CrackzDataset		required
batch_size	int		required
num_workers	int		required
shuffle_train	bool	True	

tuple[DataLoader, DataLoader, DataLoader]		
►	core/src/crackz/ai/data/dataloader.py	
	CrackzModel	
	ComboLoss	
	dataclass	
	CrackzModel	

lr	float	
backbone	str	
in_channels	int	
num_classes	int	
threshold	float	
►	core/src/crackz/ai/model/segmentation/model.py	
	staticmethod	
from_project(project, lr)		
	CrackzConfig	Project
►	core/src/crackz/ai/model/segmentation/model.py	

Module

► `core/src/crackz/ai/model/segmentation/model.py`

`forward(y_pred, y_true)`

► `core/src/crackz/ai/model/segmentation/model.py`

LightningModule

► `core/src/crackz/ai/model/segmentation/model.py`

`training_step(batch, _batch_idx)`

► `core/src/crackz/ai/model/segmentation/model.py`

`validation_step(batch, _batch_idx)`

► `core/src/crackz/ai/model/segmentation/model.py`

`test_step(batch, _batch_idx)`

► `core/src/crackz/ai/model/segmentation/model.py`

`on_train_epoch_end()`

► `core/src/crackz/ai/model/segmentation/model.py`

`on_validation_epoch_end()`

► `core/src/crackz/ai/model/segmentation/model.py`

`on_test_epoch_end()`

► `core/src/crackz/ai/model/segmentation/model.py`

```
configure_optimizers()
```

► `core/src/crackz/ai/model/segmentation/model.py`

```
get_loss_function(config)
```

► `core/src/crackz/ai/model/segmentation/model.py`

Callback

► `core/src/crackz/ai/training/callbacks.py`

```
on_train_start(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_train_batch_end(  
    trainer, pl_module, outputs, batch, batch_idx  
)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_train_epoch_end(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_validation_epoch_end(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_test_epoch_end(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_fit_end(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

Callback

► `core/src/crackz/ai/training/callbacks.py`

```
__call__(progress, message=None)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_train_start(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_train_epoch_start(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_train_batch_end(  
    trainer, pl_module, outputs, batch, batch_idx  
)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_train_epoch_end(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_fit_end(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_test_end(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

Callback

► `core/src/crackz/ai/training/callbacks.py`

```
on_validation_epoch_end(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

```
make_reporter(ui_callback)
```

► `core/src/crackz/ai/training/callbacks.py`

```
create_training_callbacks(  
    project,  
    val_ds,  
    tb_logger,  
    progress_reporter,  
    runtime_tracker,  
    cancellation_token,  
)
```

project	Project		required
val_ds	CrackzDataset		required

tb_logger	TensorBoardLogger None		required
progress_reporter	ProgressReporter None		required
runtime_tracker	RuntimeTracker		required
cancellation_token	CancellationTokens None		required

list[Any]	
▶	core/src/crackz/ai/training/callbacks.py

get_available_checkpoints(checkpoints_dir)

checkpoints_dir	Path		required

list[Path]	
▶	core/src/crackz/ai/model/checkpoints.py

has_trained_model(project)

project	Project		required

bool	
▶	core/src/crackz/ai/model/checkpoints.py

select_checkpoint(checkpoints_dir)

checkpoints_dir

Path

required

Path | None

▶ core/src/crackz/ai/model/checkpoints.py

peek_checkpoint_backbone(checkpoint_path)

▶ core/src/crackz/ai/model/checkpoints.py

get_checkpoint_compatibility(project)

▶ core/src/crackz/ai/model/checkpoints.py

load_model_from_checkpoint(
 project,
 lr,
 checkpoint_name=None,
 *,
 on_incompatible="original",
)

project

[Project](#)

required

lr

float

required

checkpoint_name

str | None

None

on_incompatible

IncompatibleMode

"original"

"raise"

"untrained"

CheckpointIncompatibleError

'original'

[CrackzModel](#)

[CheckpointIncompatibleError](#) *on_incompatible* "raise"

► `core/src/crackz/ai/model/checkpoints.py`

```
create_checkpoint_callback(project)
```

project [Project](#) *required*

ModelCheckpoint

► `core/src/crackz/ai/model/checkpoints.py`

```
prune_checkpoints(project, keep)
```

project [Project](#) *required*

keep int *required*

list[Path]

► `core/src/crackz/ai/model/checkpoints.py`


```
plot_image(img_path, out_file, pred_mask)
```

► `core/src/crackz/ai/visualization/visualization.py`

```
save_detection_figure(fig, out_file)
```

► `core/src/crackz/ai/visualization/visualization.py`

```
save_metrics_json(results, output_dir)
```

► `core/src/crackz/ai/visualization/visualization.py`

```
save_training_metrics_json(results, output_dir)
```

► `core/src/crackz/ai/visualization/visualization.py`

```
show_image(ax, index, title, img, alpha=None, cmap='gray')
```

► `core/src/crackz/ai/visualization/visualization.py`

```
to_numpy_img(tensor)
```

► `core/src/crackz/ai/visualization/visualization.py`

Enum

► `core/src/crackz/ai/device/device.py`

`staticmethod`

available_devices()

▶ core/src/crackz/ai/device/device.py

staticmethod

default()

▶ core/src/crackz/ai/device/device.py

staticmethod

test_device(device_name)

device_name

str

required

bool

str

•

dict[str, str | None]

•

tuple[bool, str, dict[str, str | None]]

•

▶ core/src/crackz/ai/device/device.py

get_accelerator()

▶ core/src/crackz/ai/device/utils.py

resolve_device_and_accelerator(project)

project

[Project](#)

required

str

► `core/src/crackz/ai/device/utls.py`

[AIPipelineError](#)

[ModelNotFoundError](#)

[DataLoadingError](#)

CrackzError

► `core/src/crackz/ai/exceptions.py`

CrackzError

► `core/src/crackz/ai/exceptions.py`

[AIPipelineError](#)

► `core/src/crackz/ai/exceptions.py`

[AIPipelineError](#)

► `core/src/crackz/ai/exceptions.py`

[AIPipelineError](#)

► `core/src/crackz/ai/exceptions.py`

[AIPipelineError](#)

► `core/src/crackz/ai/exceptions.py`

`detection_results.json`



`save_metrics.json`

`crackz.ai.eval`

[BaseModel](#)

► `core/src/crackz/ai/detection/results.py`

```
load_detection_results(project_or_path, *, strict=False)
```

► `core/src/crackz/ai/detection/results.py`

`iter_iou(results)`

► `core/src/crackz/ai/detection/results.py`

`DetectionResult`

`aggregate_iou(results)`

► `core/src/crackz/ai/detection/eval.py`

`basic_summary(results)`

► `core/src/crackz/ai/detection/eval.py`

`get_available_checkpoints(checkpoints_dir)`

<code>checkpoints_dir</code>	<code>Path</code>		<i>required</i>

`list[Path]`

► `core/src/crackz/ai/model/checkpoints.py`

`has_trained_model(project)`

```
project Project required
```

```
bool  
▶ core/src/crackz/ai/model/checkpoints.py
```

```
select_checkpoint(checkpoints_dir)
```

```
checkpoints_dir Path required
```

```
Path | None  
▶ core/src/crackz/ai/model/checkpoints.py
```

```
peek_checkpoint_backbone(checkpoint_path)
```

```
▶ core/src/crackz/ai/model/checkpoints.py
```

```
get_checkpoint_compatibility(project)
```

```
▶ core/src/crackz/ai/model/checkpoints.py
```

```
load_model_from_checkpoint(  
    project,  
    lr,  
    checkpoint_name=None,  
    *,  
    on_incompatible="original",  
)
```

project	Project		<i>required</i>
lr	float		<i>required</i>
checkpoint_name	str None		None
on_incompatible	IncompatibleMode	"original" "untrained" "raise"	CheckpointIncompatibleError 'original'

[CrackzModel](#)

[CheckpointIncompatibleError](#)

on_incompatible "raise"

► core/src/crackz/ai/model/checkpoints.py

create_checkpoint_callback(project)

project [Project](#) *required*

ModelCheckpoint

► core/src/crackz/ai/model/checkpoints.py

prune_checkpoints(project, keep)

project	Project		<i>required</i>
keep	int		<i>required</i>

--	--

list[Path]

► core/src/crackz/ai/model/checkpoints.py

► core/src/crackz/ai/pipeline/cancellation.py

`__init__()`

► core/src/crackz/ai/pipeline/cancellation.py

`cancel()`

► core/src/crackz/ai/pipeline/cancellation.py

`is_cancelled()`

--	--

bool

► core/src/crackz/ai/pipeline/cancellation.py

`check_cancelled()`

CancellationError

► core/src/crackz/ai/pipeline/cancellation.py

Handler

► core/src/crackz/log/crackz_logger.py

emit(record)

► core/src/crackz/log/crackz_logger.py

use_rich_console()

bool

► core/src/crackz/log/crackz_logger.py

setup_logging(level=None, log_dir=None, *, use_rich=None)

level str | None None

log_dir	Path None		None
use_rich	bool None		None

--	--

Logger

► core/src/crackz/log/crackz_logger.py

get_logger(name, filename=None, log_dir='logs')

	logging	InterceptHandler	filename
►	core/src/crackz/log/crackz_logger.py		

TyperGroup

► core/src/crackz/cli/crackz_cli.py

```
main(
    args=None,
    prog_name=None,
    complete_var=None,
    standalone_mode=True,
    windows_expand_args=True,
    **extra,
)
```

► core/src/crackz/cli/crackz_cli.py

list_commands(ctx)

► core/src/crackz/cli/crackz_cli.py

get_command(ctx, cmd_name)

► `core/src/crackz/cli/crackz_cli.py`

```
format_commands(ctx, formatter)
```

► `core/src/crackz/cli/crackz_cli.py`

```
root_callback(  
    ctx,  
    version=False,  
    debug=False,  
    quiet=False,  
    log_dir=None,  
)
```

► `core/src/crackz/cli/crackz_cli.py`

```
main()
```

► `core/src/crackz/cli/crackz_cli.py`

```
init_project(  
    name=NAME_OPTION,  
    image_path=IMAGE_PATH_OPTION,  
    image_mask_path=IMAGE_MASK_PATH_OPTION,  
    artifacts_path=ARTIFACTS_PATH_OPTION,  
    description=DESCRIPTION_OPTION,  
    log_dir=LOG_DIR_OPTION,  
    *,  
    force=FORCE_OPTION,  
)
```

► `core/src/crackz/cli/project_cli.py`

```
export_project_config(  
    cfg=CFG_OPTION,  
    project_path=PROJECT_PATH_OPTION,  
    out=EXPORT_OUTPUT_OPTION,  
)
```

► `core/src/crackz/cli/project_cli.py`

```
migrate_project_config(  
    cfg=CFG_OPTION,
```

```
project_path=PROJECT_PATH_OPTION,
*,
dry_run=DRY_RUN_OPTION,
no_backup=NO_BACKUP_OPTION,
migrate_files=typer.Option(
    default=False,
    help="Also migrate artifacts directory structure (checkpoints to models, etc.)",
),
)
```

► core/src/crackz/cli/project_cli.py

```
clean_project(
    project_path=PROJECT_PATH_OPTION,
    *,
    dry_run=DRY_RUN_OPTION,
    logs_only=LOGS_ONLY_OPTION,
    keep_best_model=KEEP_BEST_MODEL_OPTION,
)
```

► core/src/crackz/cli/project_cli.py

```
import_images(
    cfg=CFG_OPTION,
    project_path=PROJECT_PATH_OPTION,
    source_path=SOURCE_PATH_OPTION,
    masks_source_path=MASKS_SOURCE_PATH_OPTION,
    image_type=TYPE_OPTION,
    image_size=SIZE_OPTION,
    quality=QUALITY_OPTION,
    target_size=TARGET_SIZE_OPTION,
    *,
    overwrite=OVERWRITE_OPTION,
    split=SPLIT_OPTION,
    train_ratio=TRAIN_RATIO_OPTION,
    val_ratio=VAL_RATIO_OPTION,
    test_ratio=TEST_RATIO_OPTION,
    random_seed=RANDOM_SEED_OPTION,
)
```

cfg	Path None	CFG_OPTION
project_path	Path None	PROJECT_PATH_OPTION
source_path	Path	SOURCE_PATH_OPTION

masks_source_path	Path None	source_path	MASKS_SOURCE_PATH_OPTION
image_type	ImagesType		TYPE_OPTION
image_size	SizeType		SIZE_OPTION
quality	int		QUALITY_OPTION
target_size	int None		TARGET_SIZE_OPTION
overwrite	bool		OVERWRITE_OPTION
split	bool		SPLIT_OPTION
train_ratio	float		TRAIN_RATIO_OPTION
val_ratio	float		VAL_RATIO_OPTION
test_ratio	float		TEST_RATIO_OPTION
random_seed	int None		RANDOM_SEED_OPTION

BadParameter

Exit

None

► core/src/crackz/cli/image_import_cli.py

```
run_detection(  
    cfg=CFG_OPTION,  
    project_path=PROJECT_PATH_OPTION,  
    model=CHECKPOINT_NAME_OPTION,  
)
```

► core/src/crackz/cli/detection_cli.py

```
detection_train(  
    cfg=CFG_OPTION,  
    project_path=PROJECT_PATH_OPTION,  
    epochs=EPOCHS_OPTION,  
    batch_size=BATCH_SIZE_OPTION,  
    resume=RESUME_FROM_OPTION,  
)
```

cfg	Path None	project_path	CFG_OPTION
project_path	Path None	cfg	PROJECT_PATH_OPTION
epochs	int None		EPOCHS_OPTION
batch_size	int None		BATCH_SIZE_OPTION
resume	str None		RESUME_FROM_OPTION

```
BadParameter      cfg      project_path  
  
Exit  
  
► core/src/crackz/cli/detection_cli.py
```

```
prune_checkpoints_cmd(  
    keep=KEEP_CHECKPOINTS_OPTION,  
    cfg=CFG_OPTION,  
    project_path=PROJECT_PATH_OPTION,  
)
```

```
► core/src/crackz/cli/detection_cli.py
```

	CrackzModel	
	CrackzDataset	
	MetricsLoggerCallback	
	resolve_device_and_accelerator	
	DetectionResult	
	Pipeline	

1.	
2.	
3.	
4.	DetectionResult
5.	evaluation.py
6.	
7.	
8.	
9.	
10.	
11.	
12.	
13.	
14.	
15.	
16.	
17.	
18.	ProjectConfig
19.	
20.	
21.	
22.	Pipeline

```
Project -> load_model_from_checkpoint -> CrackzModel.eval() -> DataLoader -> forward -> sigmoid -> threshold -> mask -> metrics(json)
```

requirements.txt

CLI Reference

app

[OPTIONS] COMMAND [ARGS]...

-v, --version	Show version and exit
--debug	Enable debug logging
--quiet	Only warnings and errors
--log-dir DIRECTORY	Central log directory (precedence: env CRACKZ_LOG_DIR > --log-dir > system temp). Created if missing. Project logs are always under <project>/logs.
--install-completion	Install completion for the current shell.
--show-completion	Show completion for the current shell, to copy it or customize the installation.

dashboard

crackz dashboard --project my-project

http://localhost:8501

crackz dashboard --project my-project --port 8080

crackz dashboard

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.
- 11.
- 12.
- 13.
- 14.
- 15.
- 16.

17.
18.

```
crackz detect --project my-project
crackz dashboard --project my-project
# Navigate to "Detection Results" page
```

```
crackz train --project my-project --epochs 50
crackz dashboard --project my-project
# Navigate to "Training Metrics" page
```

```
crackz dashboard --project my-project
# Navigate to "GPS Map" page
```

```
crackz dashboard --project my-project
# Navigate to "Parameter Tuning" page
# Adjust thresholds and preview results
```

clean

```
crackz clean --project ./my_project
```

```
crackz clean -p ./my_project
```

- *.log
 -
 -
 -
 -
 -
- models/ checkpoints/

```
crackz clean --project ./my_project --dry-run
```

Dry run: The following would be cleaned:

- /path/to/project/logs/crackz.log
- /path/to/project/artifacts/training/tensorboard
- /path/to/project/artifacts/models
- /path/to/project/artifacts/detections

Re-run without --dry-run to apply changes.

```
crackz clean --project ./my_project --logs-only
```

```
*.log
```

```
models/
```

```
checkpoints/
```

```
$ crackz clean --project ./my_project
```

```
Project cleaned! Removed 5 items (logs and artifacts).
```

```
--logs-only
```

```
$ crackz clean --project ./my_project --logs-only
```

```
Project cleaned! Removed 2 items (logs).
```

```
crackz clean -p ./my_project
```

```
crackz clean -p ./my_project --logs-only
```

```
crackz clean -p ./my_project --dry-run
```

```
migrate
```

```
1      ai.training.tensorboard      false      data.mask_threshold: 1      schema_version:
      crackz_version

      training/tensorboard/      runtime/tensorboard/      checkpoints/      models/
      runtime/lightning_logs/      detections/*.png      detections/visualizations/      training/lightning_logs/
```

```
crackz migrate --project ./my_project
```

```
crackz migrate --project ./my_project --migrate-files
```

```
crackz migrate --cfg ./my_project/my_project_project.yaml --migrate-files
```

```
crackz migrate -p ./my_project --migrate-files
```

```
crackz migrate -c ./my_project/my_project_project.yaml --migrate-files
```

```
# Schema changes only
crackz migrate --project ./my_project --dry-run

# Schema + file changes
crackz migrate --project ./my_project --migrate-files --dry-run
```

would

`.v{version}.{timestamp}.bak`

```
crackz migrate --project ./my_project --no-backup
```

```
[migrated] my_project schema 0 -> 1 (current=1) steps: 0->1: added tensorboard default + schema_version=1
```

```
[up-to-date] my_project schema 1 -> 1 (current=1) steps: none
```

```
build.yml
integration-tests.yml    pytest -m integration
release.yml
docker-build.yml
smoke-tests.yml
benchmark-tests.yml     pytest -m benchmark
docs.yml
```

```
pytest -m integration -vv
```

```
pytest -m "not integration"
```

```
pytest -m benchmark --benchmark-only
```

`crackz/cli/*_cli.py`

`schema_version`

- `.bak`
- `CRACKZ_AUTO_MIGRATE=1`

```
crackz migrate --project <project-dir>
```

```
--dry-run --no-backup
```

```
migration.md
```

```
model
```

```
crackz model export --project ./my_project --out my_model.zip
```

```
crackz model export -p ./my_project -o my_model.zip
```

```
<project_name>_model.zip
```

```
crackz model export -p ./my_project -c final -o final_model.zip
```

```
best last final
```

```
crackz model import --project ./my_project --model my_model.zip
```

```
crackz model import -p ./my_project -m my_model.zip
```

```
crackz model import -p ./my_project -m my_model.zip --force
```

```
--force
```

```
# Export from training project
crackz model export -p ./training_project -o harbor_crack_v1.zip

# Import into deployment project
crackz model import -p ./deployment_project -m harbor_crack_v1.zip
```

```
# Export pre-trained model
crackz model export -p ./pretrained_project -o base_model.zip

# Import into new project for fine-tuning
crackz model import -p ./finetuning_project -m base_model.zip
```

```
# Export after successful training
crackz model export -p ./my_project -o backups/model_$(date +%Y%m%d).zip
```

<checkpoint_name>.ckpt

metadata.json

```
{
  "project_name": "harbor_detection",
  "checkpoint_name": "best.ckpt",
  "model_config": {
    "encoder": "resnet34",
    "encoder_weights": "imagenet",
    "in_channels": 3,
    "num_classes": 1,
    "threshold": 0.5,
    "loss_name": "combo",
    "dice_weight": 0.7,
    "focal_weight": 0.3
  }
}
```

Model Evaluation

Model Evaluation

```
# Run evaluation on test dataset
crackz evaluate run --project my_harbor

# Specific metrics
crackz evaluate run --project my_harbor --metrics accuracy,iou

# Export results
crackz evaluate run --project my_harbor --output results.json
crackz evaluate run --project my_harbor --output results.csv
```

```
from pathlib import Path
from crackz.ai.evaluation import evaluate_model, export_json

# Run evaluation
results = evaluate_model(
    project_path=Path("my_harbor"),
    metrics=["accuracy", "precision", "recall", "iou"]
)

# Display results
print(results)

# Export to JSON
export_json(results, Path("evaluation_results.json"))
```

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$
$$\text{Precision} = TP / (TP + FP)$$
$$\text{Recall} = TP / (TP + FN)$$
$$F1 = 2 * (Precision * Recall) / (Precision + Recall)$$
$$\text{IoU} = \text{Area of Overlap} / \text{Area of Union}$$

```
# Default threshold (0.5)
crackz evaluate run --project my_harbor
```

```
# Custom threshold
crackz evaluate run --project my_harbor --threshold 0.7
```

```
# All metrics (default)
crackz evaluate run --project my_harbor

# Specific metrics
crackz evaluate run --project my_harbor --metrics accuracy,precision,recall
crackz evaluate run --project my_harbor --metrics iou # Only IoU
```

```
accuracy precision recall f1 iou
```

```
Evaluation Results (500 images)
```

```
=====
Accuracy: 92.3%
Precision: 89.1%
Recall: 94.8%
F1 Score: 91.9%
Mean IoU: 0.847
Threshold: 0.5
```

```
{
  "num_images": 500,
  "threshold": 0.5,
  "metrics": {
    "accuracy": 0.923,
    "precision": 0.891,
    "recall": 0.948,
    "f1_score": 0.919,
    "iou": 0.847
  }
}
```

```
Metric,Value
Num Images,500
Threshold,0.5
Accuracy,0.9230
Precision,0.8910
Recall,0.9480
F1 Score,0.9190
IoU,0.8470
```

```
project/
├─ images/
│   └─ test/      # Test images
│       ├── img1.png
│       ├── img2.png
│       └─ ...
└─ masks/
    └─ test/      # Ground truth masks
        └─ img1.png
```

```
└─ img2.png
└─ ...
```

```
def evaluate_model(
    project_path: Path,
    test_images: list[Path] | None = None,
    test_masks: list[Path] | None = None,
    metrics: list[str] | None = None,
    threshold: float = 0.5,
) -> EvaluationResults:
    """Evaluate crack detection model on test dataset."""
```

```

    project_path      test_images
test_masks           threshold           metrics
    EvaluationResults
```

```
@dataclass
class EvaluationResults:
    accuracy: float
    precision: float
    recall: float
    f1_score: float
    iou: float
    num_images: int
    threshold: float
```

```
# Export to JSON
export_json(results, Path("results.json"))

# Export to CSV
export_csv(results, Path("results.csv"))
```

```
ValueError: No test images found
```

```
<project>/images/test/
```

```
ls <project>/images/test/
```

```
ValueError: Image/mask count mismatch: 100 vs 98
```

```
# Check counts
ls <project>/images/test/ | wc -l
ls <project>/masks/test/ | wc -l

# Find missing masks
diff <(ls <project>/images/test/) <(ls <project>/masks/test/)
```

```
FileNotFoundError: No trained model checkpoint found
```



```
crackz detect train --project <project>
```

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- [_____](#)
- [_____](#)

MCP Integration

MCP (Model Context Protocol) Integration

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- 1.
- 2.
- 3.
- 4.

`.github/copilot-instructions.md`

Development Agents	GitHub Copilot	End-User GUI
4 MCP Servers - filesystem - github - python - uv	× No MCP IDE context Workspace files	× No MCP Simple API Project info only
Tools: Claude Desktop, Cline	Tools: VS Code, IDE	Tools: Crackz GUI Chat Panel

-
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-

```

TOOL CALL: filesystem.readDirectory("src/")
→ returns 50 files in context

TOOL CALL: filesystem.readFile("src/file1.py")
→ returns full file in context

TOOL CALL: filesystem.readFile("src/file2.py")
→ returns full file in context

```

```

# Discover and filter in code
files = await filesystem.readDirectory("src/")
python_files = [f for f in files if f.endswith(".py")]

# Load only what's needed
for filename in python_files[:5]: # Top 5 only
    content = await filesystem.readFile(f"src/{filename}")
    # Process in code, log summary only
    print(f"{filename}: {len(content)} lines")

```

- 1.
- 2.
- 3.
- 4.

<type>/<scope>-<short-slug>

```

feat/gui-live-preview      # New GUI feature
fix/pdf-rendering-crash    # Bug fix in PDF generation
perf/dataloader-cache      # Performance improvement
docs/api-reference-update  # Documentation update
refactor/config-cleanup    # Code refactoring

```

<type>(<scope>): <description>

```

feat(gui): add real-time detection preview
fix(cli): correct path handling in Windows
perf(dataloader): implement batch caching
docs(api): update training configuration reference

```

<type>(<scope>): <description>

```

feat(gui): add export panel for detection results
fix(pdf): handle missing hero image gracefully
perf(training): optimize data loading pipeline

```

```

Fix GUI bugs and add new feature
Update documentation
Add export functionality

```

```

feat:      feat(<scope>):
perf:      perf(<scope>):

```

```

fix:      fix(<scope>):

```

```

test:      docs:
chore:

```

```

refactor:
ci:

```

```

build:

```

```
typer.secho()
crackz.cli.options

logger
crackz.log

print()
```

```
uv run task tests
```

```
uv run task mypy
```

```
crackz mcp init
```

```
.mcp/config.json
```

```
crackz mcp show
```

```
crackz mcp validate
```

```
.mcp/config.json
```

```
{
  "mcpServers": {
    "filesystem": {
      "type": "stdio",
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-filesystem", "."],
      "description": "Filesystem operations with progressive disclosure"
    },
    "github": {
      "type": "stdio",
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-github"],
      "env": {
        "GITHUB_PERSONAL_ACCESS_TOKEN": "${GITHUB_TOKEN}"
      },
      "description": "GitHub operations (issues, PRs, releases)"
    },
    "python": {
      "type": "stdio",
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-python"],
      "description": "Python code execution environment"
    },
    "uv": {
      "type": "stdio",
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-uv"],
      "description": "UV package manager operations"
    }
  }
}
```

```
{
  "metadata": {
    "project": "Crackz",
    "language": "python",
    "version": "3.13",
    "package_manager": "uv",
    "framework": "pytorch-lightning",
    "cli": "typer",
    "gui": "pyside6",
    "testing": "pytest",
    "ci": "github-actions",
    "release": "semantic-release",
    "code_execution": true,
    "progressive_disclosure": true
  }
}
```

```
{
  "patterns": {
    "code_execution": {
      "enabled": true,
      "description": "Use code execution to interact with MCP servers efficiently",
      "examples": [
        "Filter large datasets in code before returning results",
        "Loop over operations without alternating tool calls",
        "Compose multiple MCP operations in a single execution"
      ]
    },
    "progressive_disclosure": {
      "enabled": true,
      "description": "Load tool definitions on-demand",
      "workflow": [
        "List directories to discover modules",
        "Read specific files as needed",
        "Load documentation when relevant"
      ]
    }
  }
}
```

```
--output -o
```

```
.mcp/config.json
```

```
--force -f
```

```
# Create with defaults
crackz mcp init

# Create in custom location
crackz mcp init --output custom/mcp.json

# Overwrite existing config
crackz mcp init --force
```

```
--config -c
```

```
.mcp/config.json
```

```
# Show default config
crackz mcp show

# Show custom config
crackz mcp show --config custom/mcp.json
```

`--config -c``.mcp/config.json`

```
# Validate default config
crackz mcp validate

# Validate custom config
crackz mcp validate --config custom/mcp.json
```

`.mcp/config.json`

```
{
  "mcpServers": {
    "filesystem": { ... },
    "custom-api": {
      "type": "stdio",
      "command": "node",
      "args": ["/custom-mcp-server.js"],
      "description": "Custom MCP server for specialized operations"
    }
  }
}
```

```
{
  "metadata": {
    "project": "Crackz",
    "language": "python",
    "team": "ML Ops",
    "repository": "https://github.com/Anaxiatech-AB/crackz",
    "custom_field": "your_value"
  }
}
```

- 1.
- 2.
- 3.
- 4.
- 5.

```
# List Python files in src/
files = await filesystem.readDirectory("src/crackz")
py_files = [f for f in files if f.endswith(".py")]

# Read only the first 3 for initial analysis
for filename in py_files[:3]:
    content = await filesystem.readFile(f"core/src/crackz/{filename}")
    lines = len(content.split('\n'))
    print(f"{filename}: {lines} lines")
```

```
# Get recent PRs and filter in code
prs = await github.listPullRequests(repo="Anaxiatech-AB/crackz", state="open")
priority_prs = [pr for pr in prs if "feat:" in pr.title or "fix:" in pr.title]

print(f"Found {len(priority_prs)} priority PRs out of {len(prs)} total")
for pr in priority_prs[:5]: # Show top 5 only
    print(f"#{pr.number}: {pr.title}")
```

```
# Install multiple packages in one operation
packages = ["pytest", "ruff", "mypy"]
for pkg in packages:
    await uv.install(package=pkg)
print(f"✓ Installed {pkg}")
```

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logger

`print()`

- 1.
2. `CRACKZ_LOG_DIR` `--log-dir`
- 3.
- 4.
5. `<project>/logs/`
- 6.
- 7.
- 8.

Info level - operational progress

Warning level - non-critical issues

Error level - failures with details

Debug level - detailed troubleshooting info

- ---
- ---
- ---
- ---

Tracing & Observability

Tracing with OpenTelemetry

```
uv sync --extra tracing
```

```
opentelemetry-instrumentation-anthropic
opentelemetry-exporter-otlp-proto-http
opentelemetry-sdk
```

```
Ctrl+Shift+P  Cmd+Shift+P
```

```
AI Toolkit: Open Tracing
```

```
ai-mlstudio.tracing.open
```

```
export CRACKZ_TRACING_ENABLED=1
crackz-gui # Tracing auto-enabled
```

```
from crackz.tracing import setup_tracing

# Initialize with default AI Toolkit endpoint
setup_tracing()

# Or custom endpoint
setup_tracing(
    endpoint="http://custom-collector:4318/v1/traces",
    service_name="crackz-production"
)
```

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-

```
crackz
└─ anthropic.messages.create
   └─ model: claude-sonnet-4-5-20250929
      └─ input_tokens: 1245
         └─ output_tokens: 382
            └─ duration: 2.3s
```

-
-
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-

```
CRACKZ_TRACING_ENABLED 0
```

```
1 0
```

```
http://localhost:4318/v1/traces
```

```
http://localhost:4317
```

```
setup_tracing(endpoint="http://your-collector:4318/v1/traces")
```

```
# Enable tracing for all CLI operations
export CRACKZ_TRACING_ENABLED=1
crackz detect run --project my_project
```

```
# Disable for specific run
export CRACKZ_TRACING_ENABLED=0
crackz detect run --project my_project
```

```
# In main.py or startup script
from crackz.tracing import setup_tracing
```

```
# Enable before starting GUI
if os.getenv("CRACKZ_TRACING_ENABLED") == "1":
    setup_tracing()
```

```
# Run GUI
from crackz_gui.qt.app import main
main()
```

```
from crackz.tracing import setup_tracing, shutdown_tracing
```

```
# Initialize tracing
setup_tracing(service_name="crackz-batch-processor")
```

```
# Your AI operations (automatically traced)
from crackz_gui.services.ai_chat import AIChatService
chat = AIChatService(...)
chat.send_message("Analyze this crack detection result")
```

```
# Clean shutdown (flushes pending spans)
shutdown_tracing()
```

```
uv sync --extra tracing
```

Ctrl+Shift+P → AI Toolkit: Open Tracing

CRACKZ_TRACING_ENABLED=1

uv sync --extra tracing

shutdown_tracing()

```
import atexit
from crackz.tracing import shutdown_tracing
atexit.register(shutdown_tracing)
```

CRACKZ_TRACING_ENABLED=1

```
from opentelemetry import trace
tracer = trace.get_tracer(__name__)

with tracer.start_as_current_span("custom-operation") as span:
    span.set_attribute("project.name", "harbor_alpha")
    span.set_attribute("image.count", 1024)
    # Your code here
```

```
from opentelemetry.sdk.trace.sampling import TraceIdRatioBased

# Sample 10% of traces
sampler = TraceIdRatioBased(0.1)

# Pass to setup_tracing (requires custom implementation)
```

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Async Callbacks

Async Callback Architecture

-
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```
from crackz.ai.core.async_callbacks import CallbackDispatcher

dispatcher = CallbackDispatcher(maxsize=100)

# Create sync reporter for PyTorch Lightning (training thread)
progress_reporter = dispatcher.create_sync_progress_reporter()

# Use in training (synchronous context)
train(project, progress_reporter=progress_reporter)

# Consume updates asynchronously (main thread)
async for progress, message in dispatcher.iter_progress():
    update_progress_bar(progress, message)
```

```
from crackz.ai.core.async_callbacks import (
    create_async_progress_consumer,
    create_async_metrics_consumer,
)

# Define callback
def update_ui(progress: float, message: str | None = None) -> None:
    progress_bar.set_value(progress)

# Create async consumer
consumer = create_async_progress_consumer(dispatcher, update_ui)

# Run in async context
await consumer()
```

```
from crackz_gui.qt.async_bridge import (
    start_gui_callback_consumers,
    stop_gui_callback_consumers,
)

# Start consumers (manages event loop)
loop, tasks = start_gui_callback_consumers(
    dispatcher,
    progress_callback=update_progress,
    metrics_callback=update_metrics,
)

# ... run training ...
```



```
# Cleanup
stop_gui_callback_consumers(dispatcher, loop)
```

TaskSignals

async_bridge.py

```
from crackz_gui.qt.async_bridge import TaskSignals, run_in_background

def _do_work(signals: TaskSignals) -> None:
    signals.progress.emit(0.5) # Safe from any thread
    signals.status.emit("Processing...")
    result = expensive_operation()
    signals.data.emit(result)
    signals.finished.emit()

signals = run_in_background(_do_work)
signals.progress.connect(self._on_progress)
signals.finished.connect(self._on_complete)
signals.error.connect(self._on_error)
```

threading.Lock

after()

crackz_gui.qt.async_bridge

```
def progress_callback(progress: float, message: str | None = None) -> None:
    print(f"Progress: {progress:.2%}")

train(project, progress_reporter=progress_callback)
```

```
dispatcher = CallbackDispatcher(maxsize=100)

# Create sync reporters
progress_reporter = dispatcher.create_sync_progress_reporter()
metrics_reporter = dispatcher.create_sync_metrics_reporter()

# Start training in background
training_thread = threading.Thread(
    target=train,
    args=(project,),
    kwargs={
        "progress_reporter": progress_reporter,
        "metrics_callback": metrics_reporter,
    }
)
training_thread.start()

# Consume updates asynchronously
async def consume_updates():
    async for progress, message in dispatcher.iter_progress():
        print(f"Progress: {progress:.2%} - {message}")

asyncio.run(consume_updates())
```

```
from crackz.ai.core.async_callbacks import CallbackDispatcher
from crackz_gui.qt.async_bridge import (
    start_gui_callback_consumers,
    stop_gui_callback_consumers,
)

dispatcher = CallbackDispatcher(maxsize=100)

# Start async consumers with Qt-thread-safe GUI updates
loop, tasks, emitters = start_gui_callback_consumers(
    dispatcher,
    progress_callback=self.update_progress_bar,
```

```

        metrics_callback=self.update_metrics_display,
    )
    self.qt_emitters = emitters # Keep signal emitters alive

# Create reporters
progress_reporter = dispatcher.create_sync_progress_reporter()
metrics_reporter = dispatcher.create_sync_metrics_reporter()

try:
    # Run training in background thread
    training_thread = threading.Thread(
        target=train,
        args=(project,),
        kwargs={
            "progress_reporter": progress_reporter,
            "metrics_callback": metrics_reporter,
        }
    )
    training_thread.start()
    training_thread.join()
finally:
    stop_gui_callback_consumers(dispatcher, loop)

```

```

Training Loop:
  Process batch → Update metrics → **Wait for UI** → Next batch
                                ↑ 1-5ms overhead

```

```

Training Loop:
  Process batch → Put in queue → Next batch (no wait!)
                        ↓ 0.1ms
  Async Consumer: Takes from queue → Updates UI

```

```
dispatcher = CallbackDispatcher(maxsize=100)

# ... use reporters ...

# Check progress queue stats
stats = dispatcher.progress_stats
print(f"Total items put: {stats.total_put}")
print(f"Total items retrieved: {stats.total_get}")
print(f"Total items dropped: {stats.total_dropped}")
print(f"Current queue size: {stats.current_size}")
print(f"Max queue size: {stats.max_size}")

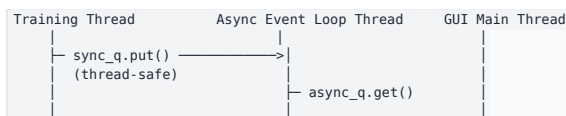
# Check metrics queue stats
metrics_stats = dispatcher.metrics_stats
```

```
total_dropped
```

```

• sync_q
•
•
•
•
• async_q
• threading.Event

```





```

# Before (still works)
train(project, progress_reporter=my_callback)

# After (optional upgrade for better performance)
dispatcher = CallbackDispatcher()
train(project, progress_reporter=dispatcher.create_sync_progress_reporter())

```

```

from crackz.ai.core.async_callbacks import CallbackDispatcher
from crackz_gui.qt.async_bridge import start_gui_callback_consumers

dispatcher = CallbackDispatcher(maxsize=100)
loop, tasks = start_gui_callback_consumers(
    dispatcher,
    progress_callback=update_progress,
    metrics_callback=update_metrics,
)

# Use dispatcher's sync reporters in training
train(
    project,
    progress_reporter=dispatcher.create_sync_progress_reporter(),
    metrics_callback=dispatcher.create_sync_metrics_reporter(),
)

stop_gui_callback_consumers(dispatcher, loop)

```

`crackz.ai.core.async_callbacks`

- `CallbackDispatcher`
- `QueueStats`
- `create_async_progress_consumer()`
- `create_async_metrics_consumer()`

`crackz_gui.qt.async_bridge`

- `AsyncEventLoop`
- `create_threadsafe_gui_callback()`
- `create_threadsafe_metrics_callback()`
- `start_gui_callback_consumers()`
- `stop_gui_callback_consumers()`

`help()`

```

from crackz.ai.core.async_callbacks import CallbackDispatcher
help(CallbackDispatcher)

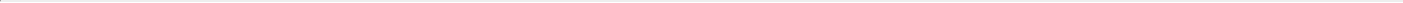
```

[docs/examples/async_callbacks_example.py](https://crackz.ai/docs/examples/async_callbacks_example.py)

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Contributing

Contributing Guide



-
- [.github/copilot-instructions.md](#)
- [.github/workflows/copilot-setup-steps.yml](#)

<type>/<scope>--<short-slug>

```
# AI creates feature branch
git checkout -b feat/gui-export-panel

# AI makes changes with proper commit messages
git commit -m "feat(gui): add export panel component"
git commit -m "feat(gui): integrate export panel with main view"

# When creating PR, AI suggests proper title
# [ ] PR Title: "feat(gui): add export panel for detection results"
# This triggers a MINOR version release (e.g., 1.2.0 → 1.3.0)
```



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```
# create & activate venv
uv venv
source .venv/bin/activate # Windows: .venv\Scripts\activate

# install with dev dependencies
uv sync --dev

# run CLI help
uv run crackz --help

# run GUI (optional)
uv run crackz-gui
```

pyproject.toml [tool.taskipy.tasks]

```
uv run task lint      # ruff lint (auto-fix)
uv run task format    # ruff format
uv run task mypy       # static type check
uv run task tests      # unit tests only
uv run task integration # integration tests
uv run task all_tests  # full test matrix
uv run task docs       # live docs (mkdocs serve)
uv run task build_docs # strict build
```

- `tests/unit` `core/tests/unit` `gui/tests/unit`
- `tests/integration` `@pytest.mark.integration`
- `backend/backend/tests/unit/crackz/cli/test_cli_benchmark.py` `@pytest.mark.benchmark`

- `tests/test_data`
- `core/tests` `gui/tests`

```
uv run task tests
```

```
uv run task all_tests
```

```
uv run task benchmark
```

```
pyproject.toml fail_under = 75
```

```
tests/integration/crackz/gui/
```

```
# Run all integration tests (includes GUI smoke tests)
uv run task integration

# Run only GUI smoke tests
uv run pytest tests/integration/crackz/gui/test_gui_smoke.py

# Run GUI in smoke mode (auto-close after initialization)
set CRACKZ_GUI_SMOKE=1 # Windows CMD
export CRACKZ_GUI_SMOKE=1 # Linux/macOS
uv run crackz-gui
```

```
CRACKZ_GUI_SMOKE=1
```

```
CI GITHUB_ACTIONS
```

```
tests/integration/crackz/gui/
├─ test_gui_smoke.py # Smoke tests for GUI initialization
└─ ... # Future GUI integration tests
```

1.

```
env = os.environ.copy()
env["CRACKZ_GUI_SMOKE"] = "1"
subprocess.run([sys.executable, "gui/src/crackz_gui/crackz_gui.py"], env=env)
```

2.

```
@pytest.mark.skipif(not has_display() or os.environ.get("CI"),
                    reason="Requires display")
def test_component(qapp):
    component = MyComponent()
    assert component is not None
```

3.

```
def test_no_unsupported_parameters():
    # Ensure widget initialization works without errors
    browser.toggle_toolbar_buttons(visible=True) # Should not raise
```

benchmark

```
# Run benchmarks - compares to last saved baseline and fails on regression
uv run task benchmark
```

```
# Update the baseline after intentional performance-impacting changes
uv run task benchmark_save
```

```
# Manual benchmark with comparison
uv run pytest -m benchmark --benchmark-only --benchmark-autosave --benchmark-compare --benchmark-compare-fail=mean:20%
```

.benchmarks/

- uv run task format lint
-
-
- loguru print
-

<type>(<optional-scope>): <description>

```
feat: feat(<scope>): fix: fix(<scope>):
perf: perf(<scope>):
docs: refactor:
test: chore: ci:
build: style:
```

```
feat(gui): add return-to-run-view functionality
fix(pdf): correct missing hero image
perf(dataloader): optimize batch processing
```

```
Fix GUI image display and add return-to-run-view functionality
Add new feature to handle user authentication
Update documentation for new feature
```

```
feat: fix:      :
feat(gui): fix(cli):      :
Feat: Fix:      :
```

```
# Install pre-commit hooks (one-time setup)
pip install pre-commit # or: uv tool install pre-commit
pre-commit install
pre-commit install --hook-type commit-msg
```

```
# Test with a valid message
echo "feat(gui): add export panel" > temp_msg.txt
cz check --commit-msg-file temp_msg.txt

# Test with an invalid message
echo "Fix bug" > temp_msg.txt
cz check --commit-msg-file temp_msg.txt
```

```
git commit --no-verify -m "your message"
```

- -
 -
 -
 -
-

```
uv run task docs # serves at http://127.0.0.1:8000
```

```
uv run task build_docs
```

```
scripts/export_pdf.py
```

```
uv run task pdf
```

- main
pyproject.toml

cz bump
CHANGELOG.md

main
 -
 -
 -
 -
-

```
logs/crackz.log
```

main

<type>/<scope>-<short-slug>

optimization

feat/gui-live-preview

docs/api-reference-update

fix/pdf-rendering-crash

perf/dataloader-refactor/config-module-cleanup

feat(gui): add live preview panel

fix/pdf-crash

feat/gui-live-preview

fix(pdf): handle null hero image

- main

<type>/<scope>-<short-slug>

feat/gui-live-preview

fix/pdf-hero-image

refactor/logging-bootstrap

perf/dataloader-prefetch

chore/ci-release-step

hotfix/0.17.1-pdf-null-crash

exp/vector-db-poc

chore

feat

refactor

ci

build

exp

fix

docs

test

style

perf

hotfix/*

main

fix:

<type>(<optional-scope>): <imperative summary>

<body – optional>

<footer – BREAKING CHANGE / issue refs>

feat(gui): add real-time detection preview panel

fix(pdf): correct missing hero when logo == cover image

refactor(logging): unify central + project logger bootstrap

chore(ci): refine release workflow version verification

feat!: remove legacy --batch flag

BREAKING CHANGE: The --batch flag was replaced by --batch-size.

feat

fix

!

BREAKING CHANGE:

-
- feat(gui): add live preview
- fix(cli): correct path handling
- docs: update installation guide
- Fix GUI bugs
- Add new feature
-
-
-
-
-

fix: docs: <type>(<scope>): <description> feat:

Dev Build <base>.devYYYYMMDD+<shortsha>
dev-<computed-dev-version> dev-

1. main
2. cz bump
3.
4. CHANGELOG.md vX.Y.Z main
5.
6.
7. :X.Y.Z :latest
8.
9.
10.
11.
12. v0.35.0 latest
13.
14.

1. hotfix/<next-patch>-<short-desc> main
2. fix:
3. cz bump

TrainingConfig
tests/unit/crackz/project/test_training_config_schema.py
.pre-commit-config.yaml

```
pip install pre-commit # or uv tool install pre-commit
pre-commit install
```

test_project.py

docs/configuration.md TrainingConfig Contract & Compatibility
CHANGELOG.md

ModelConfig DetectionConfig

```
feat/cli-batch-export
fix/train-metrics-path
perf/dataloader-cache
refactor/model-config
docs/pdf-section
test/cli-train-e2e
chore/update-ruff
ci/release-tag-fix
build/multiarch-images
style/ruff-cleanup
hotfix/0.17.2-threshold-null
exp/alt-model-head
wip/gui-layout-pass1
```

feat(gui):

BREAKING CHANGE:

- -
 -
- feat/*

typer.secho

```
# Success messages
typer.secho("✅ Operation completed successfully", fg=typer.colors.GREEN)

# Error messages
typer.secho("❌ Operation failed", fg=typer.colors.RED)

# Warning messages
typer.secho("⚠️ Warning message", fg=typer.colors.YELLOW)

# Info messages
typer.secho("ℹ️ Information", fg=typer.colors.CYAN)
```

logger

```
from crackz.log import logger

# Detailed progress, technical info, debugging
logger.info("Detailed operational information")
logger.warning("Warning about internal state")
logger.error("Error details for troubleshooting")
```

NEVER USE

- print()
- typer.echo() typer.secho()
- console.print()

-

core/src/crackz/cli/options.py

```
from crackz.cli.options import PROJECT_PATH_OPTION, FORCE_OPTION

def command_function(
    project_path: Path | None = PROJECT_PATH_OPTION,
    force: bool = FORCE_OPTION,
) -> None:
```

```
# Only for options not reused elsewhere
LOCAL_OPTION = typer.Option(
    None,
    "--local-flag",
    help="Module-specific option not reused elsewhere",
)
```

NEVER

-
- **typer.Option()** **FORCE_OPTION**
-

```
"""Module docstring explaining CLI command purpose."""

from __future__ import annotations

from pathlib import Path
import typer

from crackz.cli.exceptions import cli_failure
from crackz.cli.options import PROJECT_PATH_OPTION, FORCE_OPTION
from crackz.log import logger # Always import logger

app = typer.Typer(no_args_is_help=True)

@app.command(name="command")
def command_function(
    project_path: Path | None = PROJECT_PATH_OPTION,
    force: bool = FORCE_OPTION,
) -> None:
    """Command description."""
    try:
        # Business logic here
        logger.info("Processing started")
        typer.secho("✅ Operation completed", fg=typer.colors.GREEN)
    except Exception as e:
        cli_failure(e, "Operation name")
```

-
-
-
- **loguru** **print()**
-

-
-
-
-

```
•
•
•
•
    tests/unit/
    tests/integration/
    @pytest.mark.integration
    @pytest.mark.benchmark
```

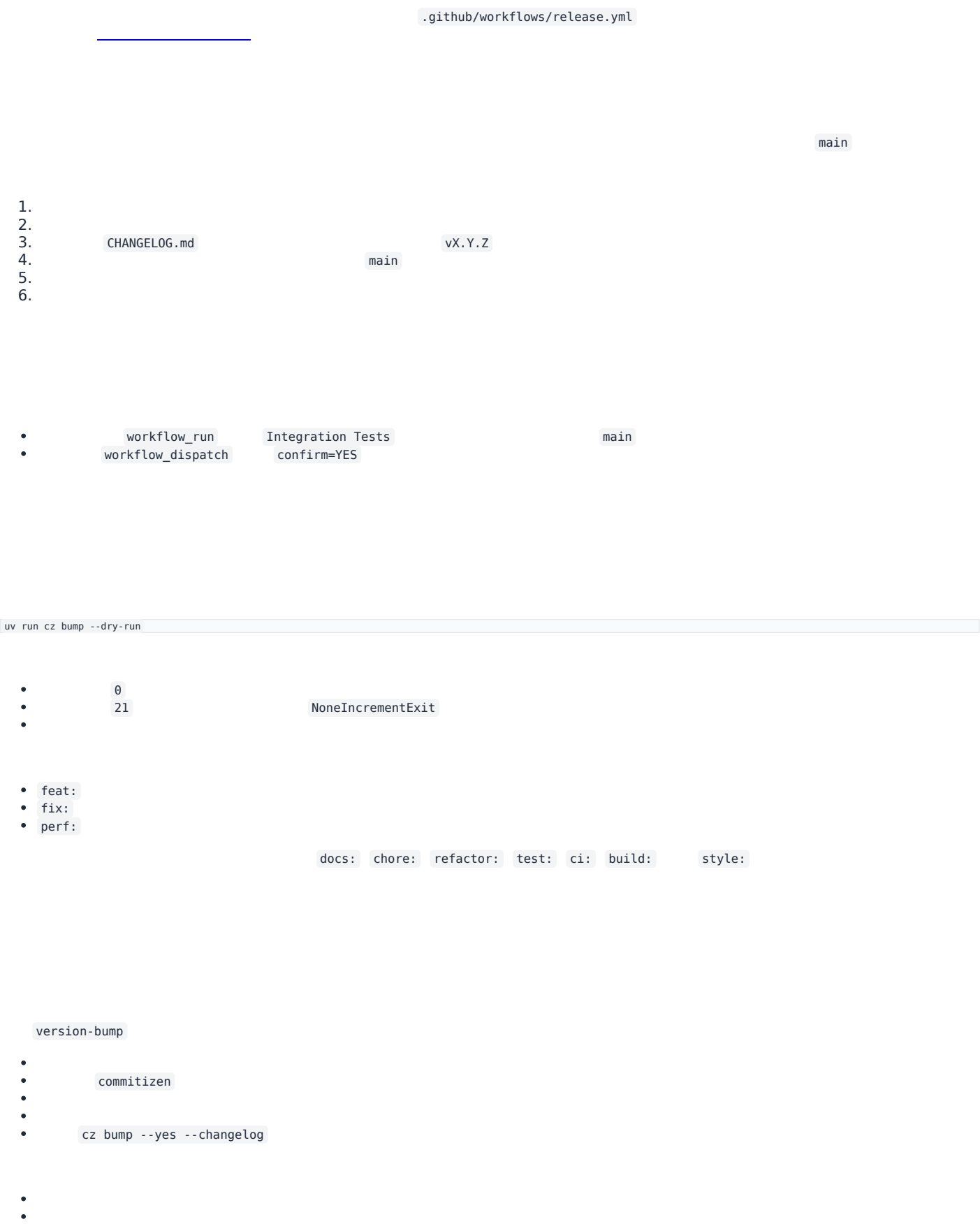
```
1.
2.
3.
4.
5.
6.
    options.py
    uv run task lint && uv run task format
    uv run task mypy
    uv run task tests
```

```
1.
2.
3.
4.
5.
    typer.secho()
    logger
    options.py
```

```
1.
2.
3.
4.
5.
6.
7.
    print()
    typer.echo()
    typer.secho()
    options.py
```

Release Flow

Release Flow



- CHANGELOG.md
-
-

uv lock

uv.lock

main

- HEAD main
-

version-bump

-
- vx.Y.Z CHANGELOG.md

- build-wheels
-
-
-
- build-sdist
- .tar.gz

crackz

release

-
-
- dist/
- site/crackz-docs.pdf .0
- CHANGELOG.md
- SHA256SUMS.txt
-
-
- --draft=false

-
- cu118 cu128
-
- CHANGELOG.md
- SHA256SUMS.txt
- .0

dist-combined

- ai-release-notes.yml
-
-
- pyinstaller-build.yml
-
-
- docker-build.yml
-
-

-
-
-
-

- feat: fix: perf: main
-

-
-
-
- .0

```
gh run list --limit 10
gh run view <run-id>
gh run watch
```

PR Review Workflow

Pull Request Review Workflow with Claude Code

- gh
-
-

```
# List open PRs
gh pr list

# View PR details
gh pr view <number>

# View PR diff
gh pr diff <number>

# Check CI status
gh pr checks <number>

# Review PR
gh pr review <number> --approve
gh pr review <number> --comment -b "your comment"
gh pr review <number> --request-changes -b "needs changes"
```

```
# List all open PRs
gh pr list

# List PRs with specific labels
gh pr list --label "ready-for-review"

# List your own PRs
gh pr list --author @me
```

```
# View PR summary
gh pr view 254

# View PR with comments
gh pr view 254 --comments

# Open PR in browser
gh pr view 254 --web
```

```
fix(docs): resolve build failures and clarify release standards #254
Open • theresiasnow wants to merge 4 commits into main from fix/docs-build-and-standards

This PR fixes critical documentation build failures...

View this pull request on GitHub: https://github.com/owner/repo/pull/254
```

```
# View full diff
gh pr diff 254

# View only changed filenames
gh pr diff 254 --name-only

# View diff in browser
gh pr diff 254 --web
```

```
# View all checks
gh pr checks 254
```

```
# View only required checks
gh pr checks 254 --required

# Watch checks until completion
gh pr checks 254 --watch

# Open checks in browser
gh pr checks 254 --web
```

```
✓ Build and Test (ubuntu-latest) – pass
✓ Build and Test (windows-latest) – pass
✓ Integration Tests – pass
✓ Lint and Format – pass
```

```
# Checkout PR to review code locally
gh pr checkout 254

# This creates/switches to the PR branch
# Now you can:
# - Run tests: uv run task tests
# - Run linting: uv run task lint
# - Build docs: uv run task build_docs
# - Run the code locally
```

```
Review the changes in scripts/validate_commit_msg.py for:
- Code quality and readability
- Error handling
- Edge cases
- Documentation completeness
```

```
Review the changes in docs/contributing.md for:
- Clarity and completeness
- Correct formatting
- Broken links
- Consistency with existing docs
```

```
Check if the changes need additional tests and identify any edge cases not covered.
```

```
gh pr review 254 --approve -b "LGTM! Code quality looks good, tests pass."
```

```
gh pr review 254 --comment -b "Good work! Minor suggestion: consider adding error handling for X."
```

```
gh pr review 254 --request-changes -b "Please address the linting issues in validate_commit_msg.py before merging."
```

```
gh pr review 254 --approve -b "$(cat <<'EOF'
Great PR! Here's my review:

☐ Code quality: Excellent
☐ Tests: All passing
☐ Documentation: Clear and complete
☐ Linting: All checks pass

Minor suggestions:
- Consider adding a note about pre-commit hook in README
- The error messages are very helpful

Approved for merge.
EOF
)"
```

```
# After review, switch back to main
git checkout main
```

```
# List all PRs and store numbers
gh pr list

# Review each PR in sequence
for pr in 254 255 256; do
  echo "Reviewing PR #$pr"
  gh pr view $pr
  gh pr diff $pr --name-only
  gh pr checks $pr
  # Ask Claude Code to review changes
done
```

```
# Get PR data as JSON
gh pr view 254 --json title,body,author,createdAt,additions,deletions,files

# Get check status as JSON
gh pr checks 254 --json name,state,completedAt
```

```
# Continuously monitor checks until completion
gh pr checks 254 --watch --interval 30

# This will update every 30 seconds until all checks complete
```

Always Check CI Before Reviewing

```
# Wait for checks to complete before reviewing
gh pr checks 254 --watch
```

Review Locally for Complex Changes

```
# For complex PRs, checkout and test locally
gh pr checkout 254
uv run task lint
uv run task tests
uv run task build_docs
```

Use Claude Code for Deep Review

Provide Constructive Feedback

```
# Instead of just "needs work"
gh pr review 254 --request-changes -b "$(cat <<'EOF'
Thanks for the PR! A few items to address:

1. Linting: Please run 'uv run task lint' to fix PLR2004 error
2. Tests: Add test case for empty commit message
3. Docs: Update contributing.md with hook installation steps

Otherwise looks great!
EOF
)"
```

Verify Release Triggers

```
# Check PR title format
gh pr view 254 --json title

# Should match: <type>(<scope>): <description>
# Examples:
```

```
# [] feat(gui): add export panel
# [] fix(docs): resolve build failures
# x Fix bugs
```

```
# 1. View the PR
gh pr view <number>

# 2. Check what files changed
gh pr diff <number> --name-only

# 3. View the actual changes
gh pr diff <number>

# 4. Checkout and test the fix
gh pr checkout <number>
# Test the fix...

# 5. Run all pre-commit checks (lint, format, mypy, tests)
uv run task pre_commit

# 6. Approve if all looks good
gh pr review <number> --approve -b "Bug fix verified, all pre-commit checks pass."
```

```
# 1. View the PR
gh pr view <number>

# 2. Checkout the branch
gh pr checkout <number>

# 3. Build docs to check for errors
uv run task build_docs

# 4. Review locally with live server
uv run task docs
# Opens at http://127.0.0.1:8000

# 5. Ask Claude Code to review
"Review the documentation changes for clarity, completeness, and broken links"

# 6. Approve or request changes
gh pr review <number> --approve
```

```
# 1. View PR and check size
gh pr view <number> --json additions,deletions,files

# 2. Check CI status
gh pr checks <number>

# 3. Checkout locally
gh pr checkout <number>

# 4. Run full test suite
uv run task all_tests

# 5. Test the feature manually
# ... manual testing ...

# 6. Ask Claude Code for code review
"Review this feature for:
- Code quality and maintainability
- Test coverage
- Documentation completeness
- Potential edge cases"

# 7. Provide detailed review
gh pr review <number> --approve -b "Feature works well, tests comprehensive."
```

- 1.
- 2.
- 3.
- 4.

```
👉 Release workflow completed successfully
👉 New release published: v0.52.1
👉 Artifacts built and uploaded
👉 CHANGELOG updated
👉 Git tag created
```

```
ⓘ Release workflow completed - No release created
👉 Reason: No conventional commits found that trigger releases
👉 This is expected behavior for documentation and maintenance commits
```

```
gh auth login
```

```
# Make sure you're in the right repository
gh repo view

# Or specify repository explicitly
gh pr view 254 -R Anaxiatech-AB/crackz
```

```
# Stash local changes first
git stash

# Then checkout PR
gh pr checkout <number>

# Restore changes later
git stash pop
```

- _____
- _____
- _____

Distribution

Distribution & Commercial Delivery

```
.github/workflows/docker-build.yml
```

```
.github/workflows/docker-build.yml
```

```
uv build --wheel
python scripts/build_cuda_wheel.py
uv build --sdist
uv run task build_exe
docker build -f Dockerfile -t crackz-cli:latest .
docker build -f Dockerfile.slim -t crackz-cli:slim .
docker build -f Dockerfile.gpu -t crackz-cli:gpu .
```

```
# Build with CUDA 11.8 (default, most compatible for Windows)
python scripts/build_cuda_wheel.py

# Build with CUDA 12.1
python scripts/build_cuda_wheel.py --cuda-version cu121

# Build with CUDA 12.8 (default for Linux)
python scripts/build_cuda_wheel.py --cuda-version cu128

# Build without cleaning artifacts first
python scripts/build_cuda_wheel.py --no-clean
```

```
# Method 1: Install with specific CUDA index
pip install dist/crackz-*.whl --extra-index-url https://download.pytorch.org/whl/cu118

# Method 2: Install with CUDA extra (uses pyproject.toml config)
pip install dist/crackz-*.whl[cu128] # For CUDA 12.8 on Linux

# Method 3: Install CPU-only version
pip install dist/crackz-*.whl[cpu]
```

```
cu118                                cu121

cu128

index-url                                pyproject.toml                                --extra-
python -c "import torch; print(f'CUDA available: {torch.cuda.is_available()}')"
```

```
# Build both CLI and GUI executables (CPU-only)
uv run task build_exe

# Build only CLI
uv run task build_exe_cli

# Build only GUI
uv run task build_exe_gui

# Clean and rebuild
uv run task build_exe_clean
```

```
# Build with CUDA 11.8 (default, recommended for Windows)
python scripts/build_cuda_exe.py
```

```
# Build with CUDA 12.8 (for newer GPUs/Linux)
python scripts/build_cuda_exe.py --cuda-version cu128

# Build only GUI with CUDA
python scripts/build_cuda_exe.py --gui-only

# Skip CUDA check if already installed
python scripts/build_cuda_exe.py --skip-cuda-check
```

dist/crackz

dist/crackz-gui

```
# Use the build script directly for more control
python scripts/build_exe.py --help
python scripts/build_exe.py --clean # Clean then build both
python scripts/build_exe.py --no-build # Only clean, don't build

# CUDA build script
python scripts/build_cuda_exe.py --help
python scripts/build_cuda_exe.py --clean --cuda-version cu128
```

- -
 -
- upload_to_pypi = false

dist/

v0.14.0

- 1.
- 2.
- 3.
- 4.

```
# Install specific version from GitHub release
pip install https://github.com/OWNER/REPO/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl

# Or download then install
wget https://github.com/OWNER/REPO/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl
pip install crackz-0.30.2-py3-none-any.whl
```

```
# Set GitHub token
export GITHUB_TOKEN="ghp_XXXXXXXXXX" # gitleaks:allow
```

```
# Download using gh CLI
gh release download v0.30.2 --pattern '*.whl' --repo OWNER/REPO
pip install crackz-*.whl

# Or use curl with authentication
curl -L \
  -H "Authorization: token $GITHUB_TOKEN" \
  -o crackz.whl \
  https://github.com/OWNER/REPO/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl
pip install crackz.whl
```

PRIVATE_INDEX_USER

__token__

PRIVATE_INDEX_PASSWORD

PRIVATE_INDEX_URL

```
pip install --index-url https://pypi.company.com/simple/ crackz
```

.github/workflows/release.yml

```
name: release
on:
  push:
    tags:
      - 'v*.*.*'
jobs:
  build-release:
    runs-on: ubuntu-latest
    permissions:
      contents: write # allow releasing
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-python@v5
        with:
          python-version: '3.13'
      - name: Install uv
        run: pip install uv
      - name: Sync deps (locked)
        run: uv sync --frozen
      - name: Build wheel + sdist
        run: uv build --wheel --sdist
      - name: Build documentation PDF
        run: |
          uv run mkdocs build --clean --strict
          uv run playwright install chromium
          uv run python scripts/export_pdf.py --outline \
            --cover-image 'assets/crackz-logo.png' \
            --logo 'assets/crackz-logo.png' \
            --out site/crackz-docs.pdf
          cp site/crackz-docs.pdf dist/
      - name: Upload to private index (twine)
        if: env.PRIVATE_INDEX_URL != ''
        env:
          TWINE_USERNAME: ${ secrets.PRIVATE_INDEX_USER }
          TWINE_PASSWORD: ${ secrets.PRIVATE_INDEX_PASSWORD }
        run: |
          pip install twine
          twine upload --repository-url ${ secrets.PRIVATE_INDEX_URL } dist/*.whl dist/*.tar.gz
      - name: Attach artifacts to Release
        uses: softprops/action-gh-release@v2
        with:
          files: dist/*
```

publish-release.yml

crackz-gui

v0.35.0

latest

https://pypi.company.com/simple/

PRIVATE_INDEX_USER

PRIVATE_INDEX_URL

__token__

PRIVATE_INDEX_PASSWORD

`.github/workflows/docker-build.yml``.github/workflows/docker-build.yml`

```

docker login ghcr.io -u USER -p TOKEN
# Build variants
docker build -f Dockerfile      -t ghcr.io/anaxiatech-ab/crackz:latest .
docker build -f Dockerfile.slim -t ghcr.io/anaxiatech-ab/crackz:slim   .
docker build -f Dockerfile.gpu  -t ghcr.io/anaxiatech-ab/crackz:gpu    .
# Push
for tag in latest slim gpu; do docker push ghcr.io/anaxiatech-ab/crackz:$tag; done

```

```

docker pull ghcr.io/anaxiatech-ab/crackz:slim

```

```

pip install --index-url https://pypi.company.com/simple/ crackz
pip install https://github.com/OWNER/REPO/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl
pip install torch torchvision --index-url <TORCH_URL> && pip install <crackz-wheel-url>
pip install --no-index --find-links wheelhouse crackz-VERSION-py3-none-any.whl
pip install --require-hashes -r requirements.txt

```

1.

```

# Download Crackz wheel from GitHub Release first
wget https://github.com/OWNER/REPO/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl

# Create wheelhouse directory and download all dependencies
mkdir wheelhouse
pip download crackz-VERSION-py3-none-any.whl -d wheelhouse

```

2.

```

(cd wheelhouse && shasum -a 256 * > SHA256SUMS.txt)

```

3.

4.

```

pip install --no-index --find-links wheelhouse crackz-VERSION-py3-none-any.whl

```

-
- `pip install pip-audit cyclonedx-bom`
-

`~/.config/pip/pip.conf`

```
[global]
extra-index-url = https://TOKEN@pkg.company.com/simple

%APPDATA%/pip/pip.ini
```

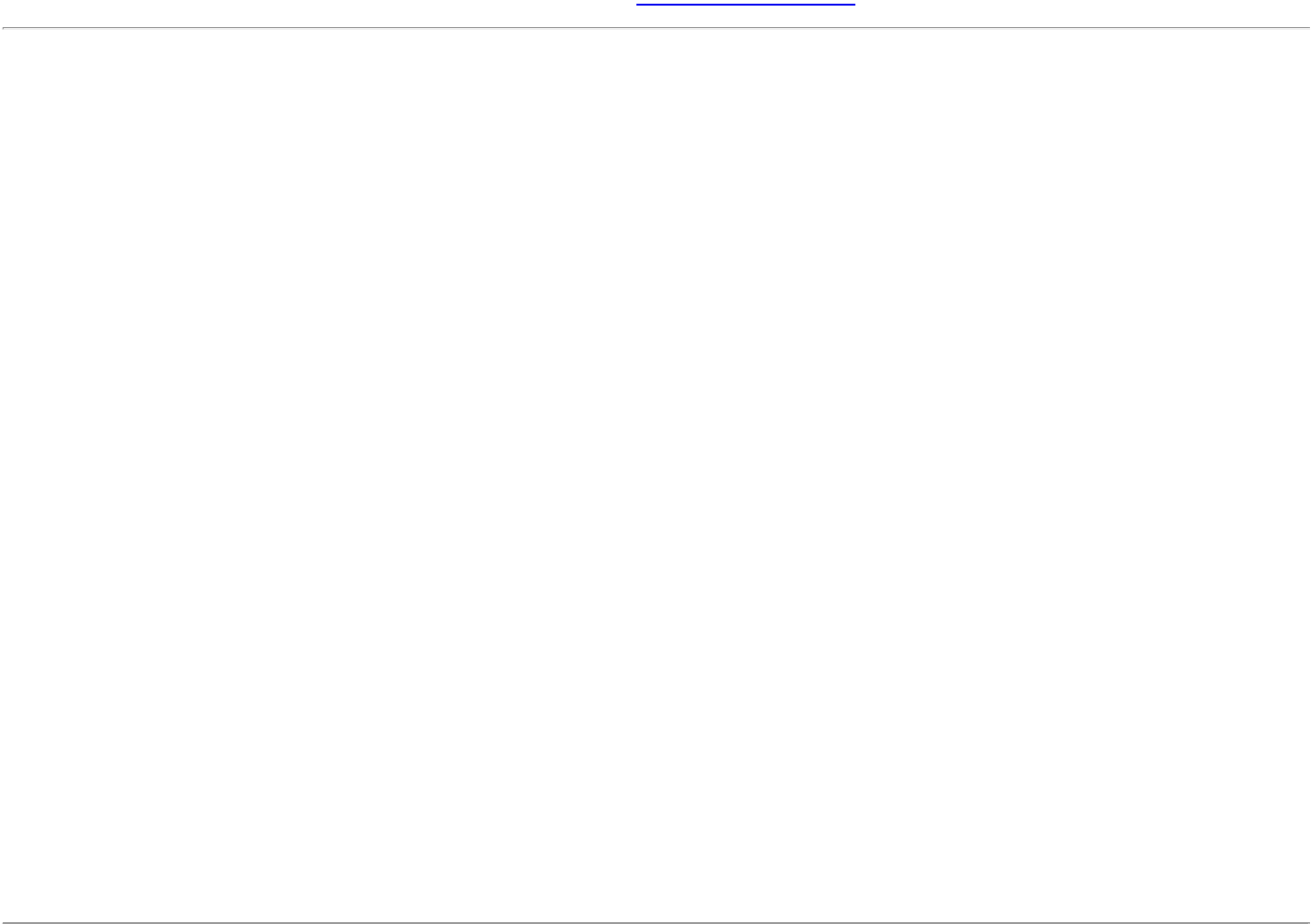
-
- `escrow-*`

-
-
-

```
# Build wheel
uv build --wheel
# Build + attach release (manual example)
git tag v0.14.0 && git push origin v0.14.0
# Install from private index
pip install --index-url https://pypi.company.com/simple/ crackz
# Install from GitHub Release
pip install https://github.com/OWNER/REPO/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl
# Offline install
pip install --no-index --find-links wheelhouse crackz-VERSION-py3-none-any.whl
```

Customer Guide

Customer Guide: Crackz for Infrastructure Management



What is a Mask?

Original Image	Mask Image	Result
[photo of concrete]	→ [white lines on black background]	= Trained model learns to detect cracks

When Do You Need Masks?

-
-
-
-
-

Recommended Tools for Creating Masks

1. CVAT (Computer Vision Annotation Tool) - Best for Teams

```
# Self-hosted option (Docker)
docker run -d -p 8080:8080 cvat/server
# Or use cloud-hosted version at app.cvat.ai
```

2. Labelme - Best for Individual Users

```
pip install labelme
labelme
```

labelme_json_to_dataset <json_file> label.png

3. Photoshop / GIMP - Best for Fine Control

masks/val/ masks/test/ masks/train/

4. Python Scripting - Best for Automation

```
# Example: Convert polygon annotations to masks
from PIL import Image, ImageDraw
import json

def polygon_to_mask(image_path, polygon_coors, output_path):
    # Load original image to get dimensions
    img = Image.open(image_path)
    width, height = img.size

    # Create black background
    mask = Image.new('L', (width, height), 0)
    draw = ImageDraw.Draw(mask)

    # Draw white polygon for crack
    draw.polygon(polygon_coors, fill=255)

    # Save mask
    mask.save(output_path)

# Usage
polygon_coors = [(100, 200), (150, 250), (200, 230), ...]
polygon_to_mask('image_01.jpg', polygon_coors, 'masks/train/image_01.png')
```

Best Practices for Mask Creation

Quality Guidelines

-
-
-
-
- image_01.jpg image_01.png

File Naming Convention

```
✓ CORRECT:
images/train/harbor_crack_001.jpg
masks/train/harbor_crack_001.png

✗ INCORRECT:
```

```
images/train/harbor_crack_001.jpg
masks/train/harbor_crack_001_mask.png ← Extra suffix
masks/train/HarborCrack001.png      ← Different name
```

Organizational Tips

- 1.
- 2.
- 3.
- 4.
- 5.

Train/Val/Test Split

```
masks/
train/  # 70% of labeled data (200-350 images)
val/    # 15% of labeled data (30-50 images)
test/   # 15% of labeled data (30-50 images)
```

images/

```
images/
train/  # Same filenames as masks/train/
val/    # Same filenames as masks/val/
test/   # Same filenames as masks/test/
```

Using Crackz with Your Masks

images/train/

```
# Automatic mask detection (if masks are in source_folder/masks/)
crackz import-images \
--project my-project \
--src ./labeled-data \
--type train \
--size 512 512

# Explicit mask source
crackz import-images \
--project my-project \
--src ./images \
--masks_src ./masks \
--type train \
--size 512 512
```

```
crackz import-images \
--project my-project \
--src ./labeled-data \
--masks_src ./labeled-data/masks \
--split \
--train-ratio 0.7 \
--val-ratio 0.15 \
--test-ratio 0.15 \
--size 512 512 \
--random-seed 42
```

Troubleshooting Mask Issues

--	--	--

masks/train/

masks/

Advanced: Semi-Automated Mask Generation

Time Estimates

Professional Labeling Services

```
inspection-campaign-2025/  
  images/      # Organized by train/val/test/raw splits  
  masks/       # Ground truth for training (if available)  
  artifacts/    # Detection results, metrics, checkpoints  
  logs/        # Complete audit trail  
  project.yaml  # Configuration snapshot
```

-
-
-
-



-
-
-
-



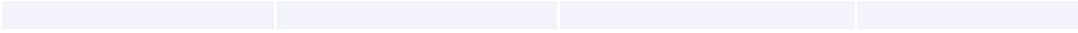


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-





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Customer Access Management

- 1.
- 2.
- 3.
- 4.

- 1.
- 2. templates/welcome-email.md
- 3. templates/access-revocation-email.md
- 4. templates/renewal-reminder-email.md
- 5.
- 6.
- 7.
- 8.

- 1.
- 2.
- 3.

Customer Access Management

- 1.
 - 2.
 3. `pip install crackz`
-

`pip install crackz`

1.

```
# Via GitHub web UI:  
# Settings > Teams > New Team  
# Team name: "Customer-Acme-Corp"  
# Privacy: Secret  
# Repository access: crackz (Read)
```

- 2.

- 3.
- 4.
- 5.

Subject: Crackz Access - Welcome to [Customer Name]

Hi [Contact Name],

Thank you for your Crackz license purchase!

To access Crackz packages and Docker images, please:

1. Create a GitHub account (if you don't have one): <https://github.com/signup>
2. Reply with your GitHub username
3. We'll add you to the private repository

Once added, you can access:

- Releases: <https://github.com/Anaxiatech-AB/crackz/releases>
- Docker images: https://github.com/anaxiatech?tab=packages&repo_name=crackz
- Documentation: <https://github.com/Anaxiatech-AB/crackz/tree/main/docs>

License: [LICENSE-KEY or CONTRACT-REF]

Now the `release.yml` workflow will automatically publish to Azure Artifacts when configured!

Best regards,
Theresia

Subject: Crackz Access Confirmed

Hi [Contact Name],

You now have access to the Crackz repository!

Installation instructions:
<https://github.com/Anaxiatech-AB/crackz/blob/main/docs/installation.md>

Getting started:
<https://github.com/Anaxiatech-AB/crackz/blob/main/docs/getting-started.md>

Docker registry:
ghcr.io/anaxiatech-ab/crackz:latest

Need help? Email theresia.sofia.snow@gmail.com

Best regards,
Theresia

```
# Find latest version at: https://github.com/Anaxiatech-AB/crackz/releases
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl
```

```
# Download from GitHub Releases page
wget https://github.com/Anaxiatech-AB/crackz/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl

# Install locally
pip install crackz-0.30.2-py3-none-any.whl
```

```
# Install GitHub CLI: https://cli.github.com/
gh auth login

# Download latest release
gh release download --repo anaxiatech/crackz --pattern '*.whl'

# Install
pip install crackz-*.whl
```

```
# Generate GitHub Personal Access Token

1. Go to: https://github.com/settings/tokens/new
```

2. Note: "Crackz Docker Access"
3. Expiration: 1 year (or custom)
4. Scopes: Select ONLY:
 - ☐ read:packages
5. Click "Generate token"
6. Copy the token (starts with ghp_)

```
# Customer runs this once
export CR_PAT=ghp_XXXXXXXXXXXXXXXXXXXX # Their PAT # gitleaks:allow
echo $CR_PAT | docker login ghcr.io -u GITHUB_USERNAME --password-stdin
```

```
# Now they can pull
docker pull ghcr.io/anaxiatech-ab/crackz:latest
docker pull ghcr.io/anaxiatech-ab/crackz:slim
docker pull ghcr.io/anaxiatech-ab/crackz:gpu
```

```
# Use as normal
docker run --rm ghcr.io/anaxiatech-ab/crackz:latest --version
```

- 1.
- 2.
- 3.
- 4.

```
# Via Azure DevOps UI:
# Artifacts > Create Feed
# Name: crackz-packages
# Visibility: Private (organization)
# Upstream sources: Enable PyPI (optional)
```

- 5.
- 6.
- 7.
- o
 - o
- 8.

```
https://pkgs.dev.azure.com/YOUR-ORG/_packaging/crackz-packages/pypi/simple/
```

```
# Settings > Secrets and variables > Actions > New repository secret
```

```
PRIVATE_INDEX_URL: https://pkgs.dev.azure.com/YOUR-ORG/_packaging/crackz-packages/pypi/upload
PRIVATE_INDEX_USER: azure
PRIVATE_INDEX_PASSWORD: <Azure-PAT-with-Packaging-Write-scope> # gitleaks:allow
```

```
publish-release.yml
```

```
# Option 1: Install with explicit index
pip install crackz \
  --index-url https://pkgs.dev.azure.com/YOUR-ORG/_packaging/crackz-packages/pypi/simple/

# Option 2: Configure pip globally
# Linux/macOS: ~/.config/pip/pip.conf
# Windows: %APPDATA%\pip\pip.ini
```

```
[global]
index-url = https://YOUR-PAT@pkgs.dev.azure.com/YOUR-ORG/_packaging/crackz-packages/pypi/simple/
```

```
pip install crackz
crackz --version
```

-
-
-

```
# Install devpi
pip install devpi-server devpi-web

# Initialize server
devpi-init

# Start server
devpi-server --start --host=0.0.0.0 --port=3141

# Create user for yourself
devpi use http://localhost:3141
devpi user -c admin password=SECRET # gitleaks:allow

# Create index
devpi index -c crackz bases=root/pypi
devpi use crackz
```

```
# Install devpi client
pip install devpi-client

# Login
devpi use http://YOUR-SERVER:3141
devpi login admin --password=SECRET # gitleaks:allow
devpi use admin/crackz

# Upload wheel
devpi upload dist/crackz-*.whl
```

```
# Customer configures pip
pip install --index-url http://YOUR-SERVER:3141/admin/crackz/+simple/ crackz
```

```
docker run -p 3141:3141 mucgcg/devpi
```

- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.

```
Subject: Crackz License Expired - Access Revoked

Hi [Contact],

Your Crackz license expired on [DATE].
Repository and package access has been revoked.

To renew, please contact theresia.sofia.snow@gmail.com

Thank you for using Crackz.
```

- 1.
- 2.
- 3.

- 1.
 - 2.
 - 3.
 - 4.
-

```
# Ask customer to verify:
1. Go to: https://github.com/Anaxiatech-AB/crackz
2. If you see "404 Not Found", you don't have access yet
3. Reply with your GitHub username for access
```

```
read:packages
```

```
# Ask customer to:
1. Generate new PAT: https://github.com/settings/tokens/new
2. Select ONLY: read:packages
3. Run: echo PAT | docker login ghcr.io -u USERNAME --password-stdin
4. Try again: docker pull ghcr.io/anaxiatech-ab/crackz:latest
```

```
# Recommend:
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl
```

```
# You prepare:
pip download crackz -d wheelhouse
tar -czf crackz-offline-bundle.tar.gz wheelhouse/

# Customer installs:
tar -xzf crackz-offline-bundle.tar.gz
pip install --no-index --find-links wheelhouse crackz
```

1.

```
# Customer pulls and retags
docker pull ghcr.io/anaxiatech-ab/crackz:latest
docker tag ghcr.io/anaxiatech-ab/crackz:latest registry.company.com/crackz:latest
docker push registry.company.com/crackz:latest
```

2.

```
# Get release download counts
gh api repos/anaxiatech/crackz/releases \
--jq '.[] | {name: .name, downloads: [.assets[].download_count] | add}'
```

```
{"name": "v0.30.2", "downloads": 47}
{"name": "v0.30.1", "downloads": 23}
```

```
# Requires GitHub GraphQL API (complex)
# Easier to use web UI for now
```

```
# Crackz Usage Report - October 2025
```

```
## Active Customers: 5
```

Customer	License	Status	Last Access
Acme Corp	LIC-001	Active	2025-10-08
Beta Inc	LIC-002	Active	2025-10-05
Gamma LLC	LIC-003	Expired	2025-09-30

```
## Download Statistics
```

```
- Total releases: 12
- Total downloads (wheels): 156
- Docker pulls (latest): 89
- Docker pulls (gpu): 34
```

```
## Top Versions
```

```
1. v0.30.2: 47 downloads
2. v0.30.1: 23 downloads
```

3. v0.30.0: 18 downloads

Action Items

- [] Contact Gamma LLC for renewal
- [] Send v0.31.0 release announcement

- 1.
2. PRIVATE_INDEX_*
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.
- 11.
- 12.

- 1.
- 2.
3. read:packages
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.
- 11.
- 12.


```
#!/bin/bash
# scripts/customer-access/onboard-customer.sh

set -euo pipefail

CUSTOMER_NAME="$1"
GITHUB_USERNAME="$2"
LICENSE_KEY="$3"

echo "👤 Onboarding customer: $CUSTOMER_NAME"
echo "GitHub username: $GITHUB_USERNAME"
echo "License: $LICENSE_KEY"

# Create GitHub team (requires gh CLI with admin access)
TEAM_SLUG="customer-$(echo $CUSTOMER_NAME | tr '[:upper:]' '[:lower:]' | tr ' ' '-')"
gh api orgs/YOUR-ORG/teams -f name="Customer-$CUSTOMER_NAME" -f privacy=secret

# Add member to team
gh api orgs/YOUR-ORG/teams/$TEAM_SLUG/memberships/$GITHUB_USERNAME -X PUT

# Add team to repository
gh api orgs/YOUR-ORG/teams/$TEAM_SLUG/repos/YOUR-ORG/crackz -X PUT -f permission=pull

echo "👤 Customer onboarded successfully!"
echo ""
echo "Next steps:"
echo "1. Send welcome email to customer"
echo "2. Add to tracking spreadsheet"
echo "3. Schedule renewal reminder for 1 year from now"
```

```
#!/bin/bash
# scripts/customer-access/revoke-customer.sh

set -euo pipefail

CUSTOMER_NAME="$1"
REASON="${2:-License expired}"

echo "👤 Revoking access for: $CUSTOMER_NAME"
echo "Reason: $REASON"

TEAM_SLUG="customer-$(echo $CUSTOMER_NAME | tr '[:upper:]' '[:lower:]' | tr ' ' '-')"

# Remove team from repository
gh api orgs/YOUR-ORG/teams/$TEAM_SLUG/repos/YOUR-ORG/crackz -X DELETE

# Optionally delete team entirely
read -p "Delete team entirely? (y/n) " -n 1 -r
echo
if [[ $REPLY =~ ^[Yy]$ ]]; then
    gh api orgs/YOUR-ORG/teams/$TEAM_SLUG -X DELETE
    echo "👤 Team deleted"
else
    echo "👤 Team kept (access removed from repository only)"
fi

echo "👤 Access revoked successfully!"
echo ""
echo "Next steps:"
echo "1. Send termination notice to customer"
echo "2. Update tracking spreadsheet"
echo "3. Archive customer records"
```

```
pip install crackz
```

```
PRIVATE_INDEX_*
```

Printable PDF

Printable / PDF Export

site/crackz-docs.pdf

The cover automatically injects the current project version from `pyproject.toml` and © Anaxiatech AB.

```
uv run task pdf
```

```
mkdocs build --clean --strict python scripts/export_pdf.py
```

site/crackz-docs.pdf

```
make docs-pdf
```

```
uv run task pdf_fast
```

```
site/crackz-docs-fast.pdf
```

```
pwsh scripts/windows/export_pdf.ps1 -Outline -CoverTitle "Crackz Documentation"
```

```
-NoCover
```

```
-NoHeader
```

```
uv run mkdocs build --clean --strict
uv run playwright install chromium # one time
uv run python scripts/export_pdf.py \
  --outline \
  --cover-image 'assets/crackz-logo.png' \
  --logo 'assets/crackz-logo.png' \
  --cover-title 'Crackz Documentation' \
  --cover-subtitle 'AI Image Analysis' \
  --out site/crackz-docs.pdf
```

```
uv run python scripts/export_pdf.py --out site/my-custom-docs.pdf
```

nav :

--no-bookmarks

<article>

```
- name: Build docs + PDF
run: |
  uv sync --dev
  uv run mkdocs build --clean --strict
  uv run playwright install chromium
  uv run python scripts/export_pdf.py --outline --cover-image 'assets/crackz-logo.png' --logo 'assets/crackz-logo.png' --out site/crackz-docs.pdf
- uses: actions/upload-artifact@v4
  with:
    name: crackz-docs-pdf
    path: site/crackz-docs.pdf
```

https://<your-user>.github.io/crackz/crackz-docs.pdf

License

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